Home Range Estimation

Continuous-Time Animal Movement Modeling

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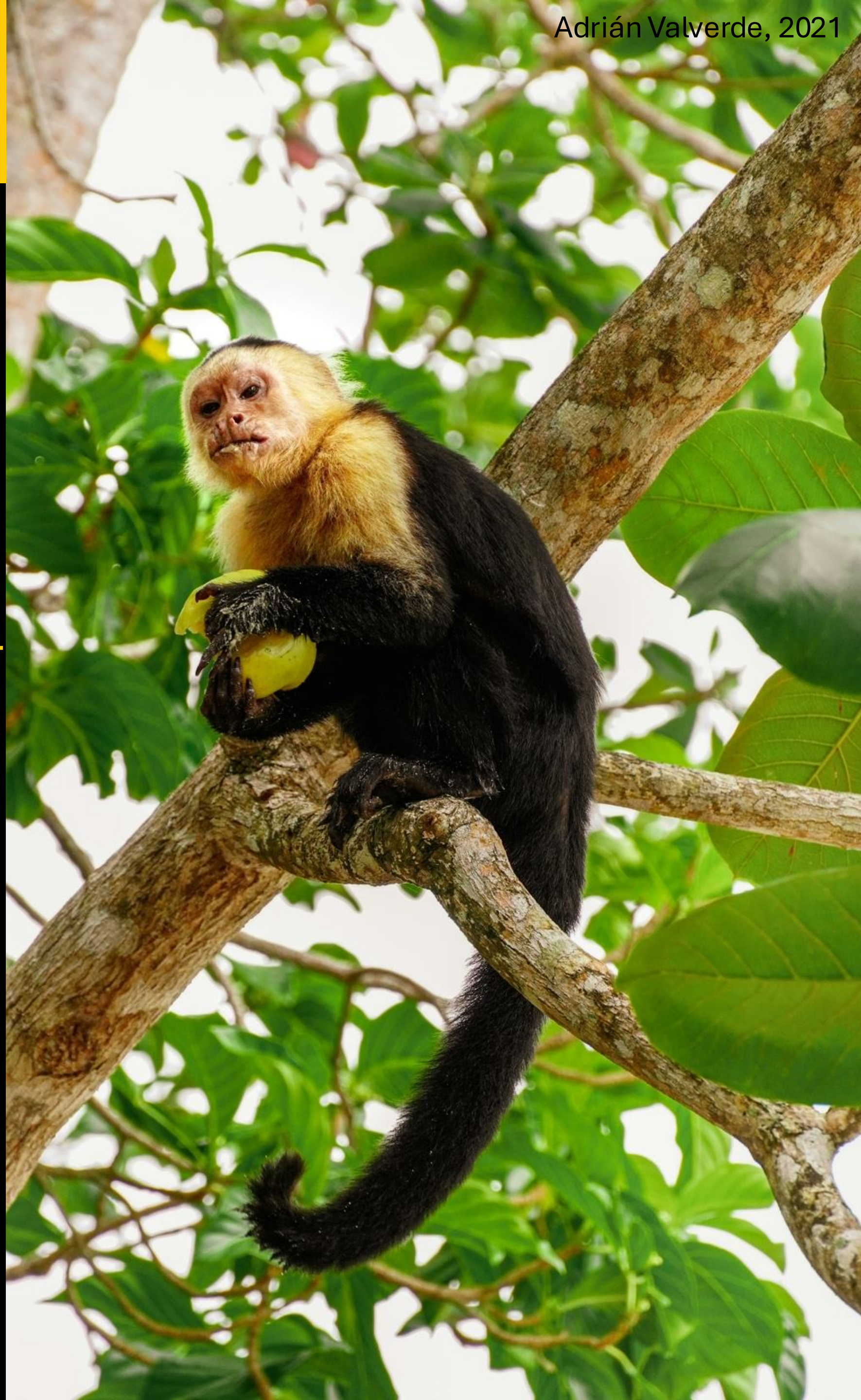

Overview

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1) What is a home range?
2) Why do these concepts matter?
3) Range vs occurrence distributions
4) Methods for home range estimation
5) Effective sample size
6) HR estimation workflow
7) Mitigating bias

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1) **What is a home range?**
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Analyzing movement behaviors

Geoffrey Reynaud

Which areas of a landscape contain high-priority resources (e.g., migratory corridors/stopover sites, prey or foraging grounds)?

Analyzing movement behaviors

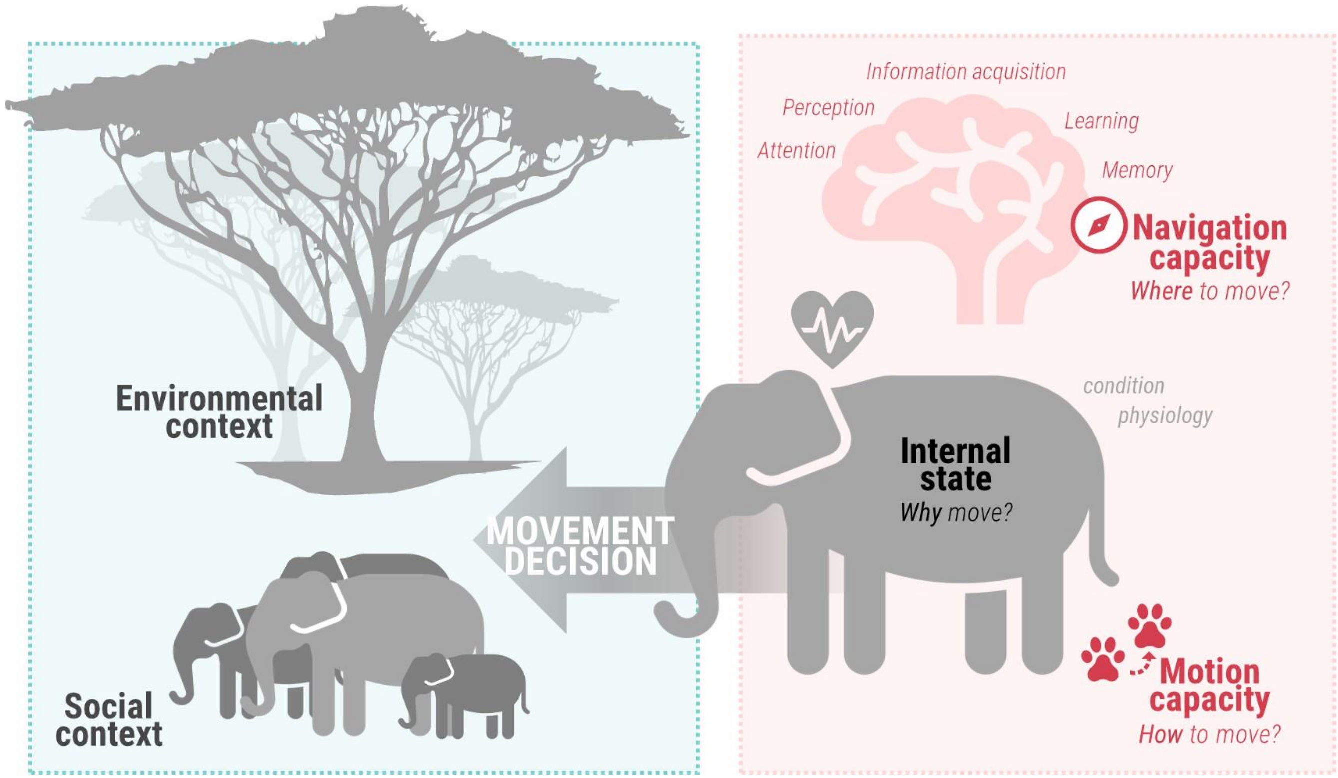
Nick Upton

Which areas of a landscape contain high-priority resources (e.g., migratory corridors/stopover sites, prey or foraging grounds)?

How likely is it to visit a location of interest (e.g., wildlife crossing sites, wind farms)?

Analyzing movement behaviors

- Characterizing spatial-temporal patterns of space use is fundamental



Adapted from **Lewis *et al.* (2021)**
DOI: 10.3389/fevo.2021.681704

What is a home range?

- **Home ranging behavior** is a prevalent pattern in space use

Konrad Wothe

“

(...) it may be here remarked that most animals and plants keep to their proper homes, and do not needlessly wander about; we see this even with migratory birds, which almost always return to the same spot.

Darwin (1861)

Size and configuration of an animal’s home range are key to understanding its *space-use requirements*

What is a home range?

First defined as:

> (...) the area traversed by the individual in its normal activities of food gathering, mating, and caring for young. Occasional sallies outside the area, perhaps exploratory in nature, should not be considered as in part of the home range.
>
> Burt (1943)

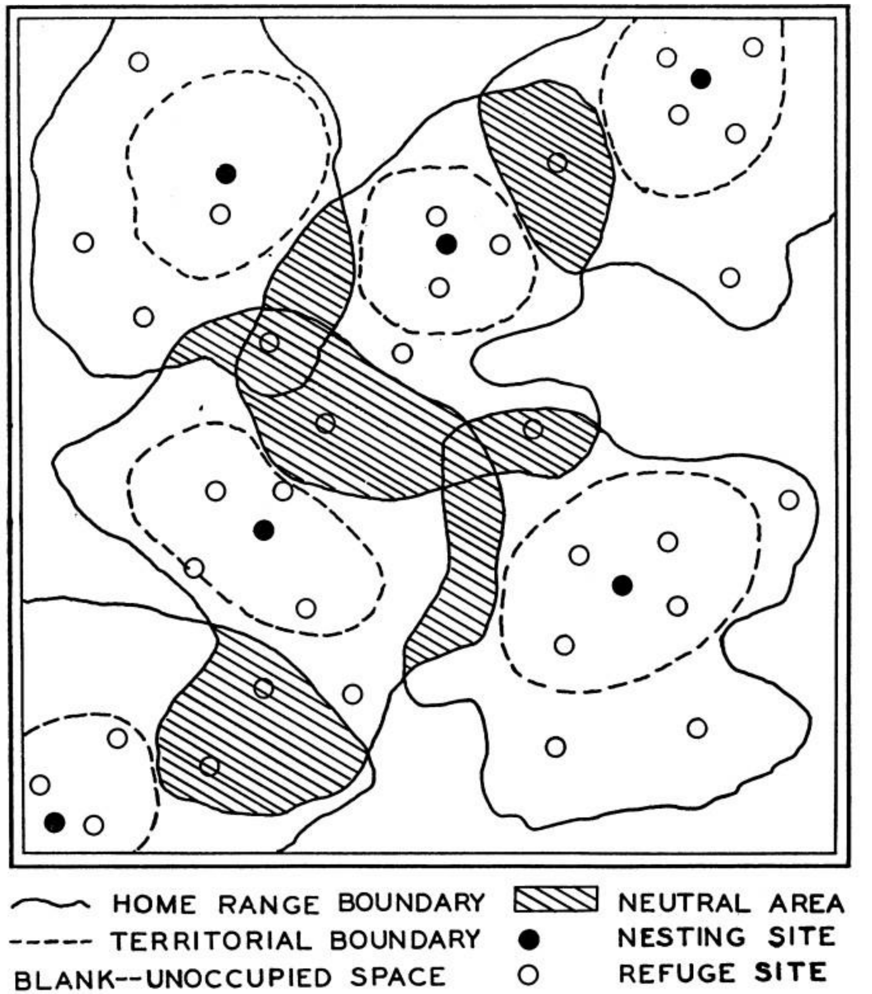


FIG. 1. Theoretical quadrat with six occupants of the same species and sex, showing territory and home range concepts as presented in text.

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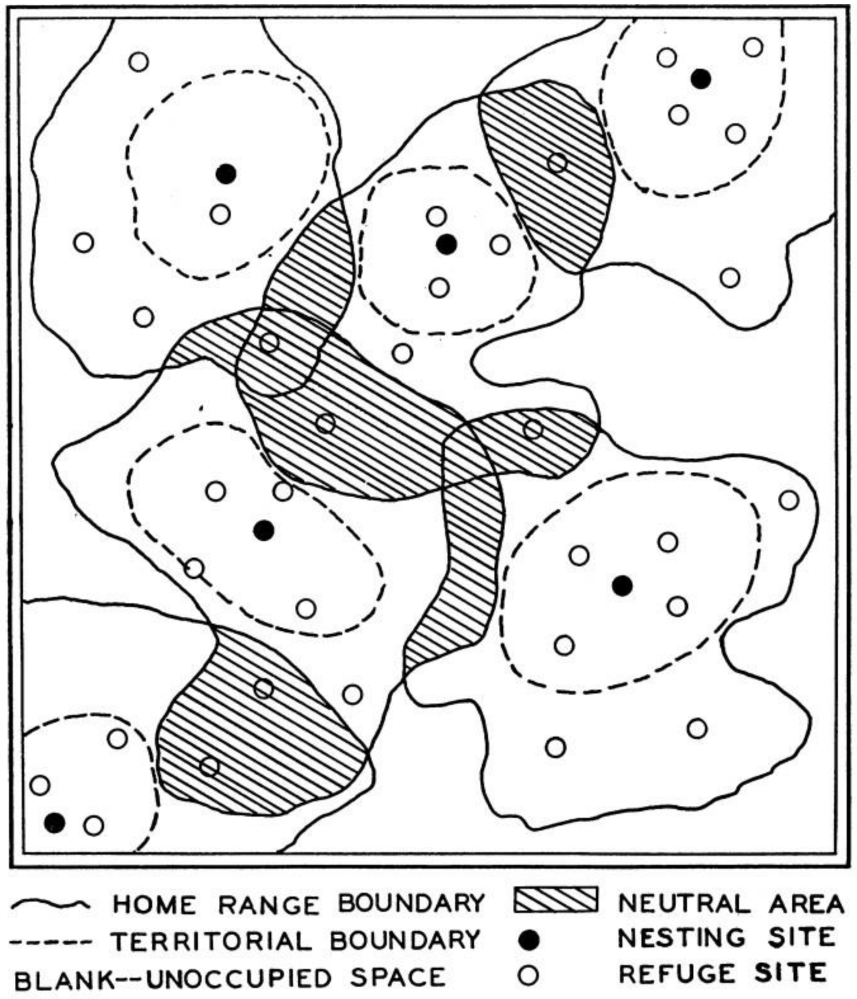
Home range ≠ *Territory*

not actively defended — actively defended



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- How to quantify a home range area?
- What constitutes an exploratory movement?
- How to identify these exploratory movements?

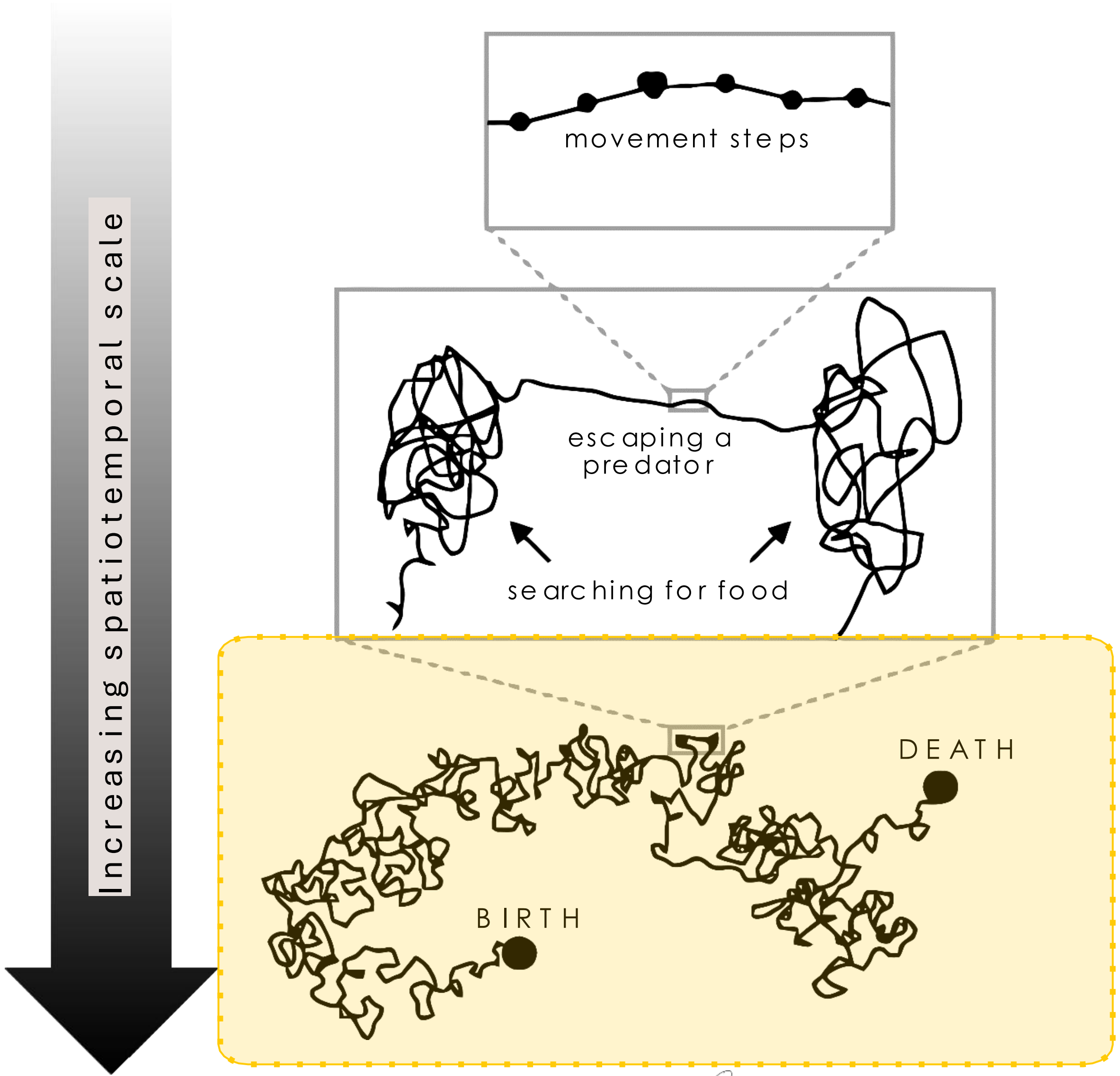
In practice, it is hard to define when a movement is purely exploratory

Excursion of infrequent *outlying relocation points.*

What is a home range?

Home range:

- The area repeatedly used throughout an animal's **lifetime** for all its *normal behaviors and activities*, excluding occasional exploratory excursions



Adapted from **Nathan *et al.* (2008)**

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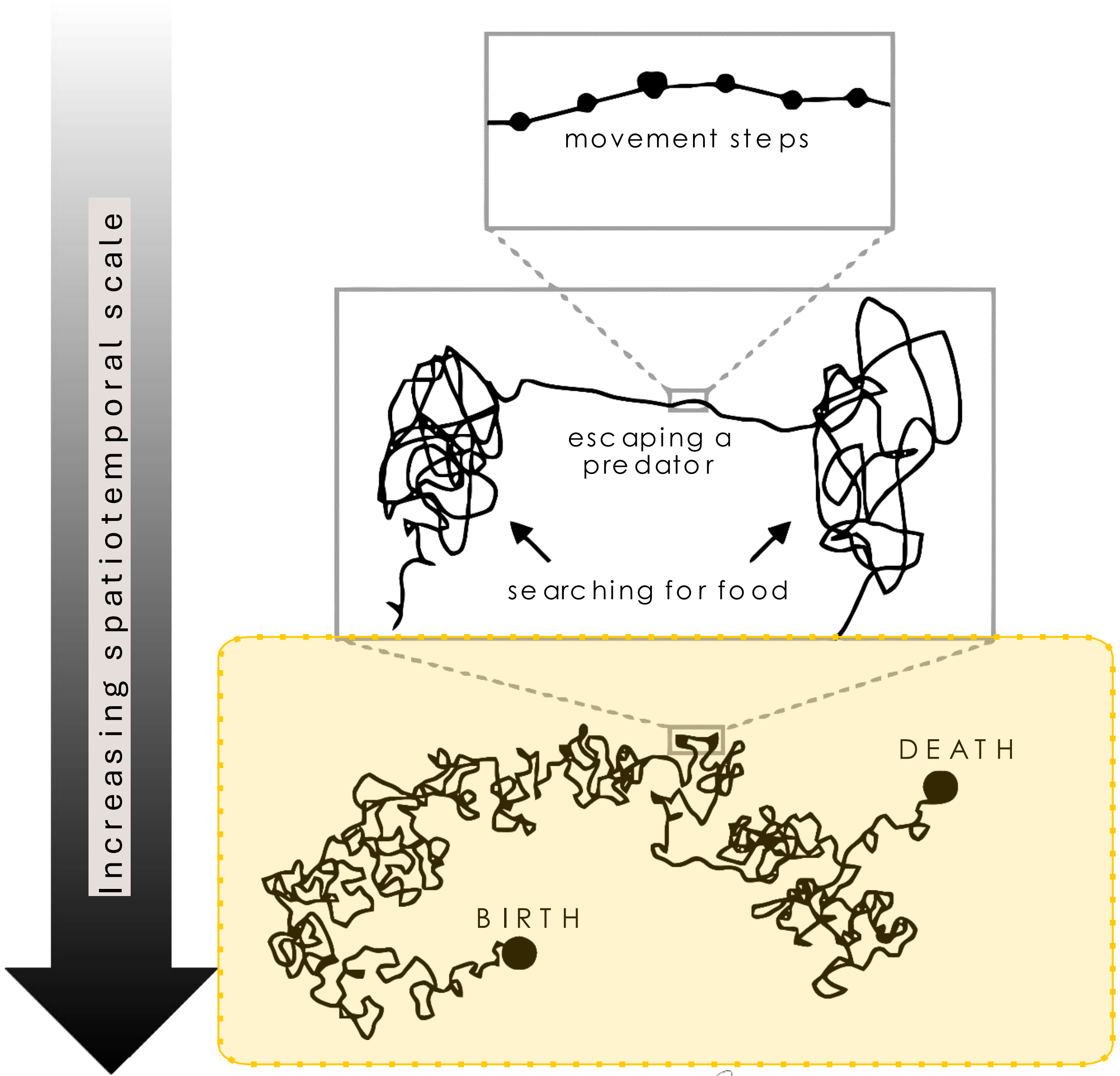
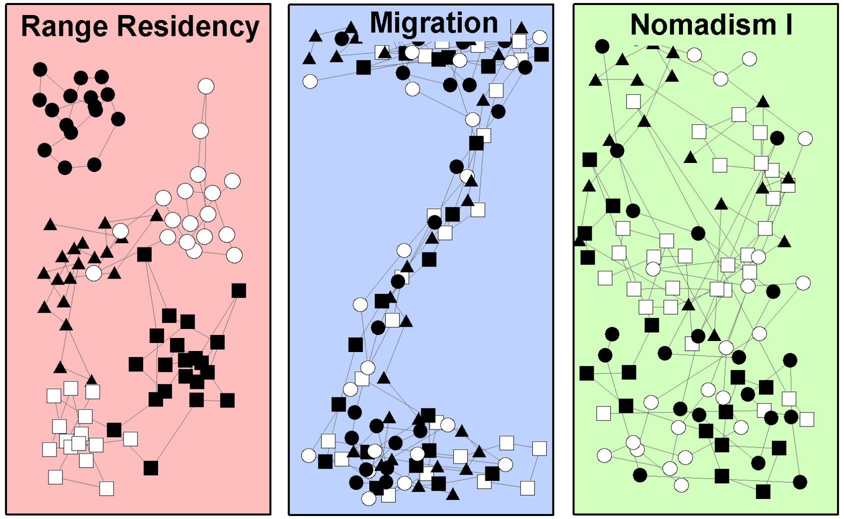
Home range area can be expected to include future locations.



Adapted from **Nathan *et al.* (2008)**

What is a home range?

- However, not all animals have home ranges…
- 4 main behaviors that animals may exhibit:
  - Occupies the same area throughout its lifetime (i.e., range resident)
  - Regular movement to and from spatially disjoint ranges (migratory)
  - Does not follow regular temporal and spatial patterns (nomadic)
  - Permanently moves from one area to another (dispersive)



Mueller *et al.* (2008)

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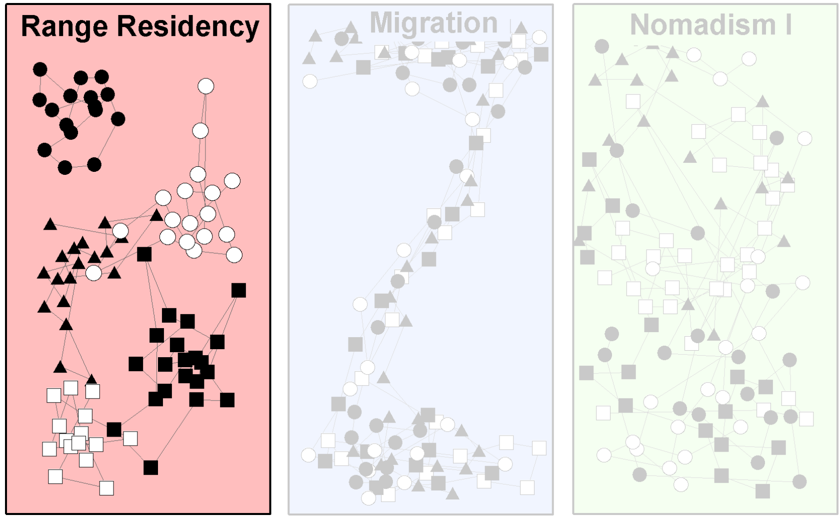
Intra-individual variation
Inter-individual variation
Inter-population variation

Or

Sampling schedule



Mueller *et al.* (2008)

Applications

Home-range estimates may inform:

- Protected area delineation
- Land-use decisions
- Conservation policy and initiatives (e.g., related to human-wildlife conflict)

It is vital to accurately capture the area repeatedly used throughout an *animal's lifetime.*

Dennis Hamilton

Avijan Saha

Aleksei Volkov

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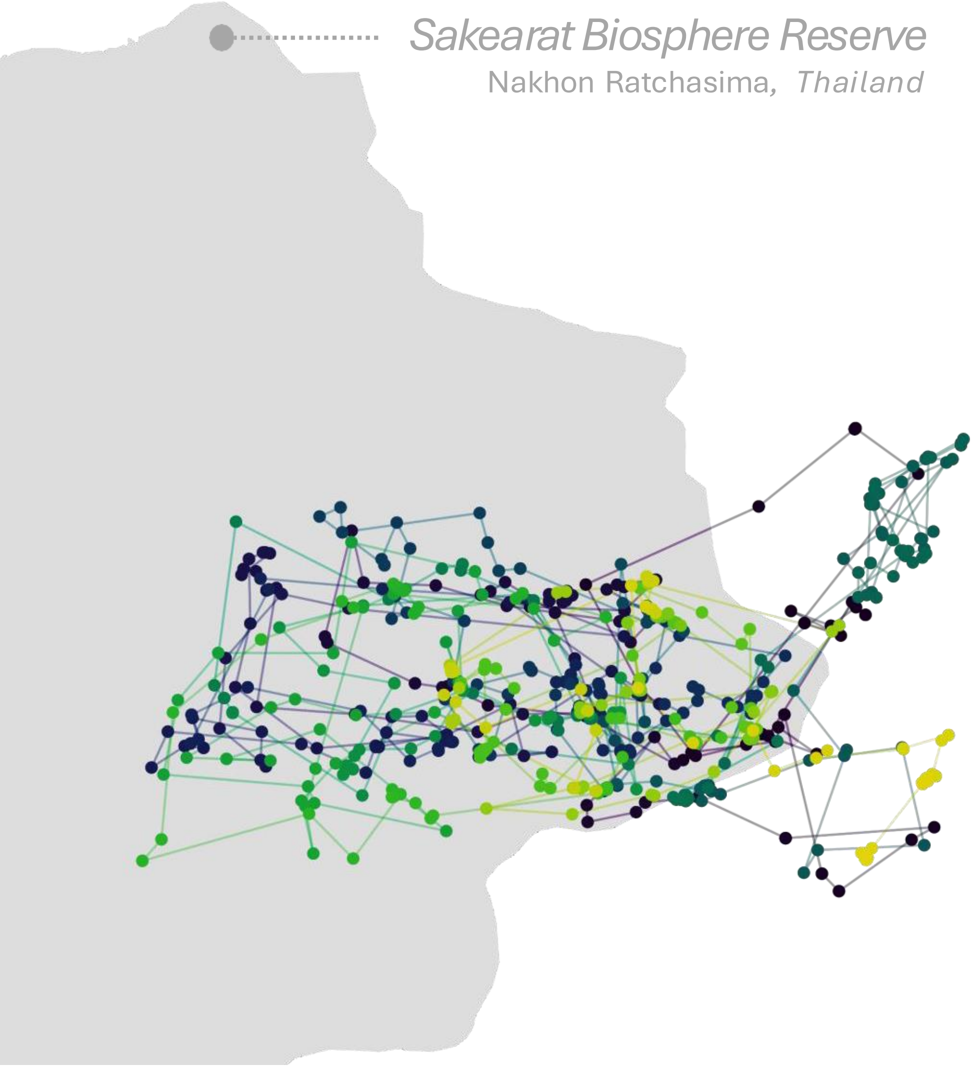
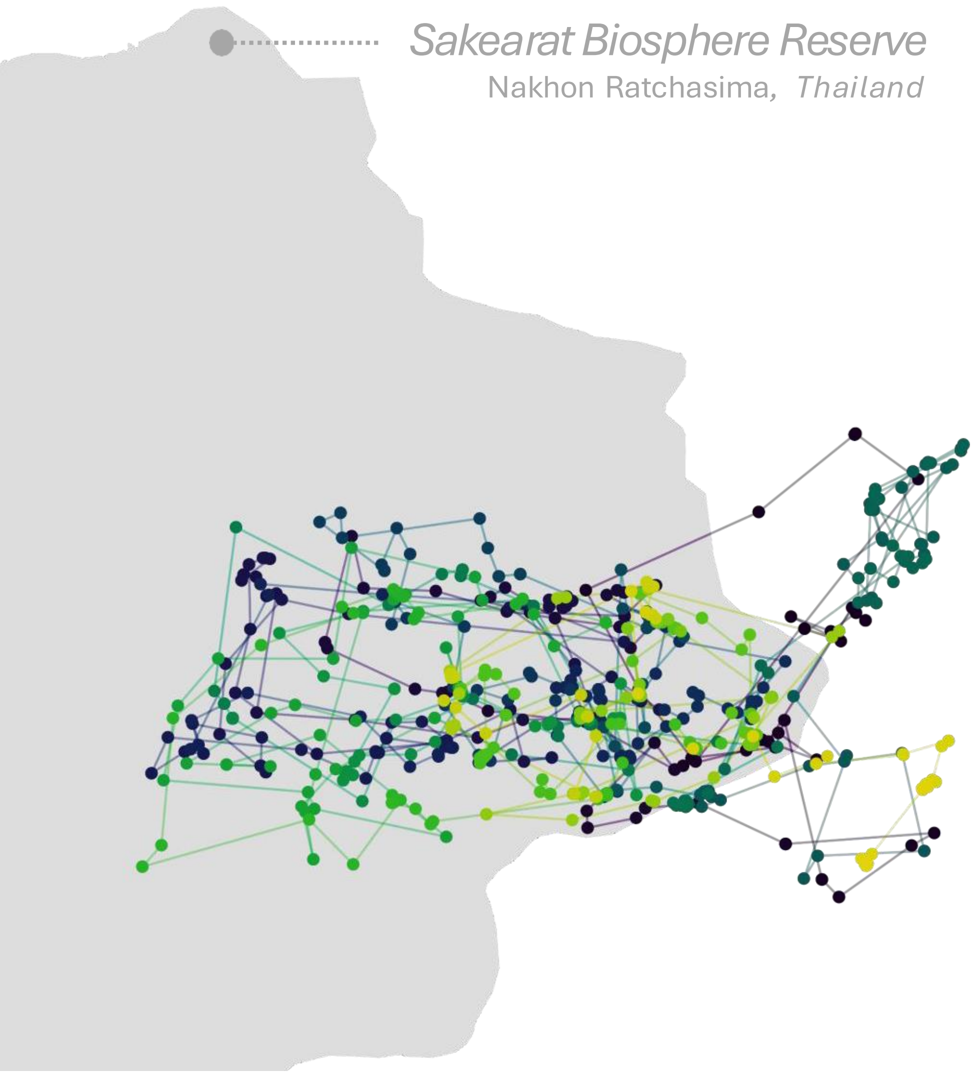
Why do these concepts matter?



Benjamin Marshall

Northern king cobra
(Ophiophagus hannah)

Why do these concepts matter?



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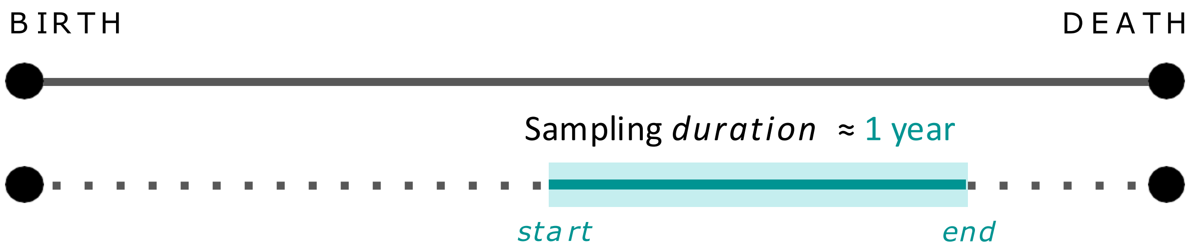
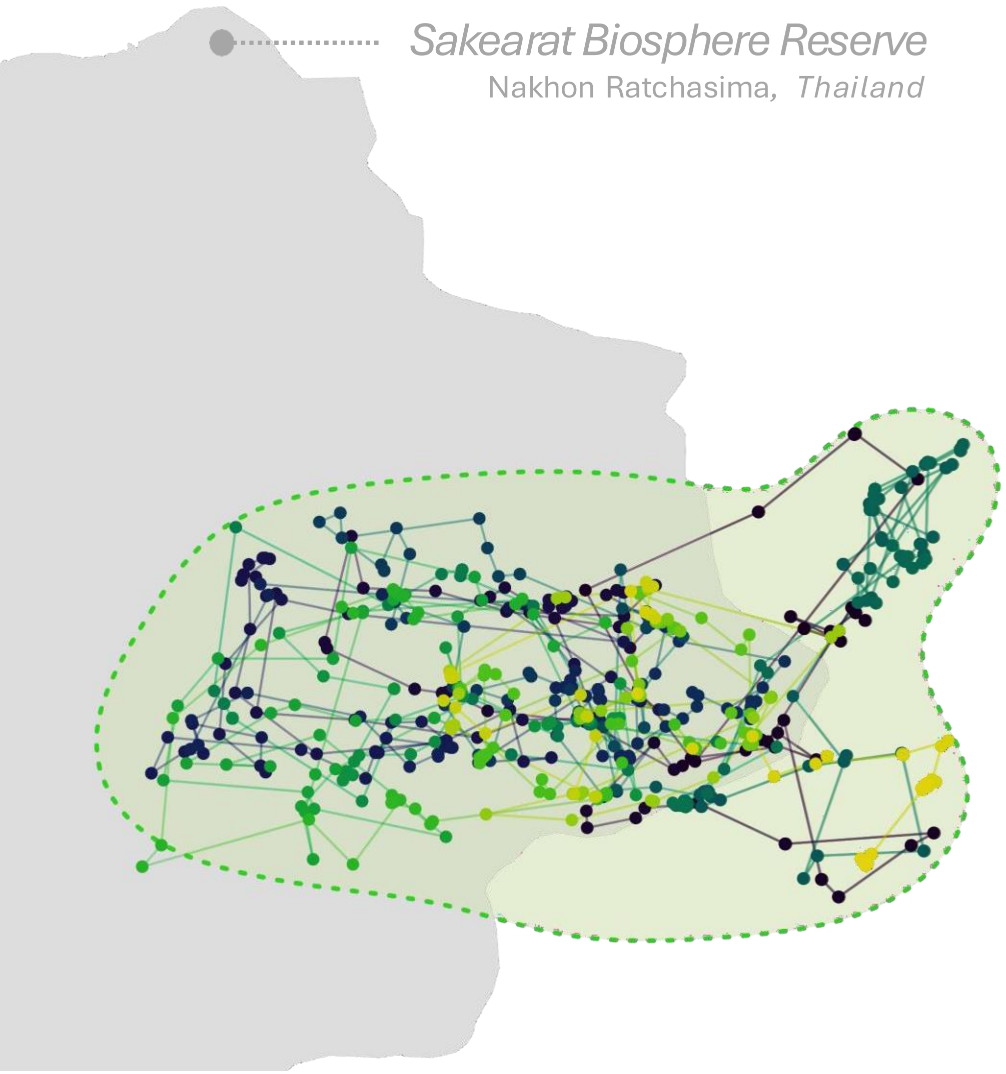


How representative is this period?
Is the movement behavior stationary?

Why do these concepts matter?



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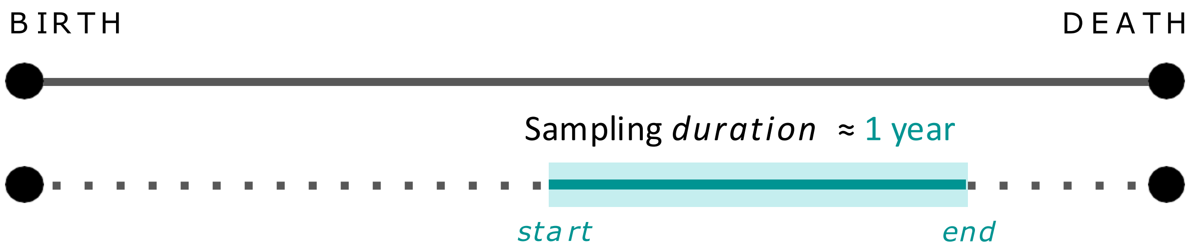

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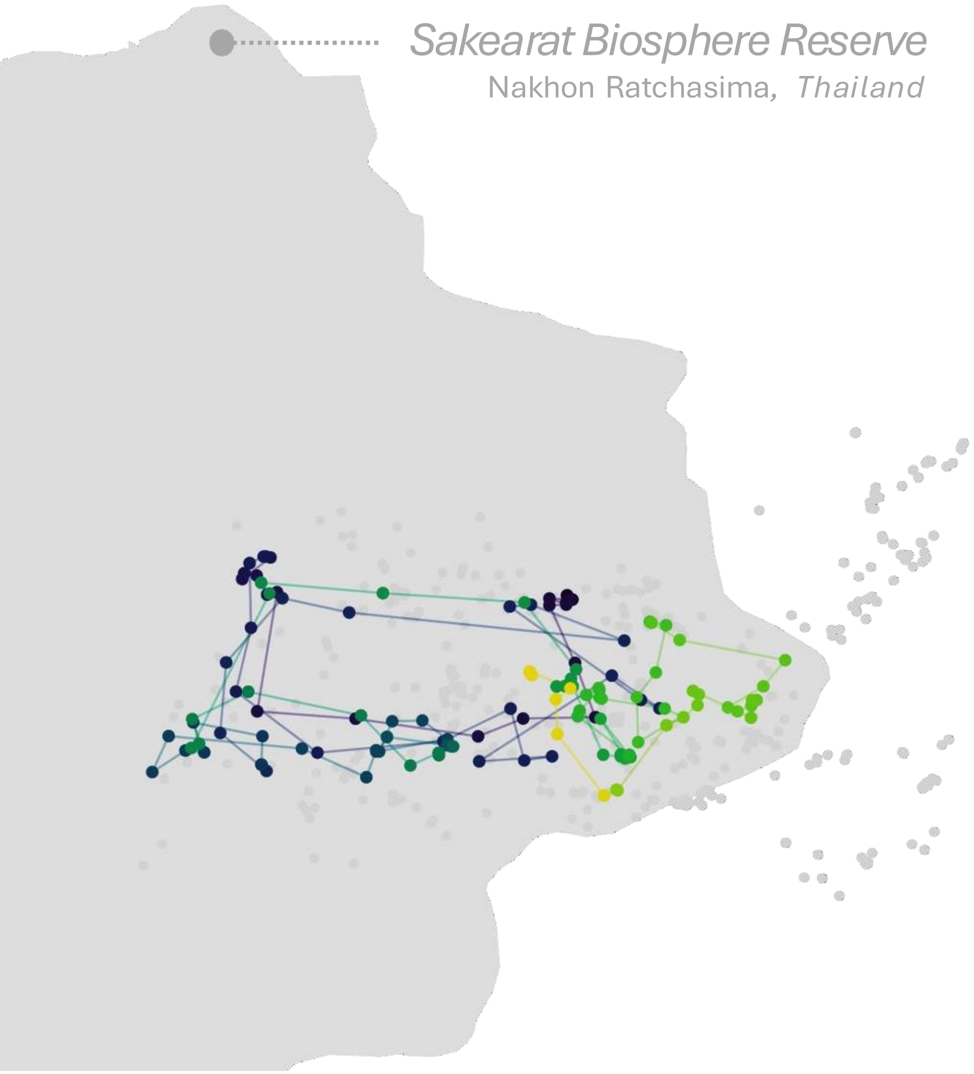

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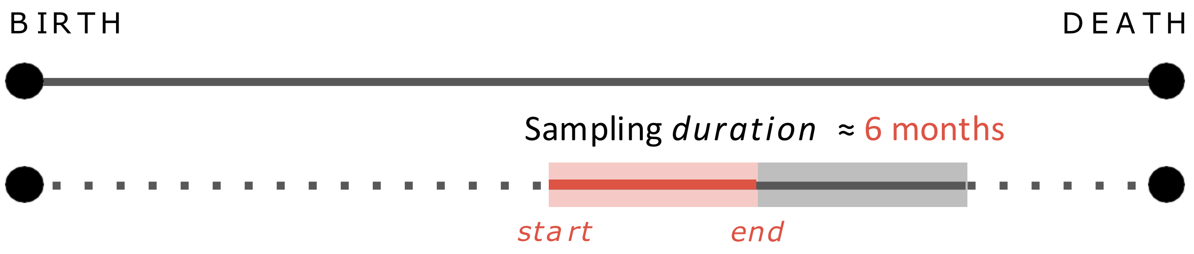

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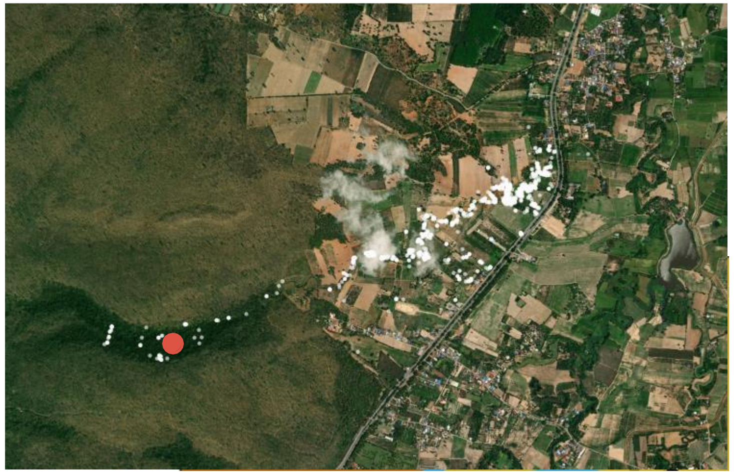
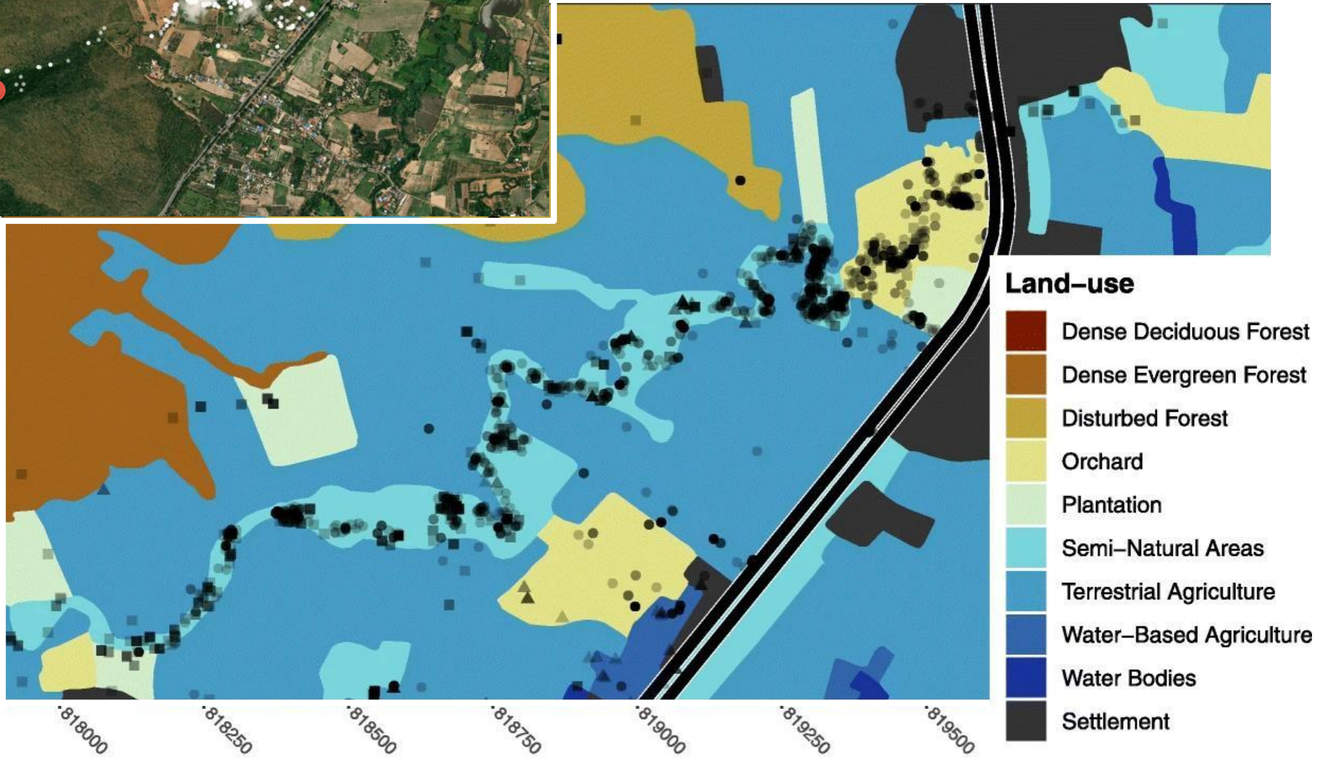


Marshall et al. (2020)

doi.org/10.1186/s40462-020-00219-5



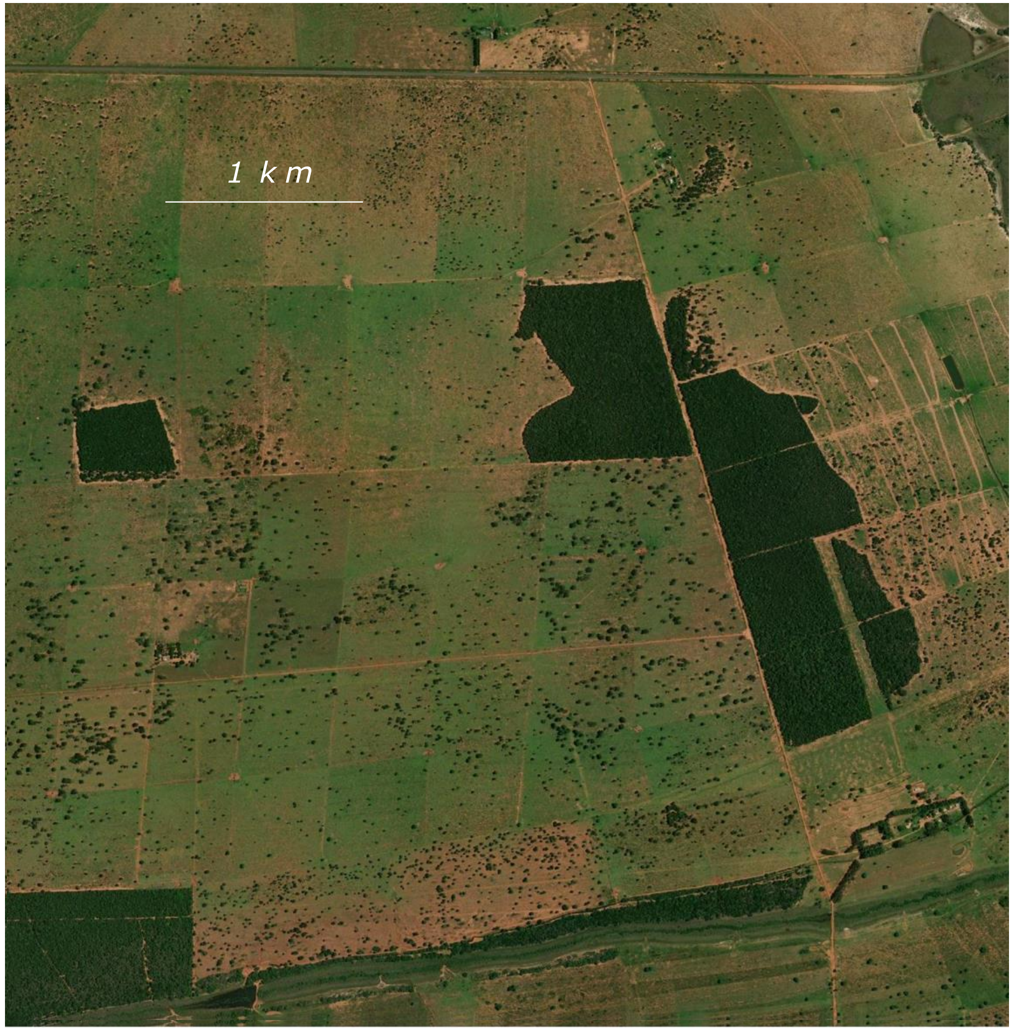
Movements are largely occurring within semi-natural areas.

Females cobras almost exclusively utilize agricultural canals as *movement corridors*.

Why do these concepts matter?

João Vitor Andriola

Giant anteater
(Myrmecophaga tridactyla)



Why do these concepts matter?

Fragmentation creates patches of suitable habitat surrounded by unsuitable habitat.

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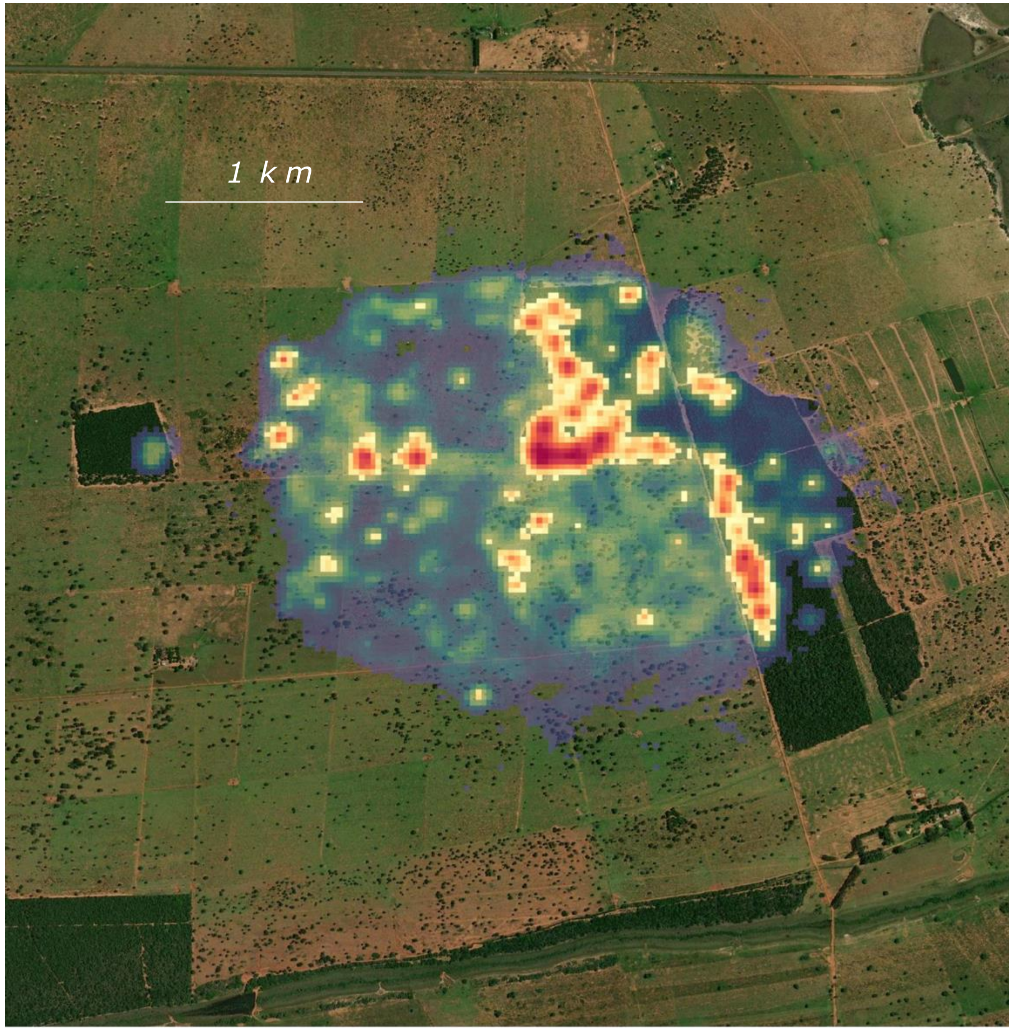
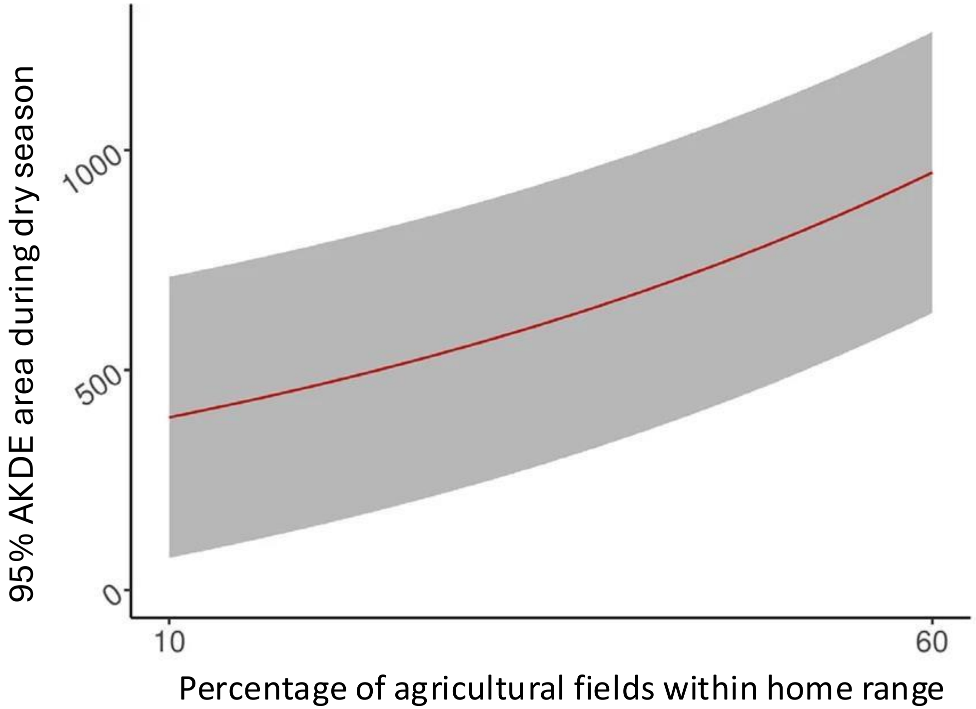


Why do these concepts matter?

- Space-use requirements increase during the *dry season* (when human-elephant interactions are the most likely to occur).

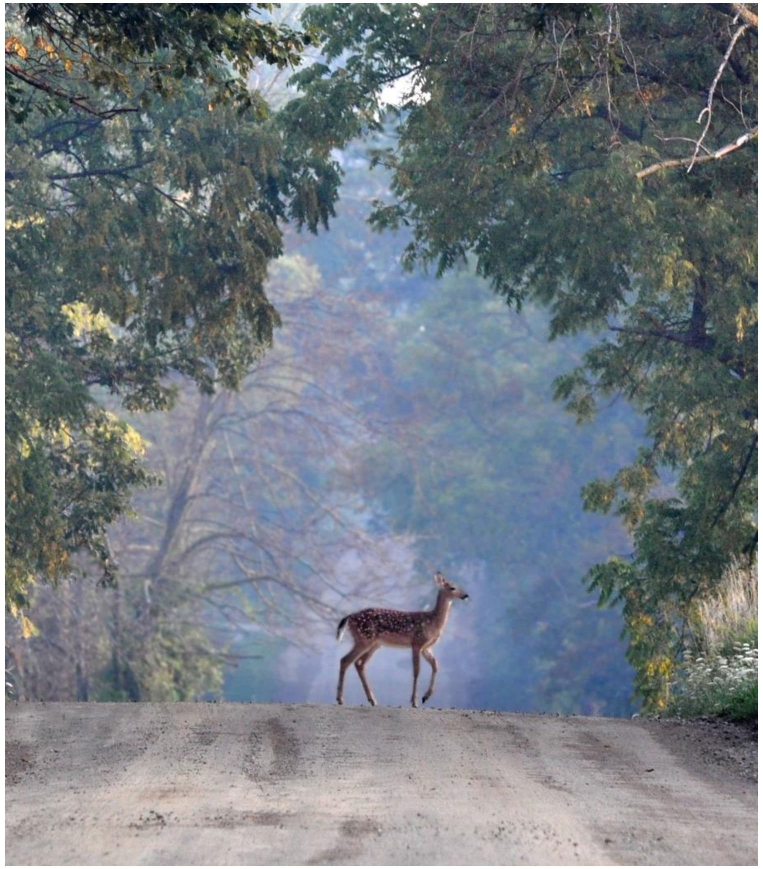
Asian elephant
(Elephas maximus)

Yathin S. Krishnappa



Analyzing movement behaviors

Julie Falk



- **Roads** have significant impacts on ecological systems, increasing the extinction risk of threatened species

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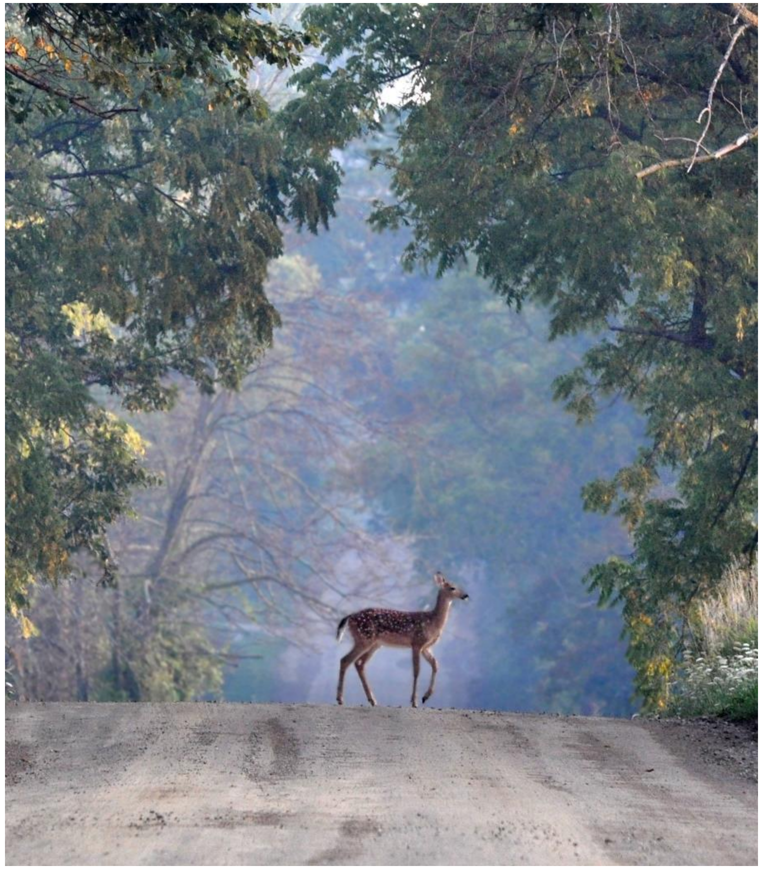

- **Roads** have significant impacts on ecological systems, increasing the extinction risk of threatened species

Where and when did an animal cross a linear feature or barrier (e.g., road- or railways)?

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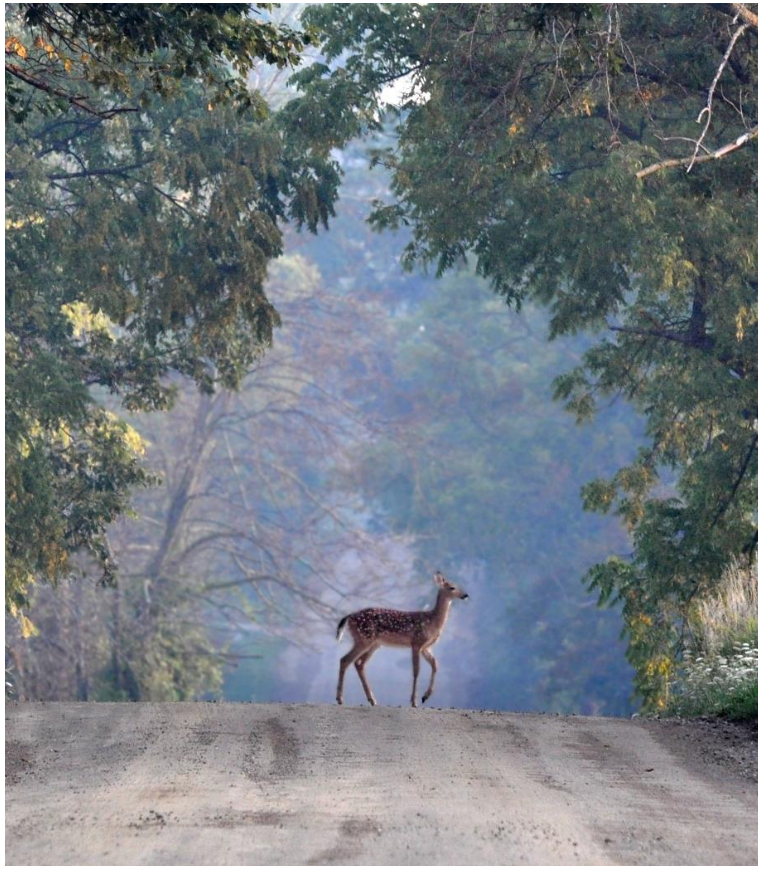

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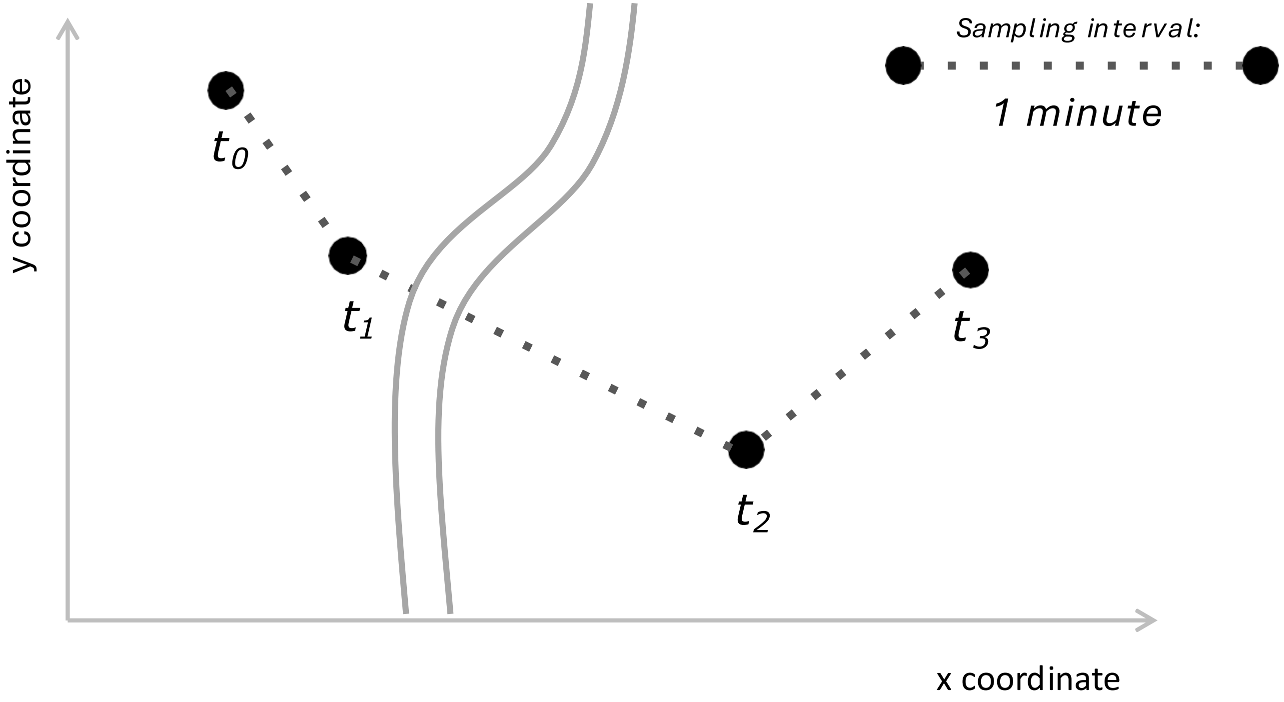

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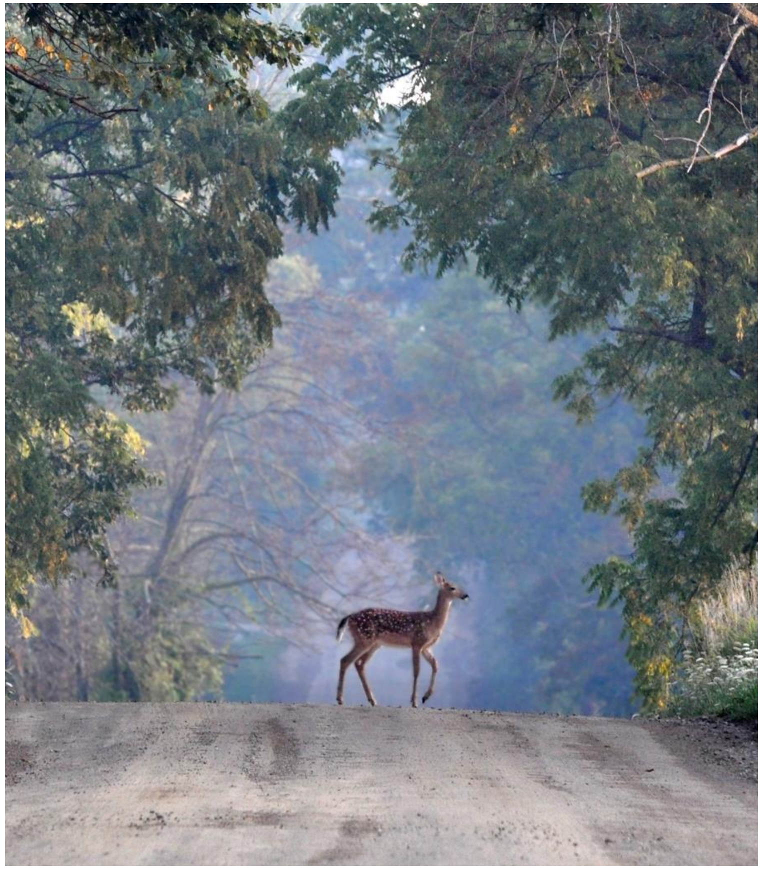

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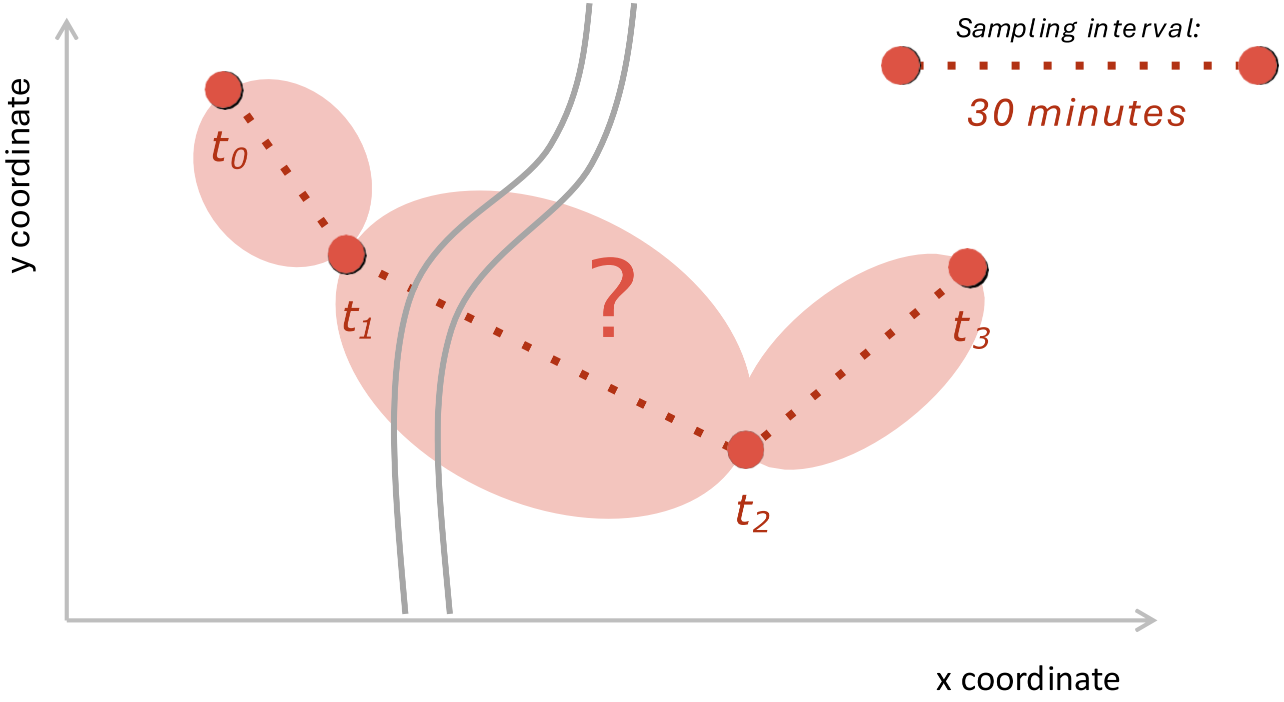

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Utilization distributions

Alston *et al.* (2022)

A **utilization distribution** (UD) represents an animal's distribution and the probability of use throughout an area.

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Range distribution

- *Extrapolate space use into the future*

Occurrence distribution

- *Interpolate between data points in the past*

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- *Extrapolate space use into the future*

- How much space does an animal need over long term?
  - What is an animal's home range?
  - What is the population range area?
  - Are protected areas sufficiently large and appropriately established?

Occurrence distribution

- *Interpolate between data points in the past*

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Occurrence distribution

- *Interpolate between data points in the past*
- Where did the animal go during a period of observation?
  - Where did an animal cross a linear feature?
  - How likely is it to visit a location of interest?
  - How much time did it spend in a specific habitat?

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Range distribution ≠ **Occurrence** distribution

Range distribution

- *Extrapolate space use into the future*

Estimates **home ranges**

Considers all realizations of the movement process that could occur

Occurrence distribution

- *Interpolate between data points in the past*

Estimates **occurrence regions**

Pinpoints the one realization of the movement process that did occur

Estimating occurrence

Fleming *et al.* (2016)

- Generalized time-series **Kriging** framework
  - "*Kriging* is a widely used tool in geostatistics, engineering, and computer science. It is a statistically optimal framework for *interpolating between discrete locations*, with well-known statistical properties."

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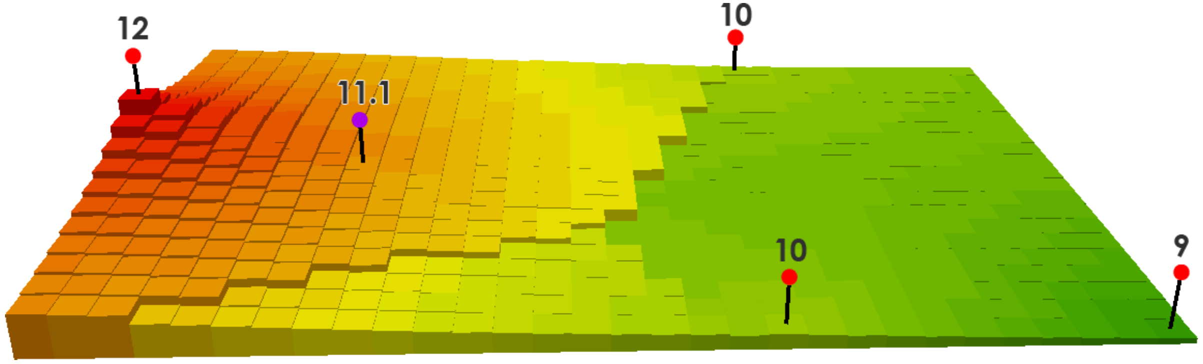
Fig. spatially interpolated temperature from weather stations.

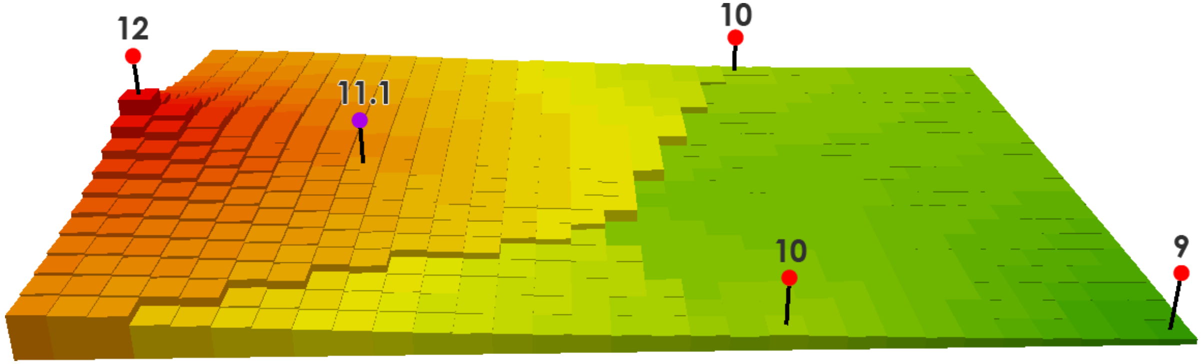


Estimating occurrence

Fleming *et al*. (2016)

- Generalized time-series **Kriging** framework
  - "*Kriging* is a widely used tool in geostatistics, engineering, and computer science. It is a statistically optimal framework for *interpolating between discrete locations*, with well-known statistical properties."
- Encompasses both:
  - *Brownian Bridge Movement Model* (BBMM; Horne et al., 2017)
  - *Correlated Random Walk library* (CRAWL; Johnson et al., 2008)

Fig. spatially interpolated temperature from weather stations.



Estimating occurrence

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  - Not only does Kriging provide an optimal *prediction surface*, but it also delivers a *measure of confidence* of how likely that prediction will be true.

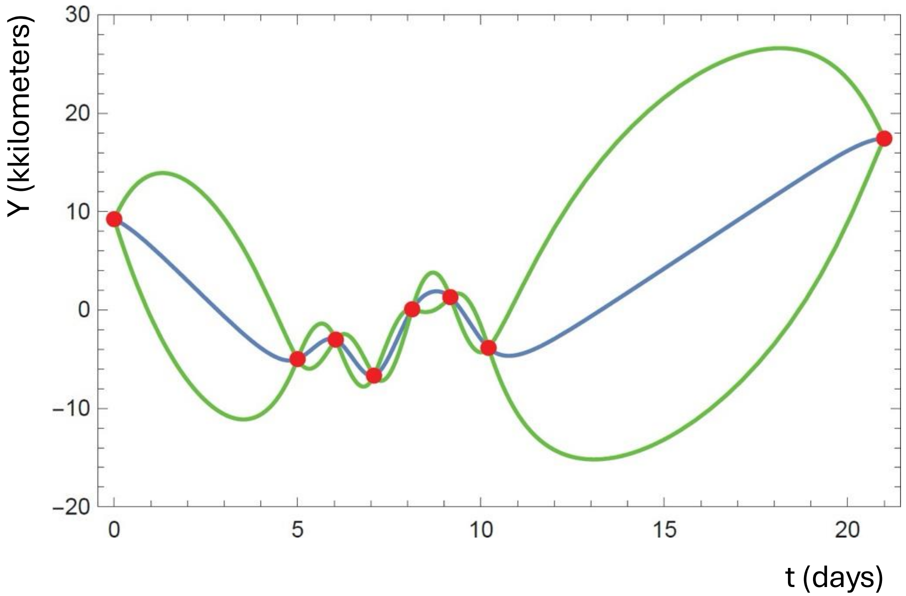
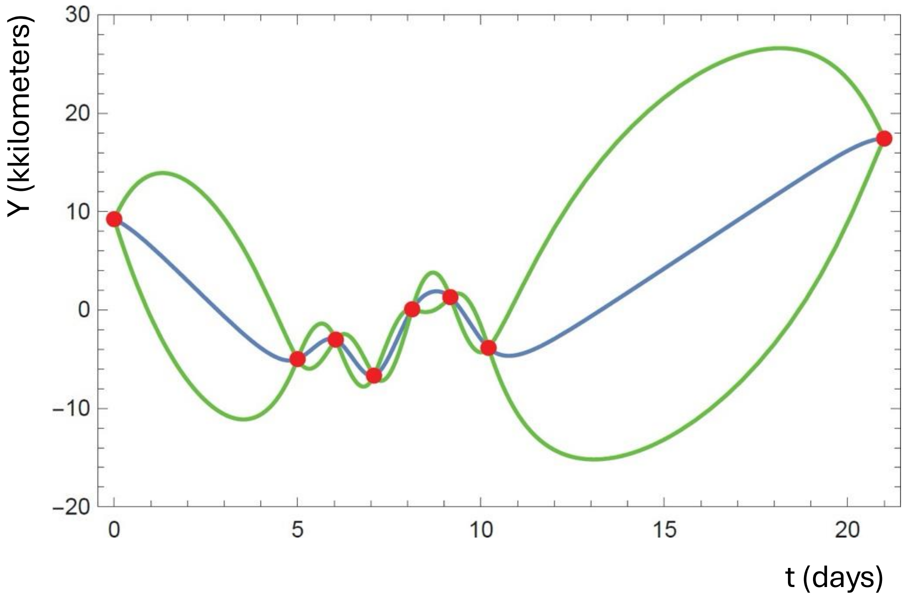


Fig. Time series of Mongolia Gazelle locations (red), with Krige interpolated locations (blue) and 95% contour intervals (green).

Estimating occurrence

Fleming *et al*. (2016)

- Generalized time-series **Kriging** framework
  - Not only does Kriging provide an optimal *prediction surface*, but it also delivers a *measure of confidence* of how likely that prediction will be true.
- Two-step process:
  1) *Select a movement model* (describing an animal's movements)
  2) *Solve for an animal's location at time t* (conditional upon the data and fitted model)



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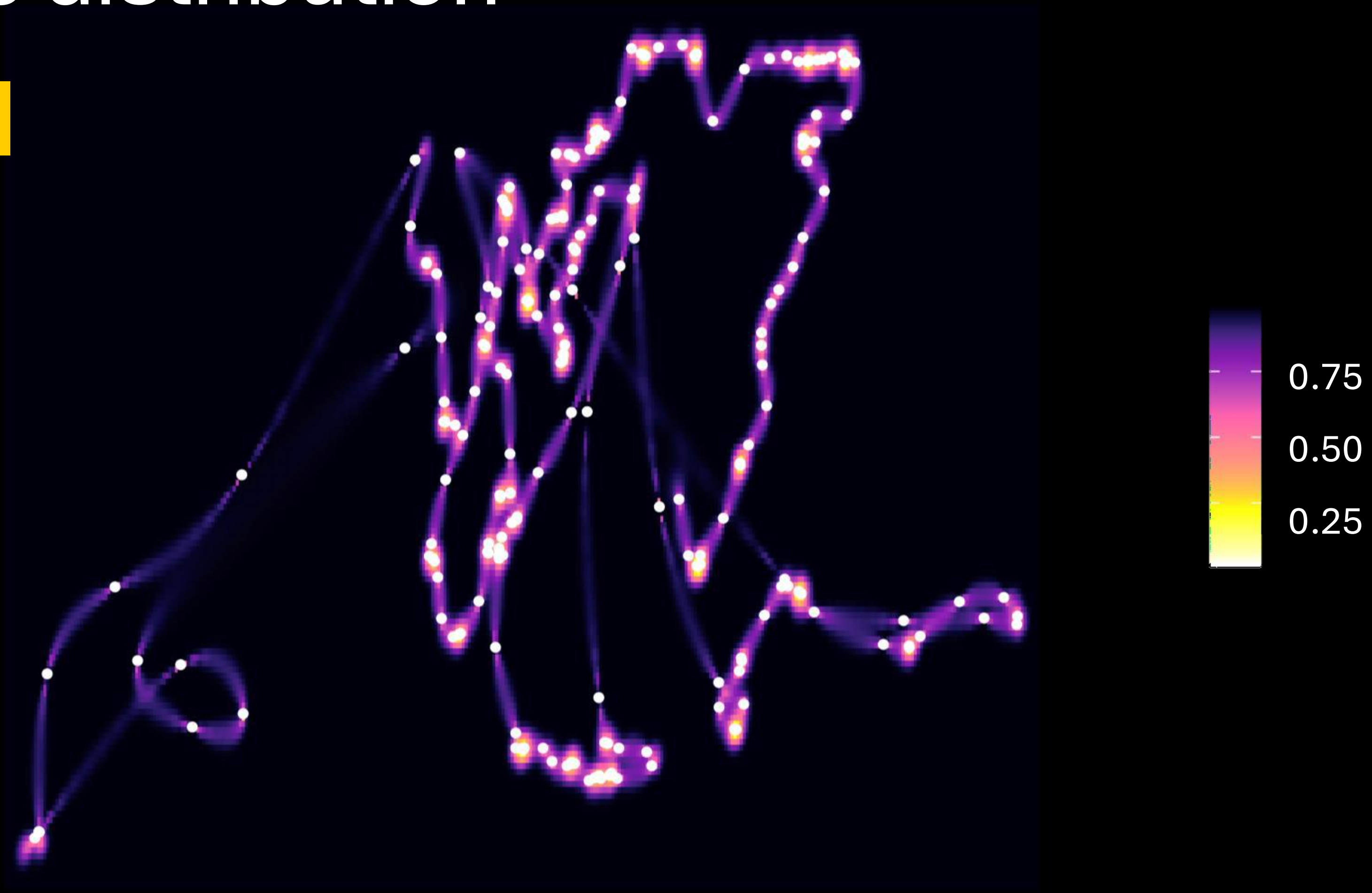
Occurrence distribution

Sampling interval

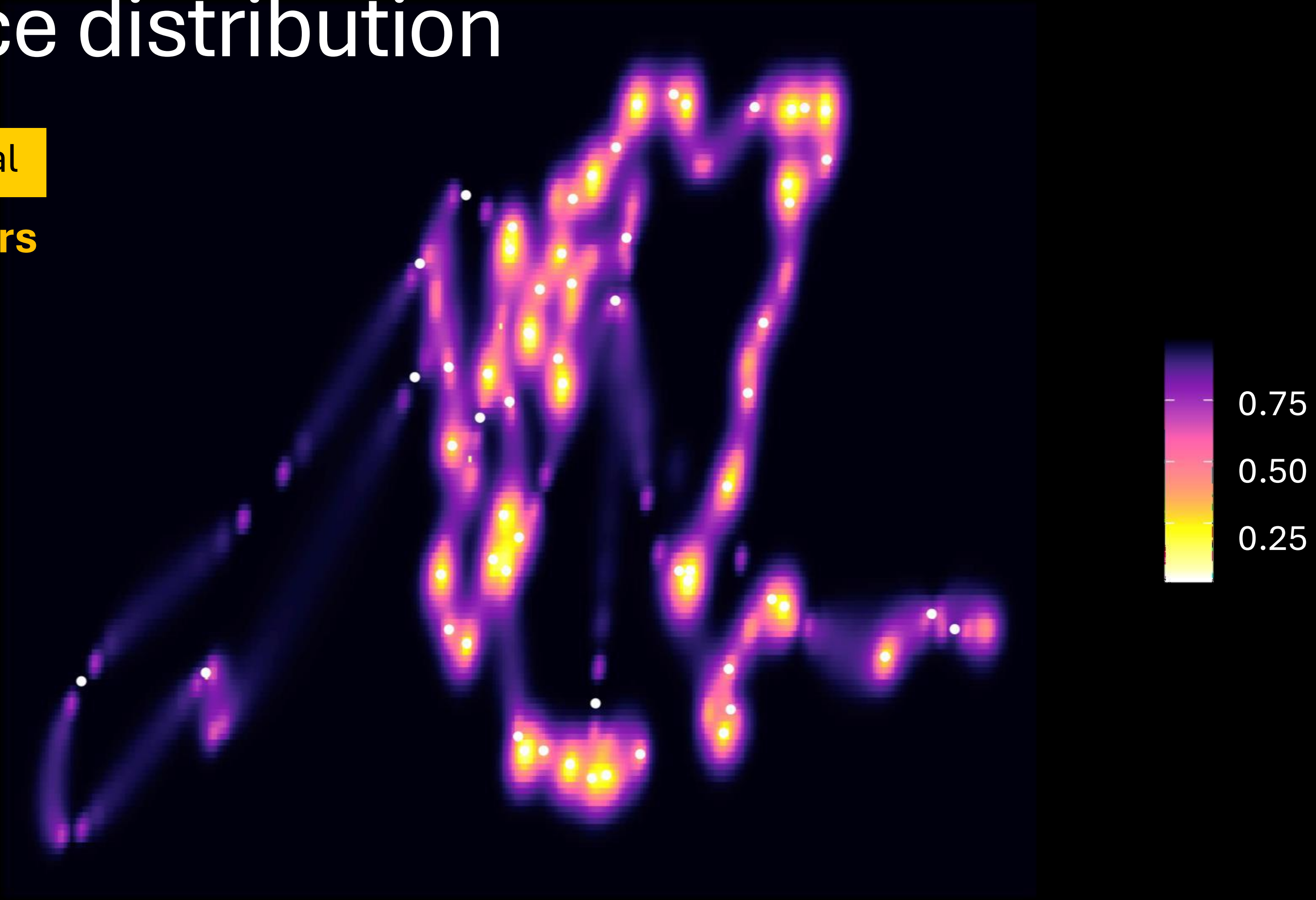
Δt = 1 hour



Occurrence distribution

Sampling interval

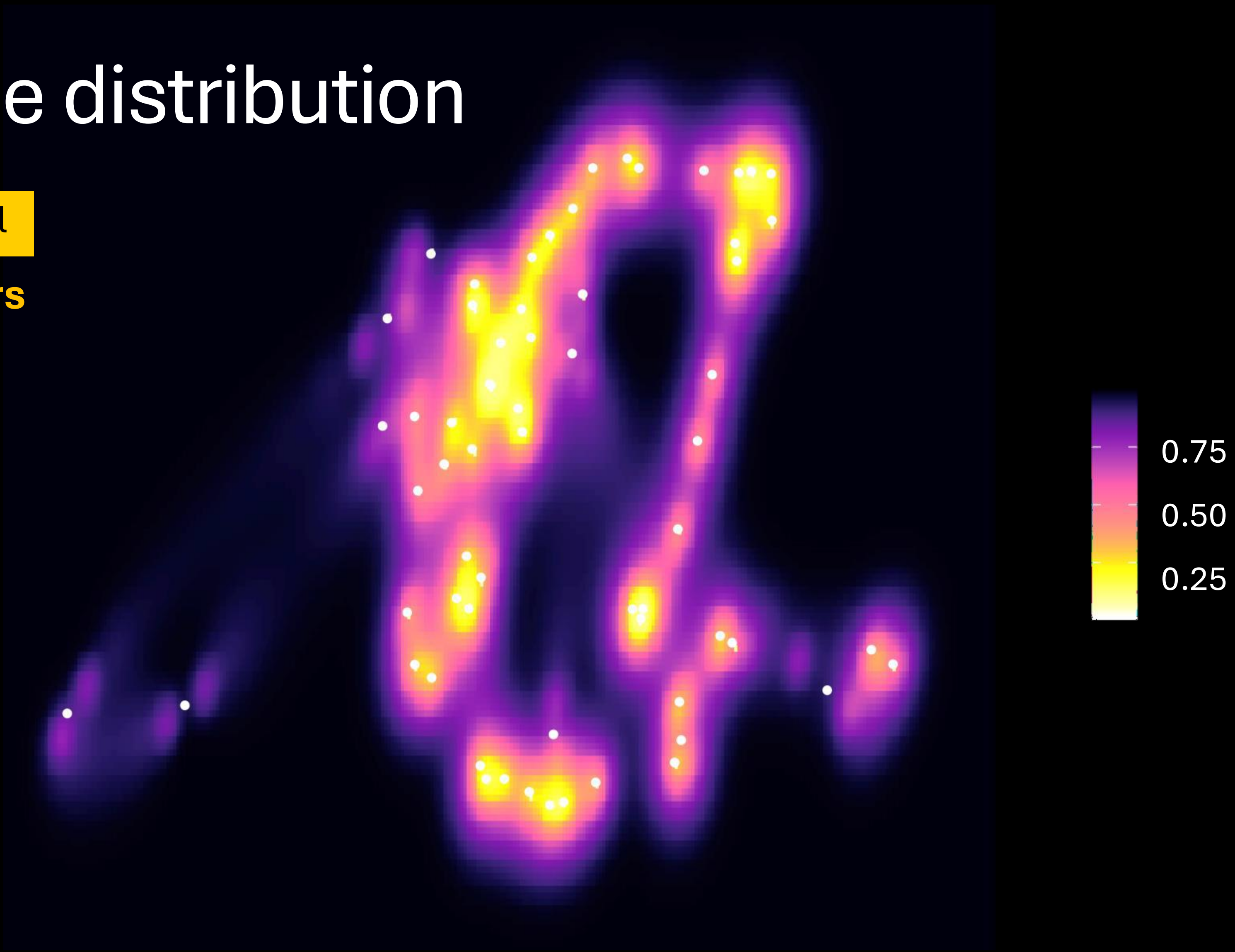
Δt = 2 hours



Occurrence distribution

Sampling interval

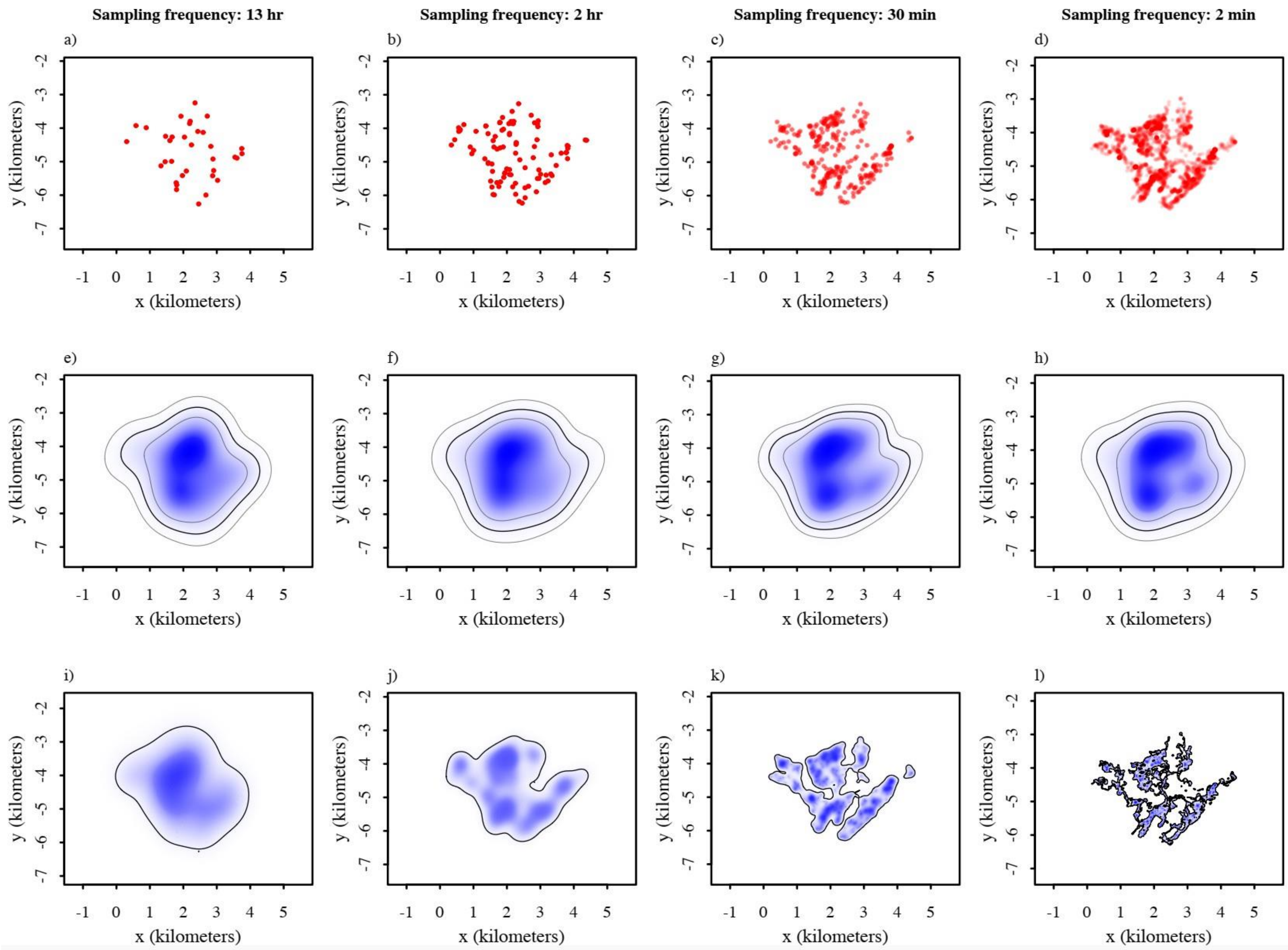
Δt = 4 hours



Data

Range

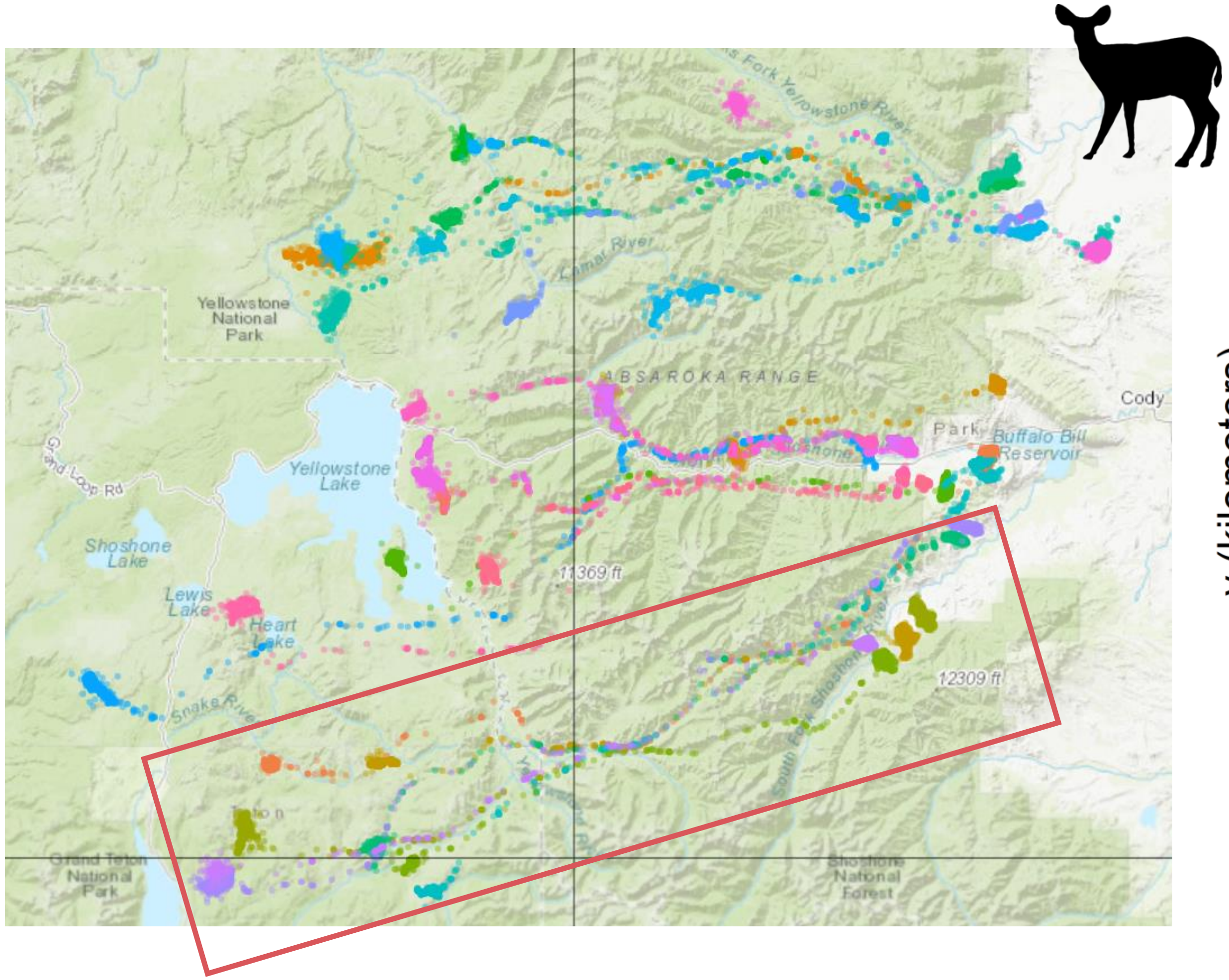
Occurrence

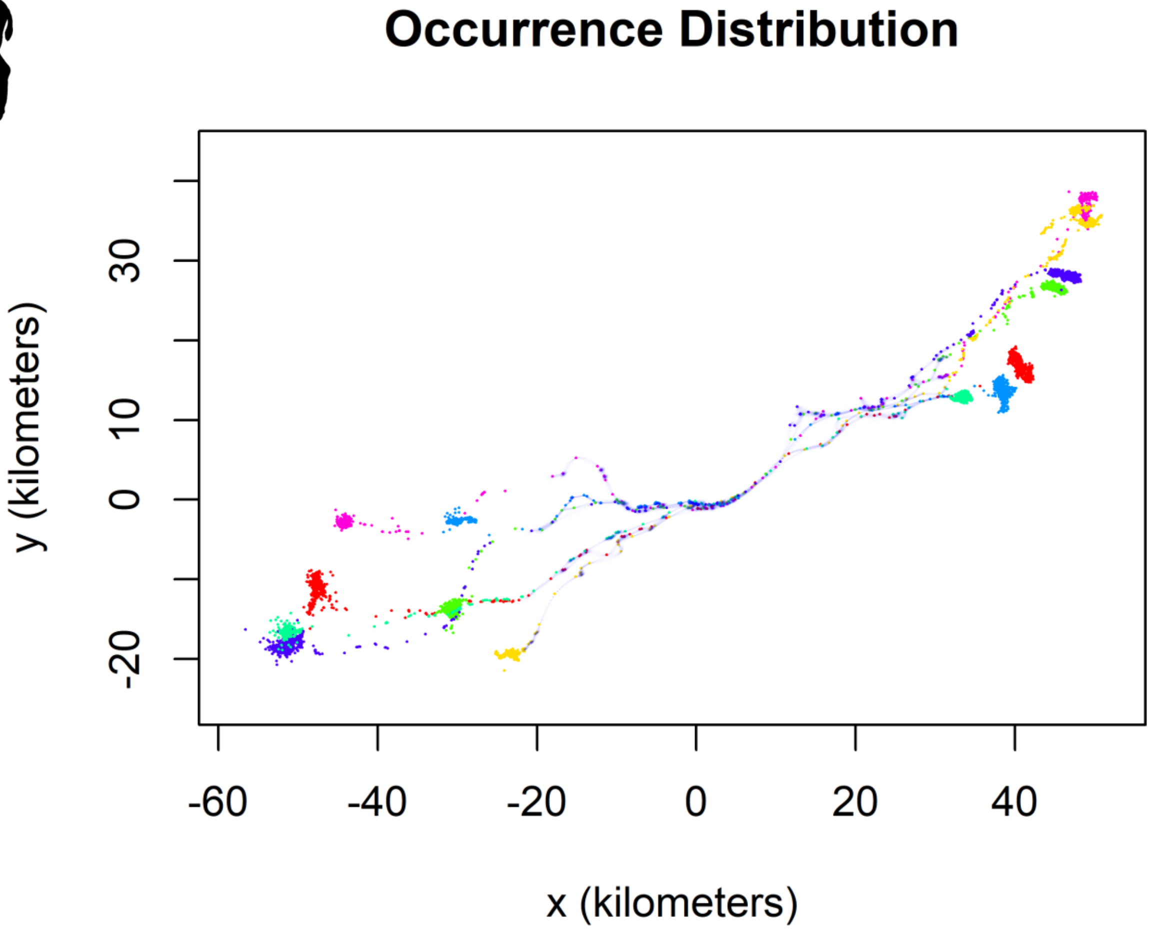


Alston *et al.* (2026)

doi.org/10.1002/ecy.70300

Corridor estimation





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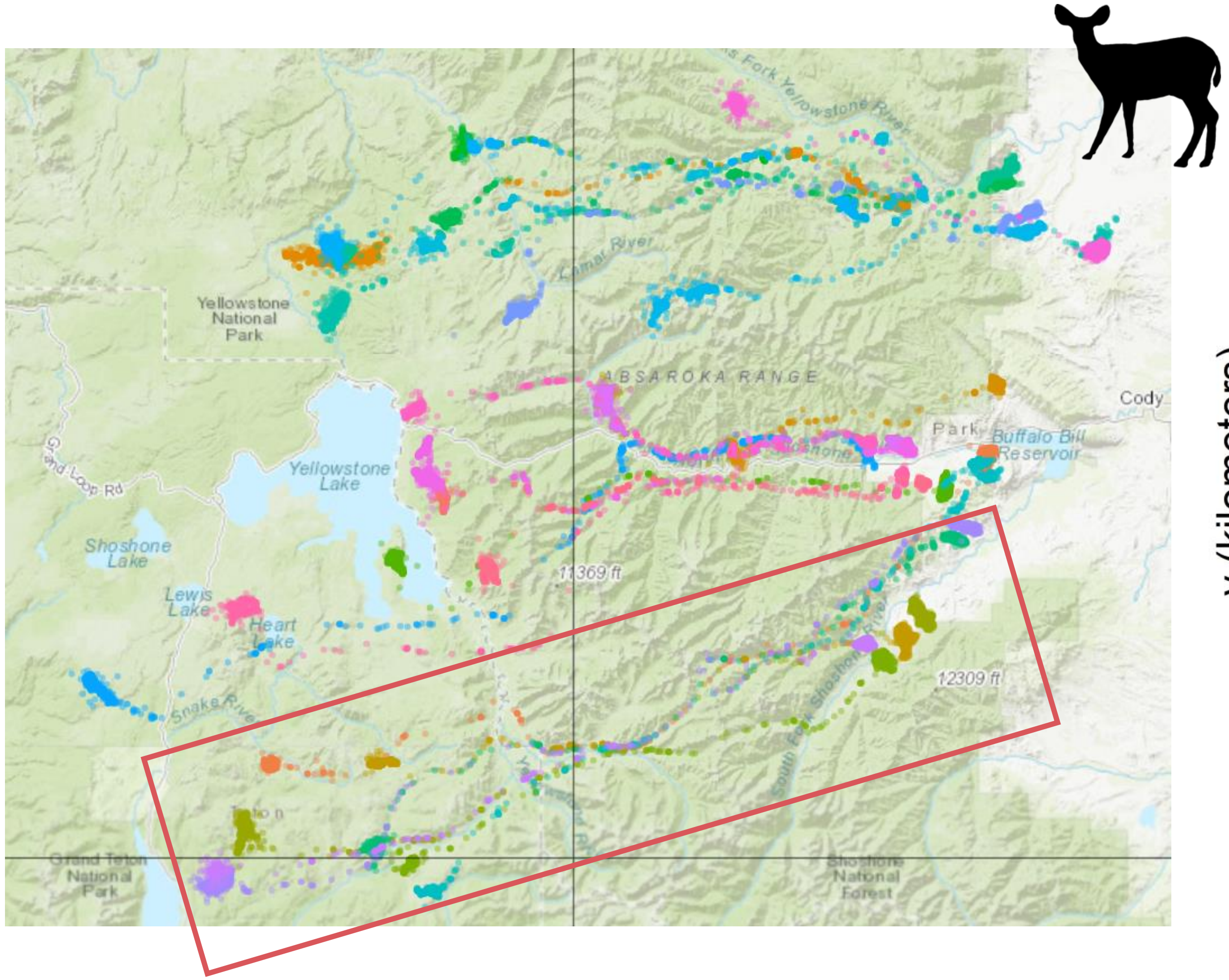
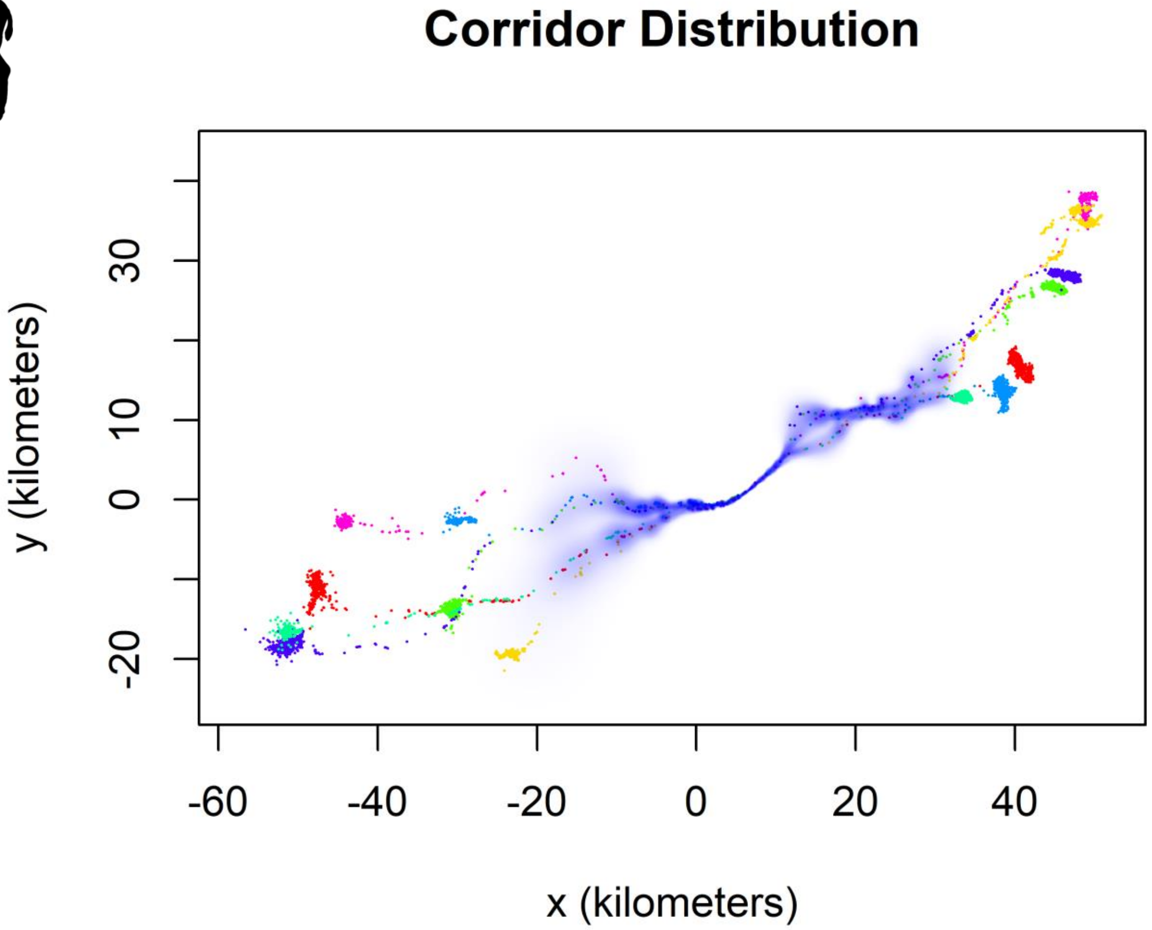
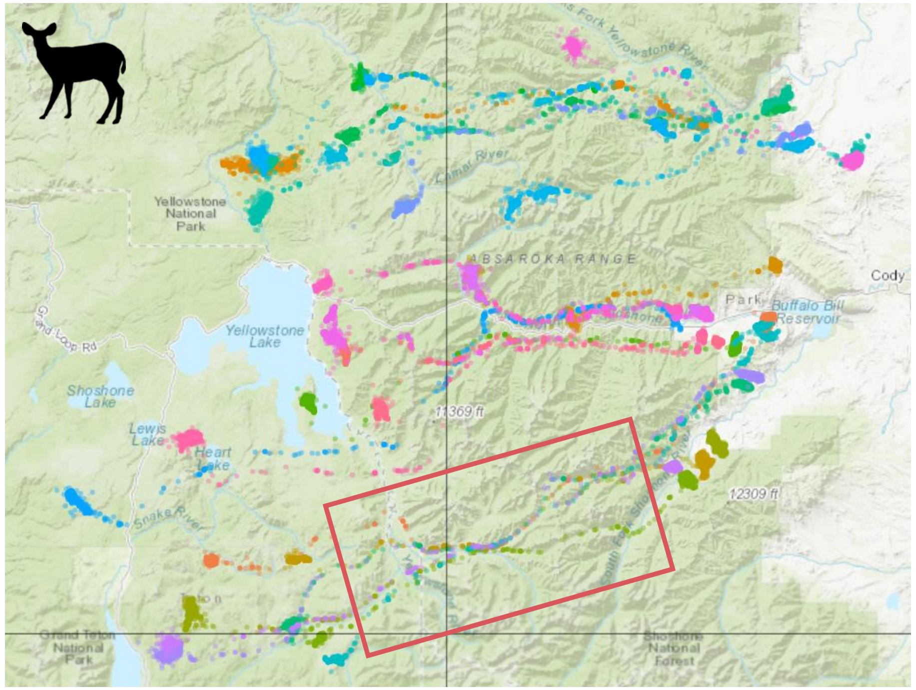




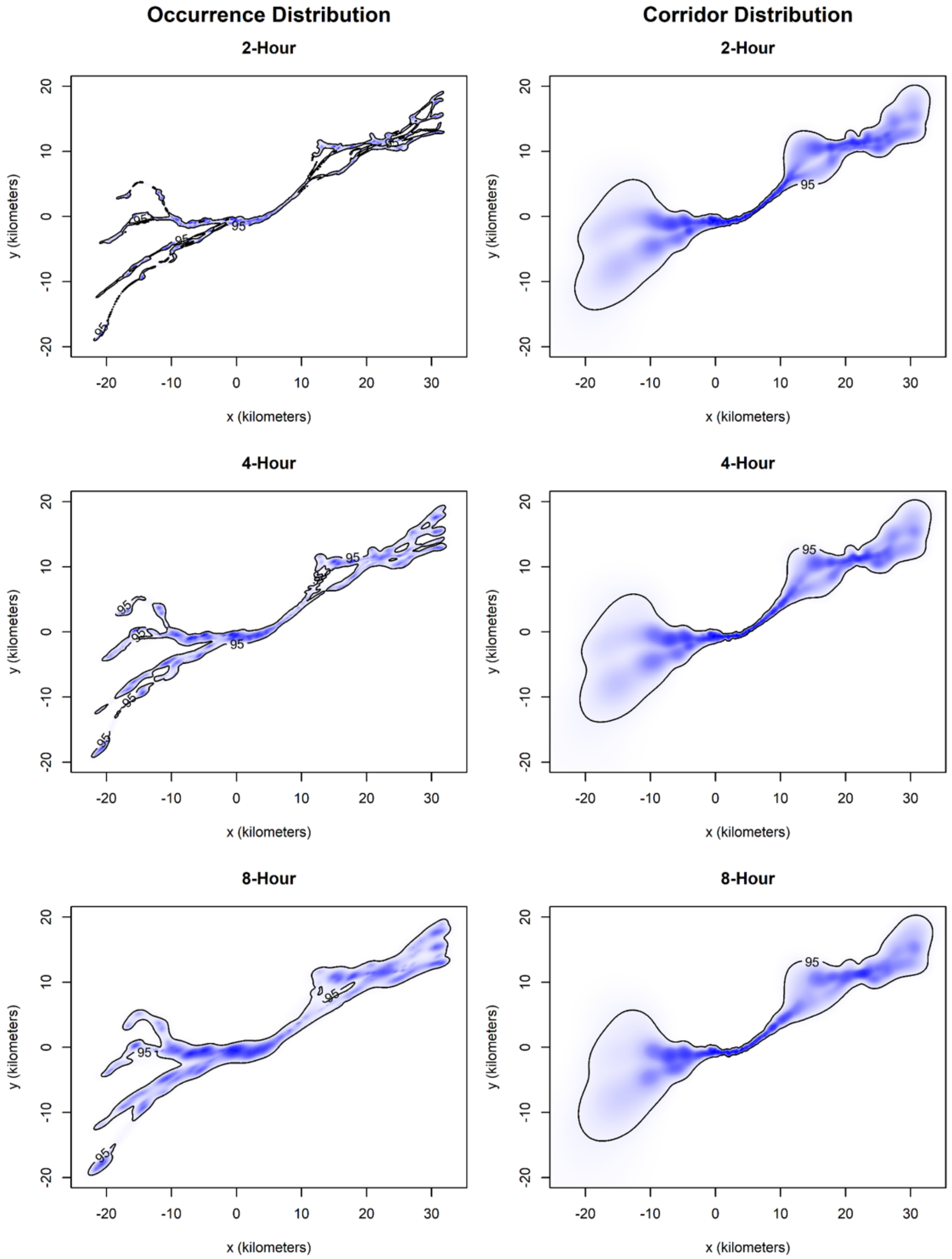
Sensitivity analysis: sampling frequency

Mule deer (*Odocoileus hemionus*)



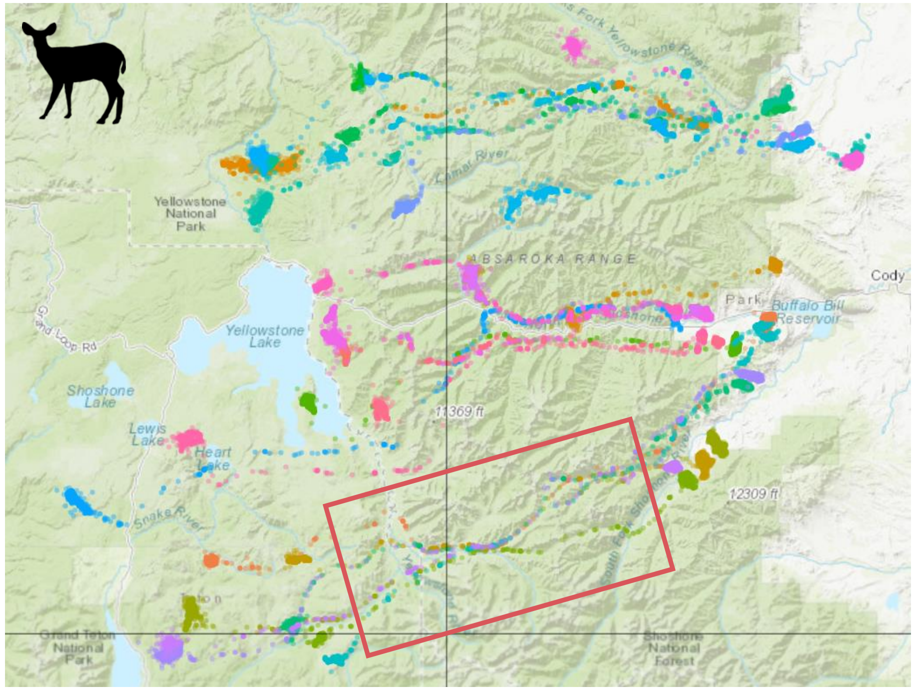
Increasing sampling interval

Less data



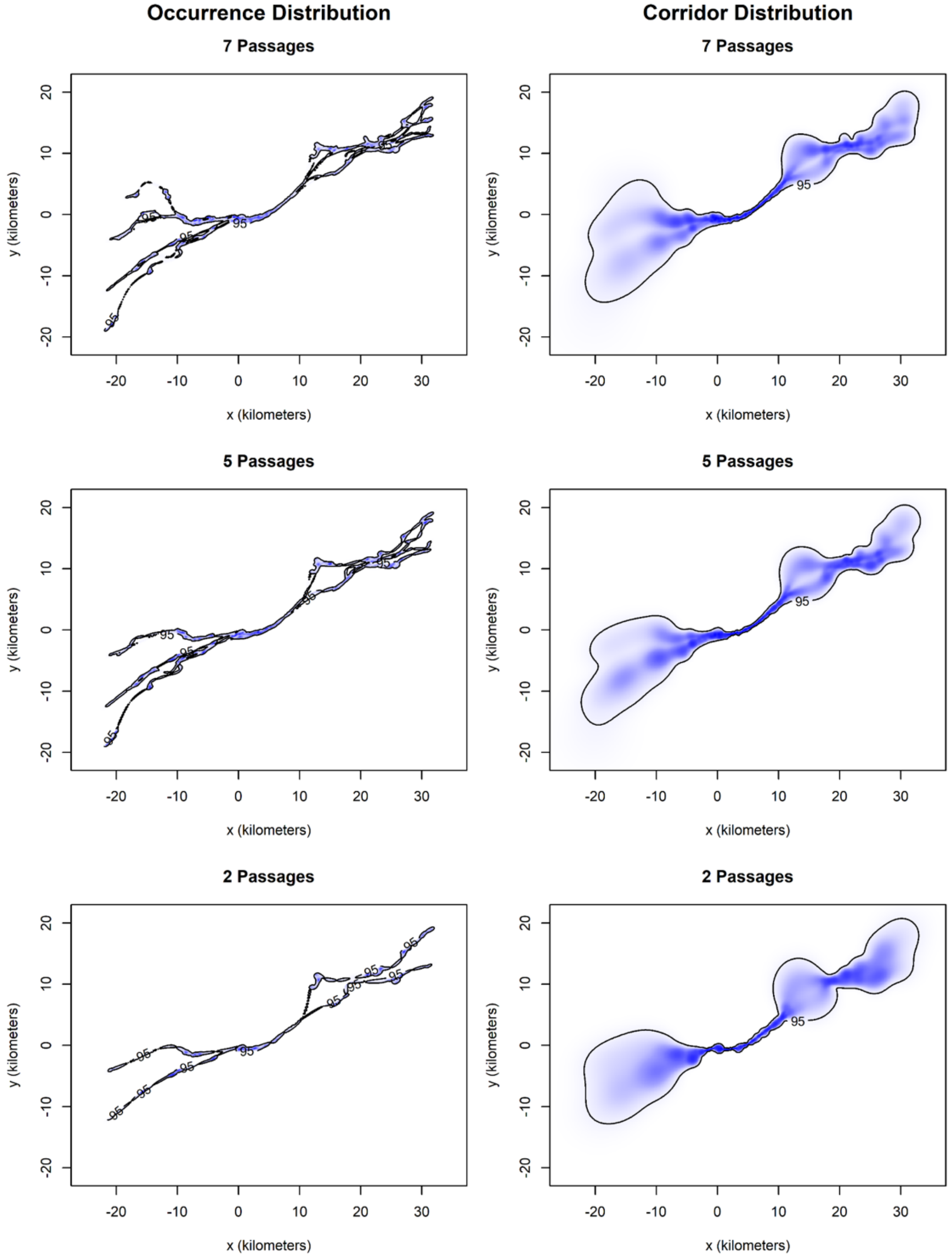
Sensitivity analysis: traversal count

Mule deer (*Odocoileus hemionus*)



Removing traversals

Less data



Utilization distributions

Alston *et al.* (2026)

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Range distribution ≠ **Occurrence** distribution

- *Extrapolate space use into the future*

Estimates **home ranges**

Sampling-independent!

- *Interpolate between data points in the past*

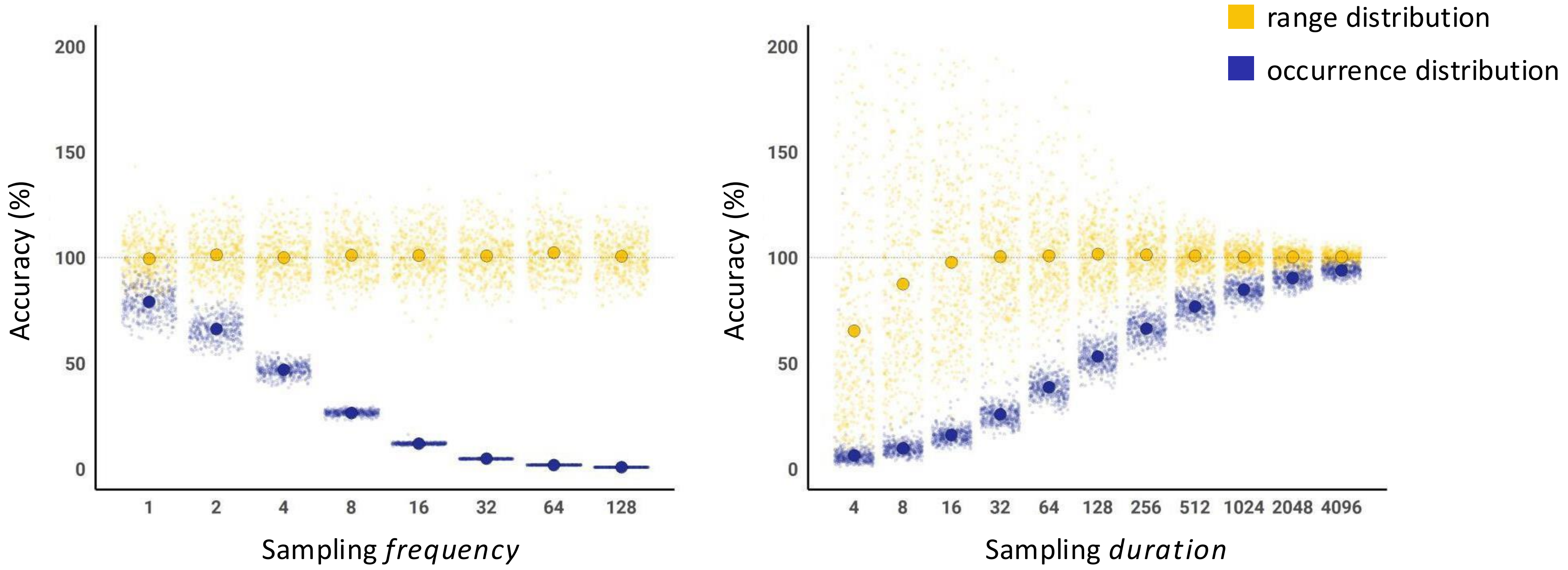
Estimates **occurrence regions**

Sampling-dependent!

Utilization distributions

Alston *et al.* (2026)
doi.org/10.1002/ecy.70300

- Simulated examples
  - Size of occurrence estimates are a function of sampling interval, while size of range estimates are not
  - ODs have sampling dependency



Utilization distributions

Alston *et al.* (2026)
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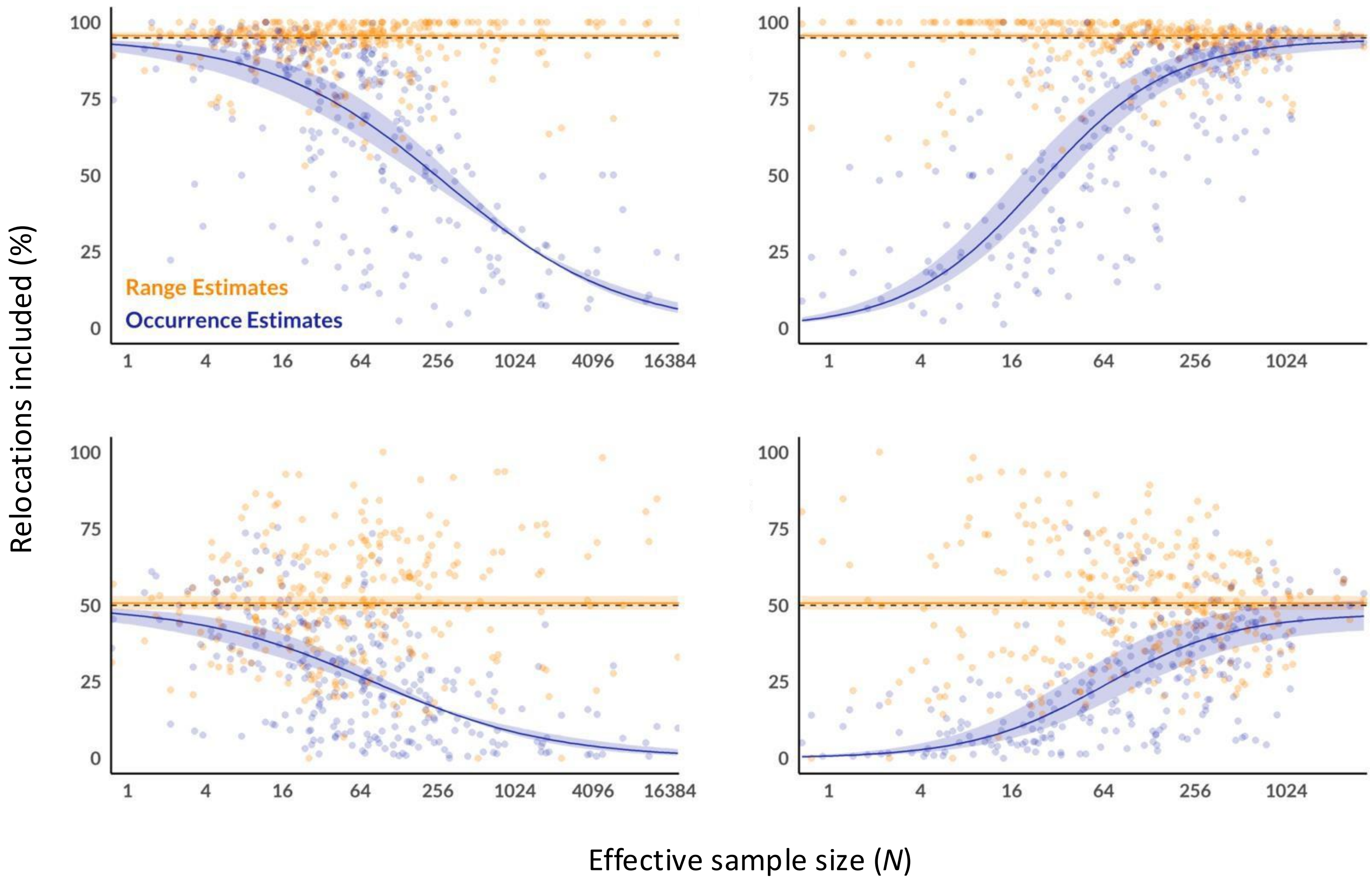
- Empirical examples
  - Holds up in real world data!



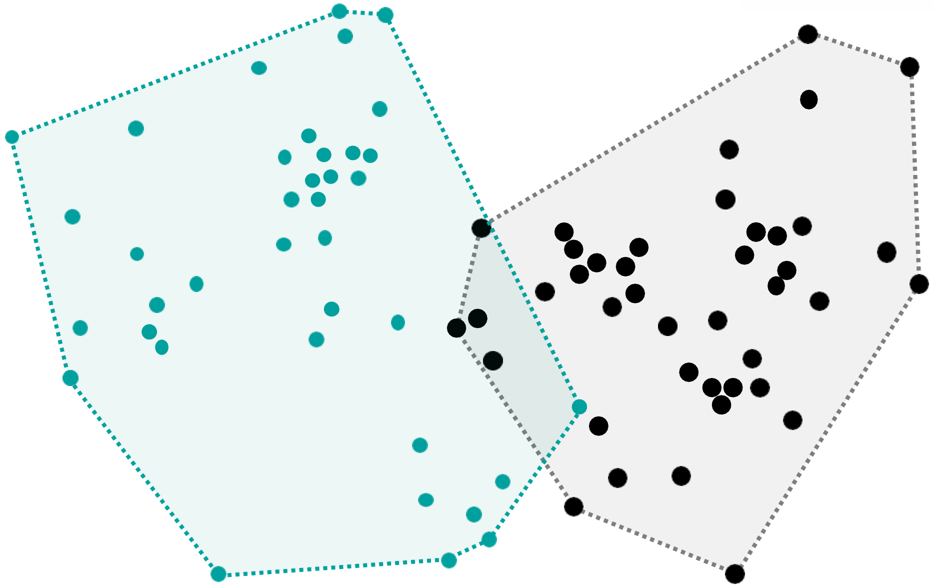
Overview

1) What is a home range?
2) Why do these concepts matter?
3) Range vs occurrence distributions
4) **Methods for home range estimation**
5) Effective sample size
6) HR estimation workflow
7) Mitigating bias

HR estimation methods

- **Minimum convex polygon** (MCP):
  - The smallest polygon drawn around tracking locations with all interior angles less than 180 degrees

Mohr (1947)



Burt (1943)

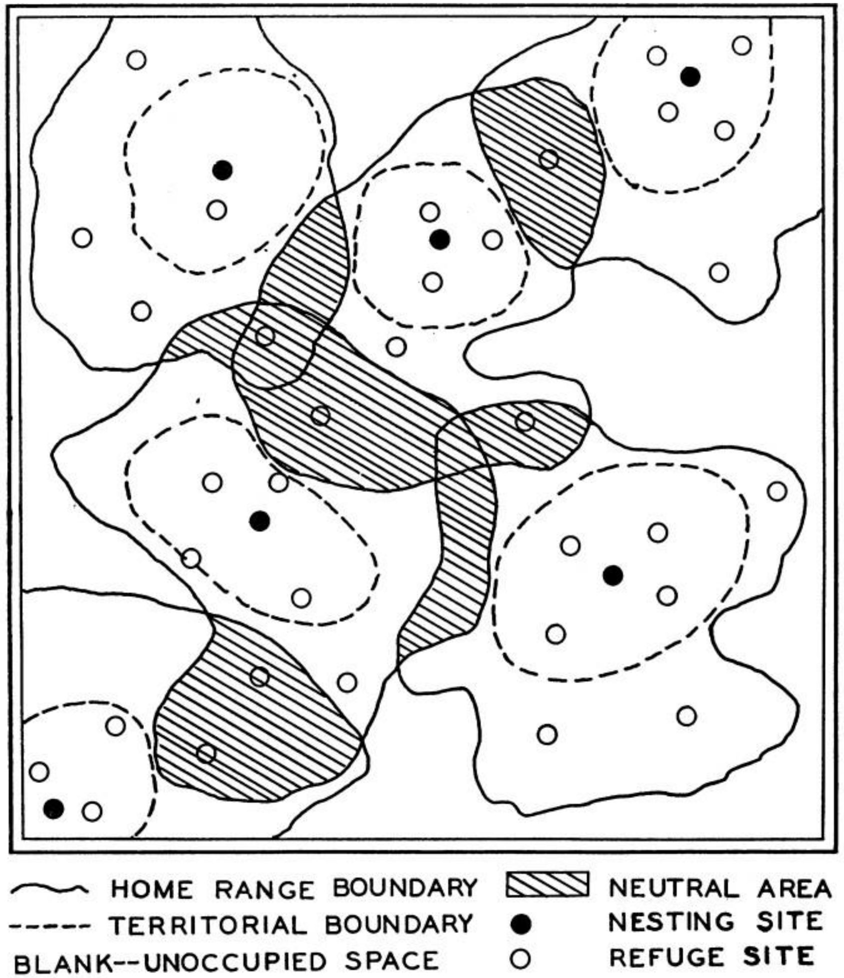
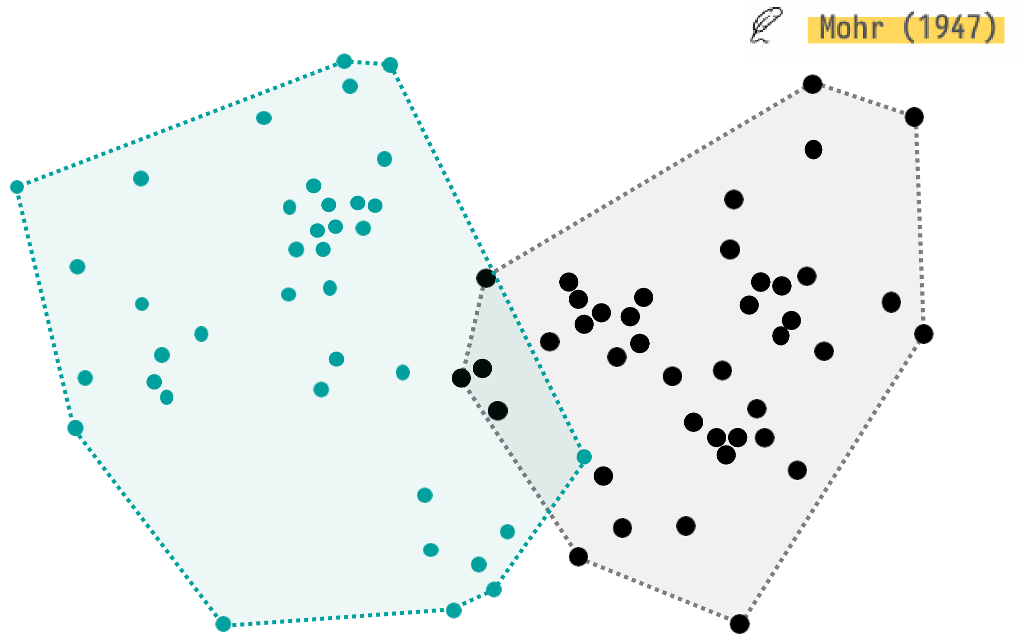


FIG. 1. Theoretical quadrat with six occupants of the same species and sex, showing territory and home range concepts as presented in text.

HR estimation methods

- **Minimum convex polygon** (MCP):
  - The smallest polygon drawn around tracking locations with all interior angles less than 180 degrees



- Assumes uniform space use
- Assumes locations are independent
- Sensitive to outliers and point geometry
- No measure of uncertainty
- Estimates 100% area only

HR estimation methods

- The use of frequently visited areas within an animal's home range should be statistically clustered

> It seems that an understanding of the biological significance of an animal's home range must include some knowledge of the intensity of use, by the animal, of various parts of the area.

Hayne (1949)

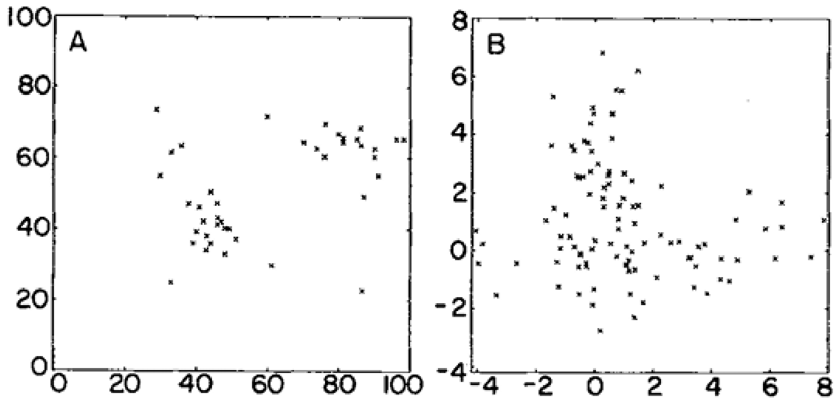


FIG. 1. Plots of (A) the DC data set and (B) the SIM data set.

Worton (1989)

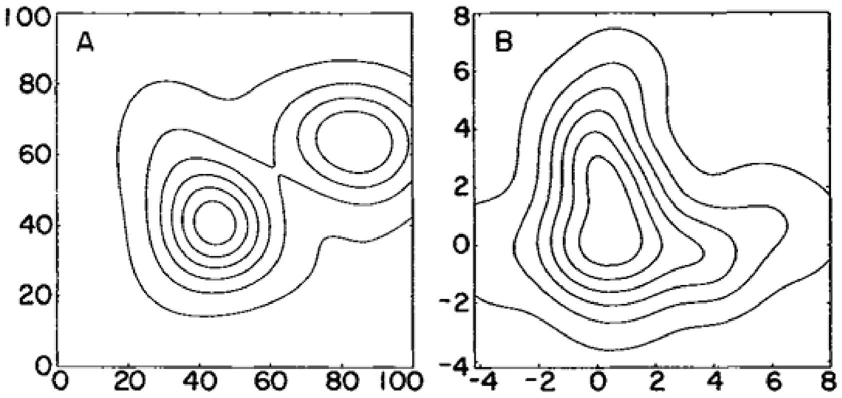
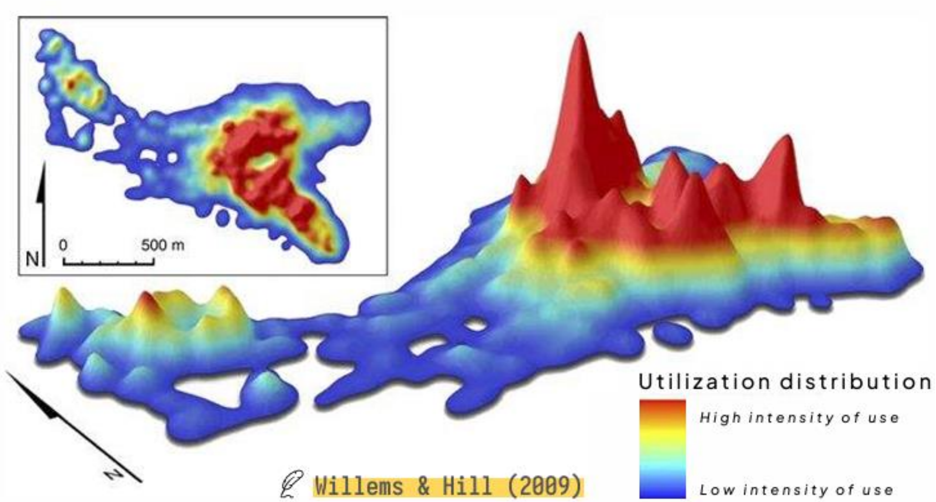


FIG. 2. Fixed kernel density estimates of the UD densities with the ad hoc choice of smoothing parameters for (A) the DC data set (h = 10.) and (B) the SIM data set (h = 1.).

HR estimation methods



Willems & Hill (2009)

> “It seems that an understanding of the biological significance of an animal's home range must include some knowledge of the intensity of use, by the animal, of various parts of the area.
>
> Hayne (1949)

Utilization distributions (UDs):

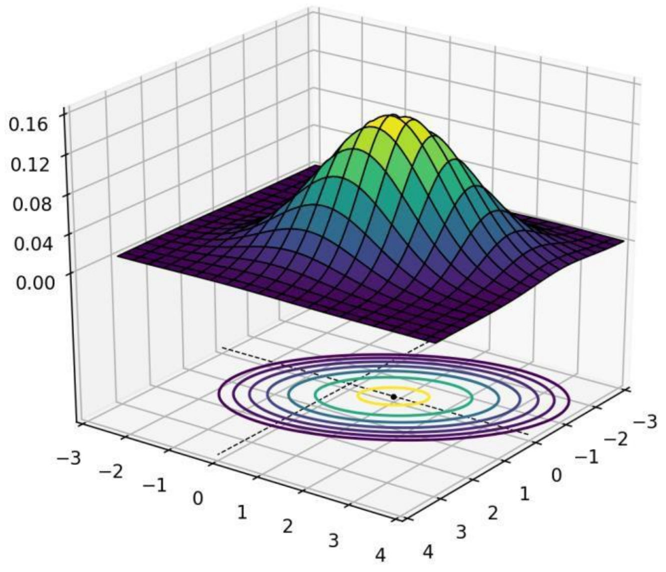
- Characterizing space-use as a density function (accounts for intensity of use)

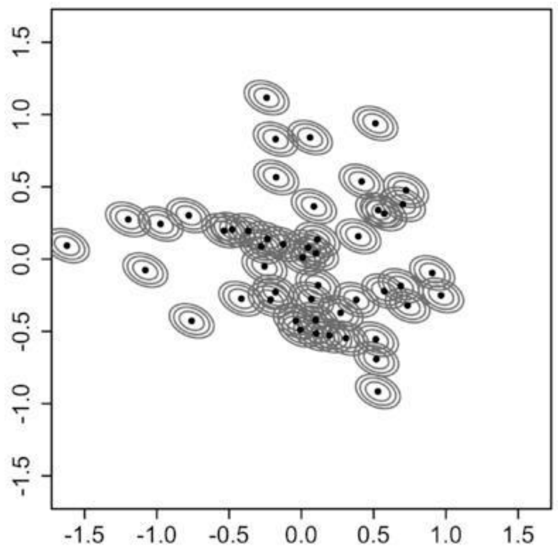
HR estimation methods

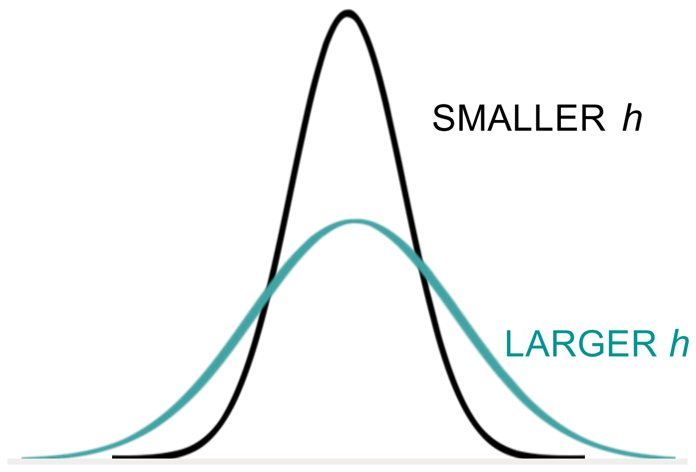
Worton *et al.* (1989)

Kernel density estimator (KDE):

- Describes not just the home range borders, but the *intensity of use*
- On an x-y plane, each location has a 3-dimensional "hill," the *kernel*
  - Shape & width of the kernel, the *bandwidth* (h), can be selected and optimized







HR estimation methods

Worton *et al.* (1989)

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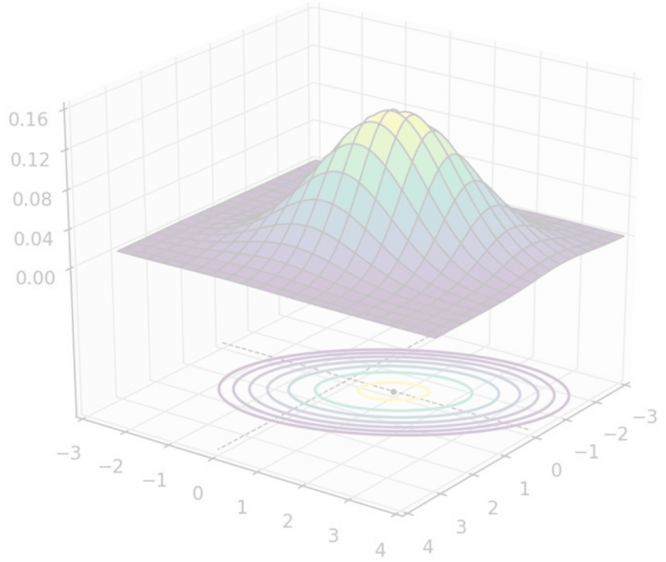
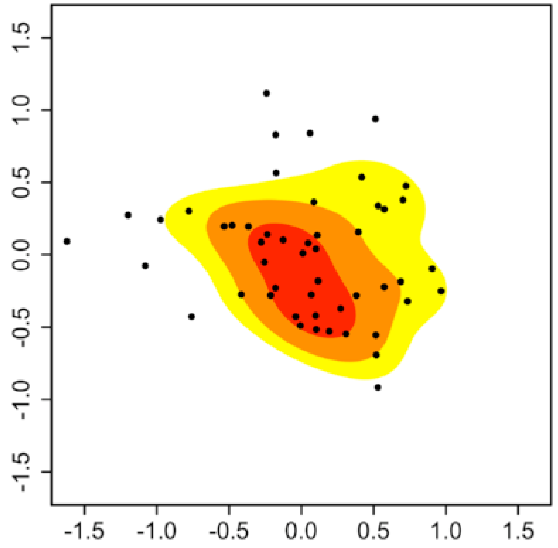
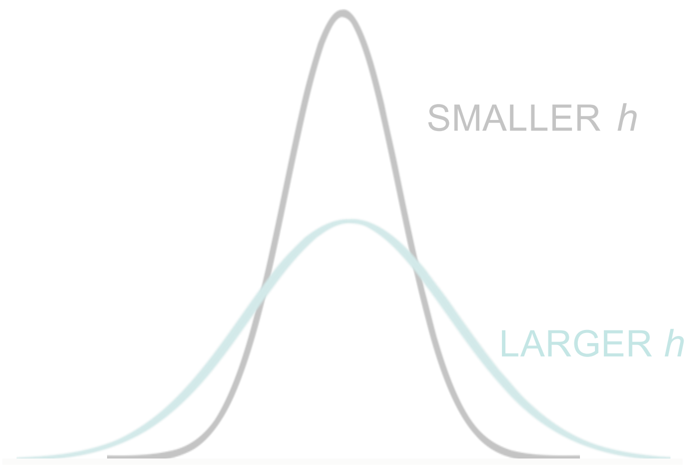


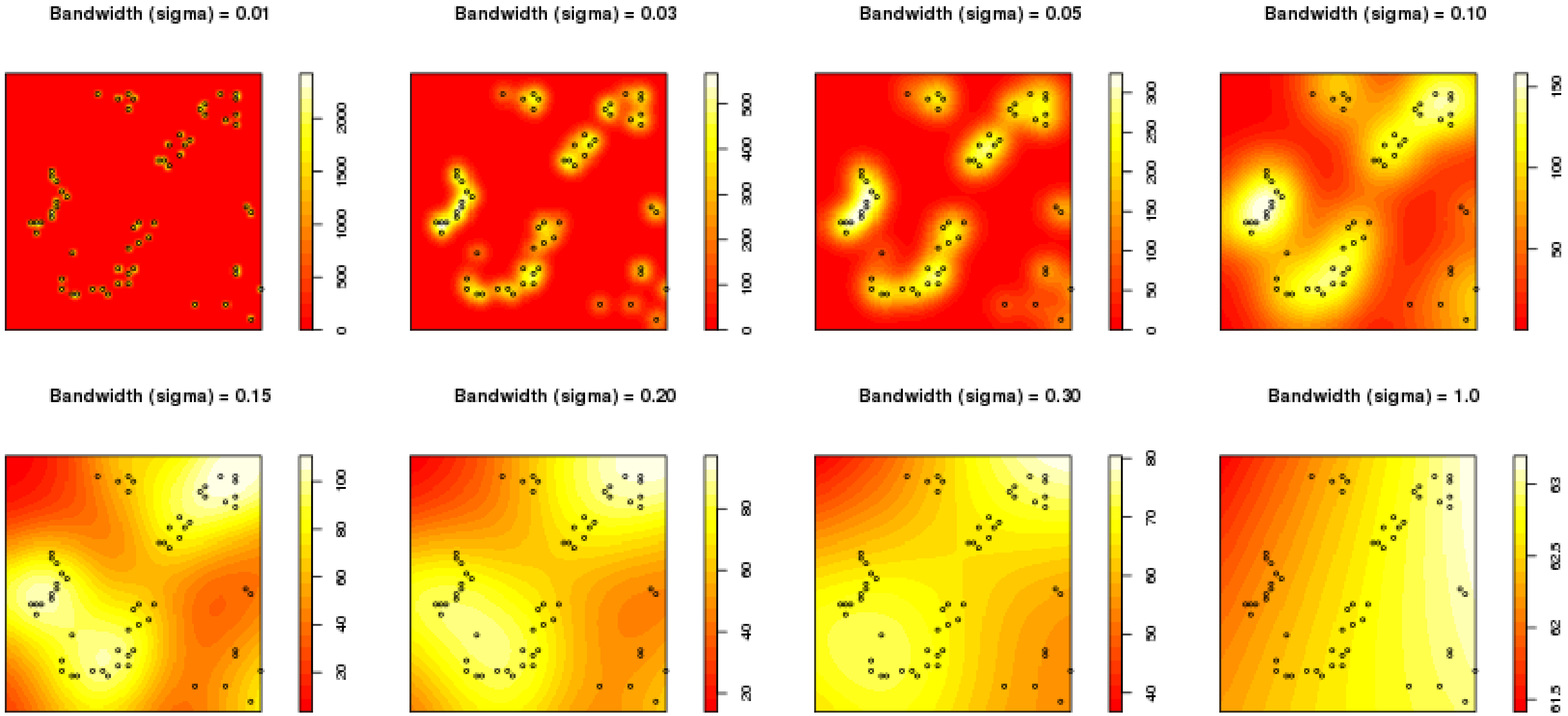




HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):



bandwidth (h), or *smoothing parameter*

HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):

- The most statistically efficient *non-parametric distribution estimator*
  - Nonparametric: not explicitly modeling the causes of space use

HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):

- The most statistically efficient *non-parametric distribution estimator*
  - Nonparametric: not explicitly modeling the causes of space use
- The objective is typically to minimize the 'mean integrated square error'

$$\mathrm{MISE}(H) = \mathrm{E}\left[\iint \left(\hat{p}(x,y|H) - p(x,y)\right)^2 dx\, dy\right]$$

... with respect to *bandwidth* or *smoothing*, *H*

HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):

- The most statistically efficient *non-parametric distribution estimator*
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$$\mathrm{MISE}(H) = \mathrm{E}\left[\iint (\hat{p}(x, y|H) - p(x, y))^2 \, dx \, dy\right]$$

... with respect to *bandwidth* or *smoothing*, h

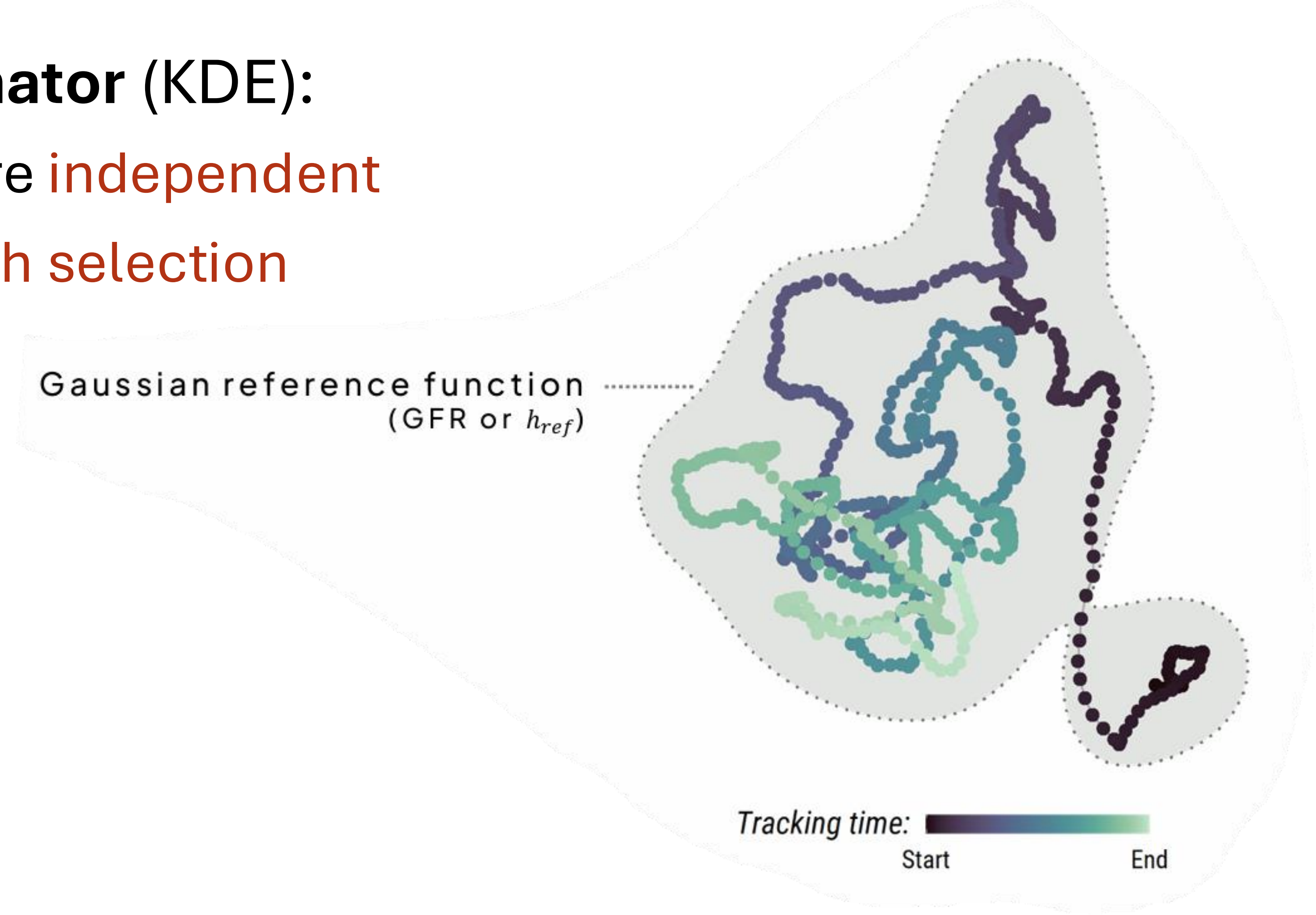
h is not a model parameter— there is no true value of *h* that best characterizes an animal! You do not choose your bandwidth. *You choose your bandwidth optimizer.*

HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):

- Assumes locations are independent
- Sensitive to bandwidth selection



HR estimation methods

Worton *et al.* (1989)

Kernel density estimator (KDE):

- Assumes locations are independent
- Sensitive to bandwidth selection

Least-squares cross-validation (LSCV)
Attempts to prevent oversmoothing...

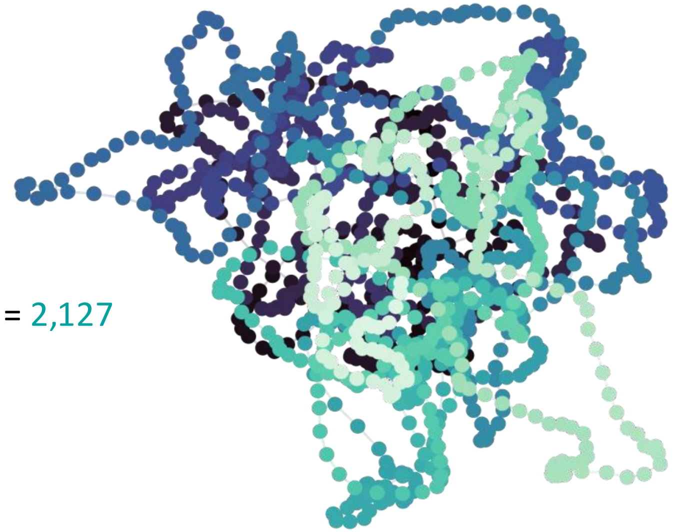
This algorithm performs poorly with *large sample sizes*, and does not account for the locations' temporal structure.

HR estimation methods

Kernel density estimator (KDE):

- *Thinning* the data to achieve *independence*…

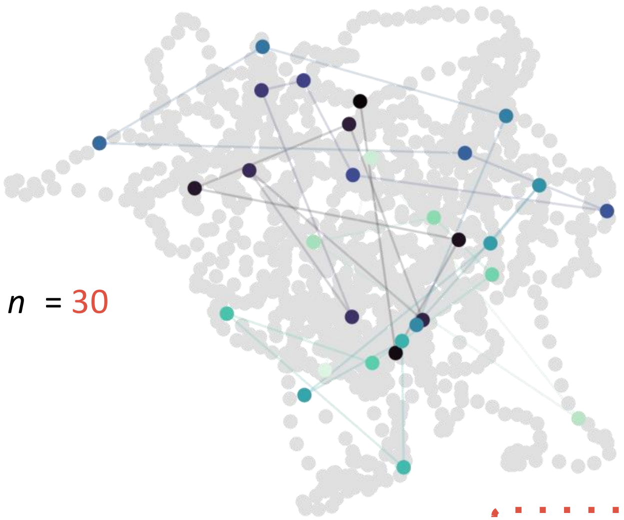
Tracking data representing
hourly locations over **one month**.



n = 2,127

Tracking time: Start End

Tracking data subsampled so there is only
one location per day.



n = 30

Data loss of ≈ 98.6%

HR estimation methods

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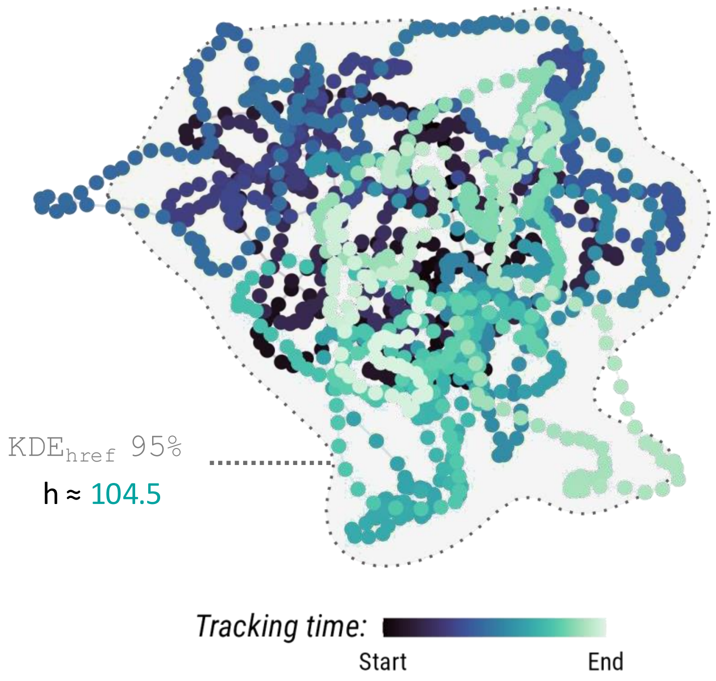
Tracking data representing
hourly locations over **one month**.



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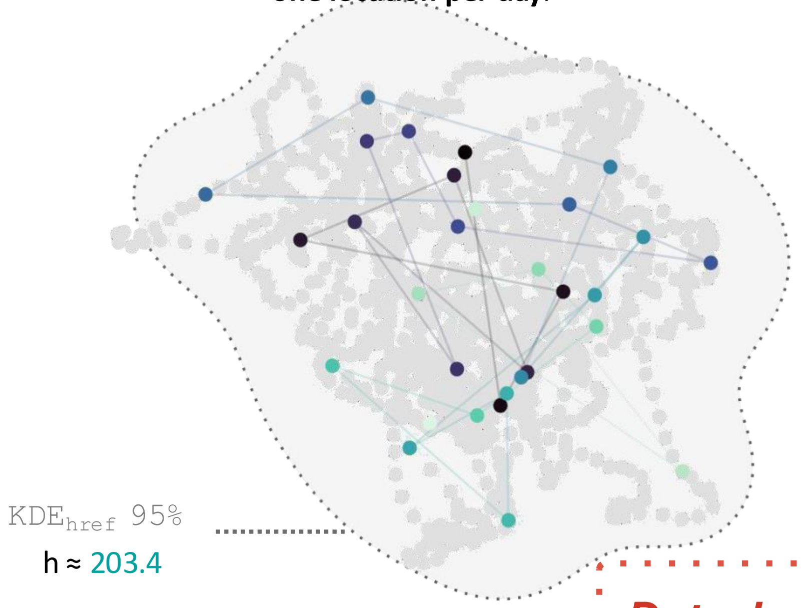


Data loss of ≈ 98.6%

HR estimation methods

Kernel density estimator (KDE):

- *Thinning* the data to achieve *independence*…

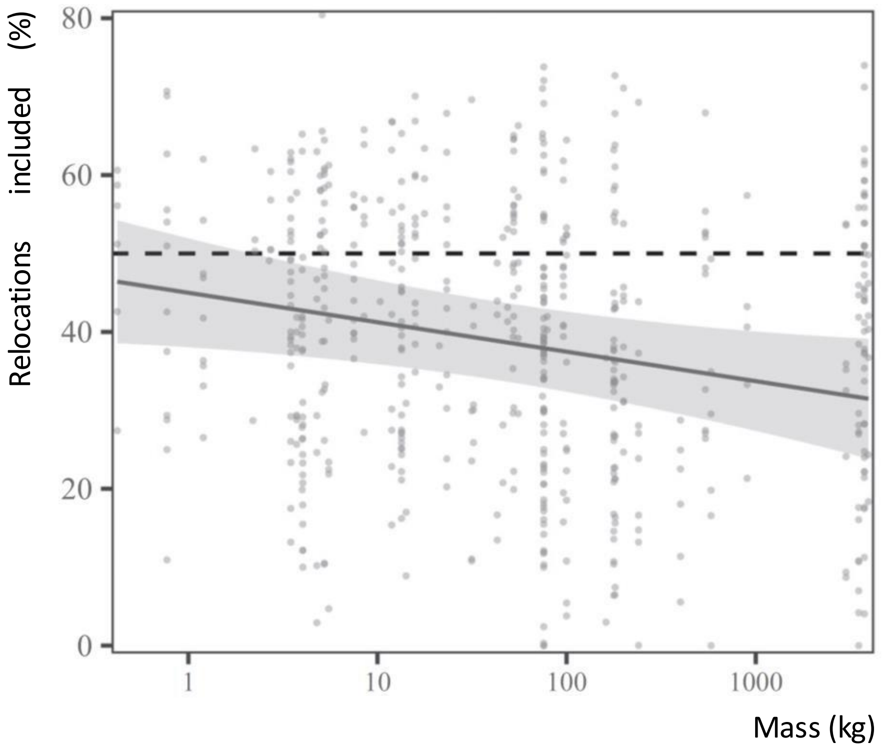
Why intentionally discard data that is costly to collect?

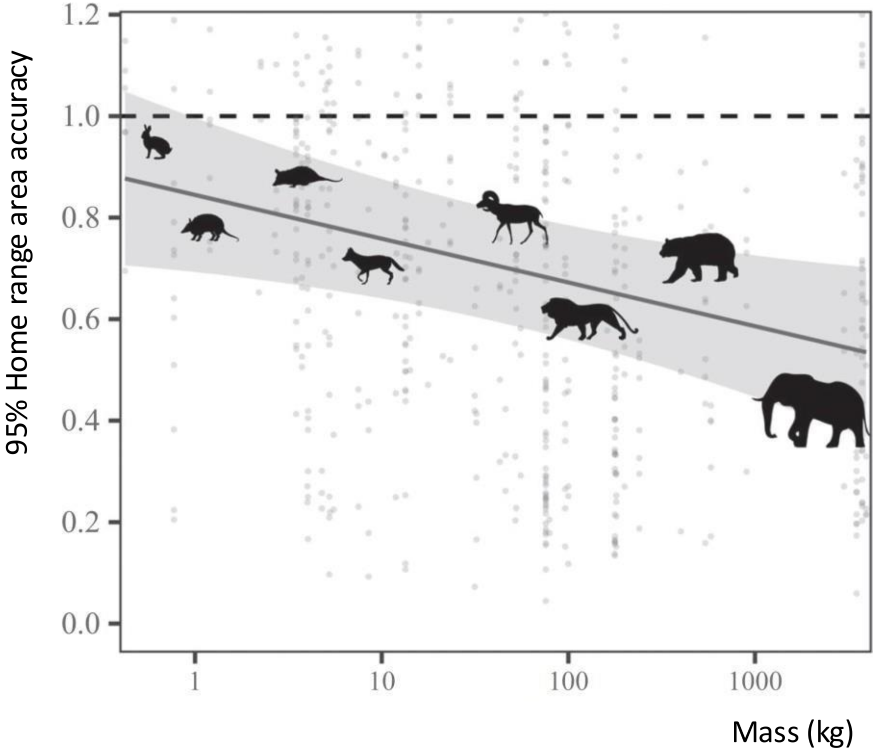
And how confident are we in these estimates anyway?

HR estimation methods

Kernel density estimator (KDE):

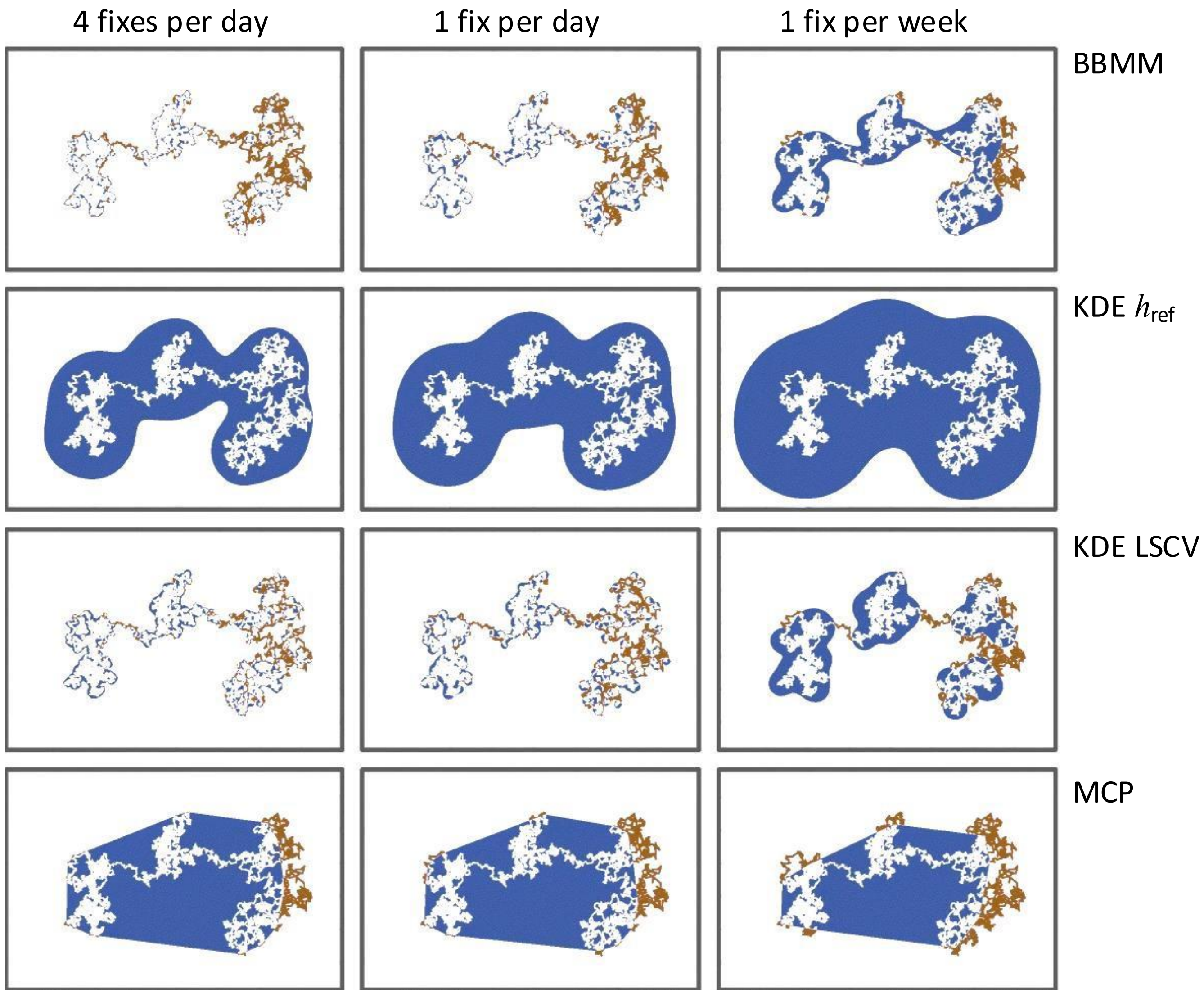
- The magnitude of KDE's underestimation worsened as body mass increased





HR estimation methods

What is each method actually estimating?

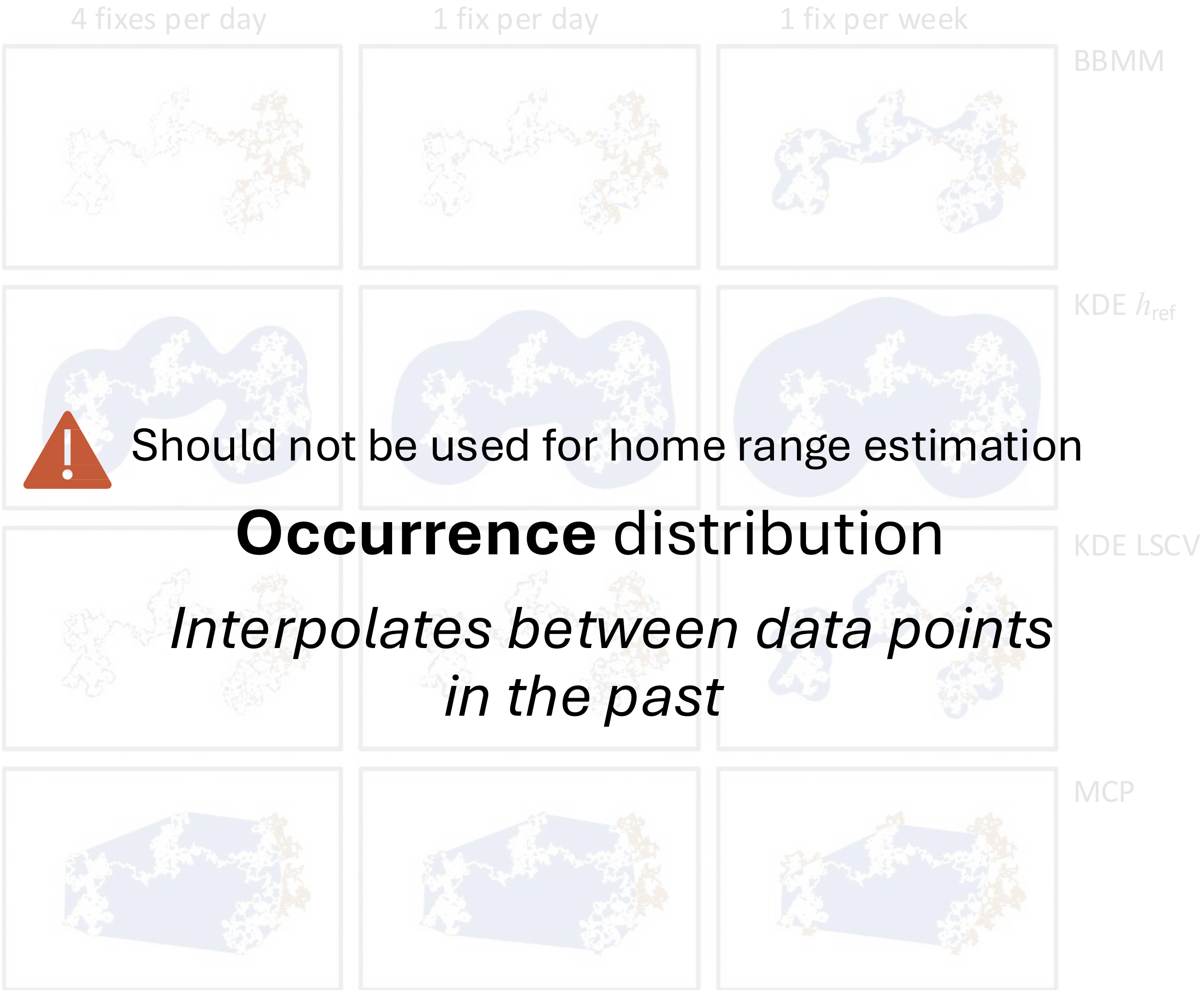


Silva *et al*. (2021)

HR estimation methods

What is each method actually estimating?

- Not all methods are answering the same questions…



Should not be used for home range estimation

Occurrence distribution

Interpolates between data points in the past

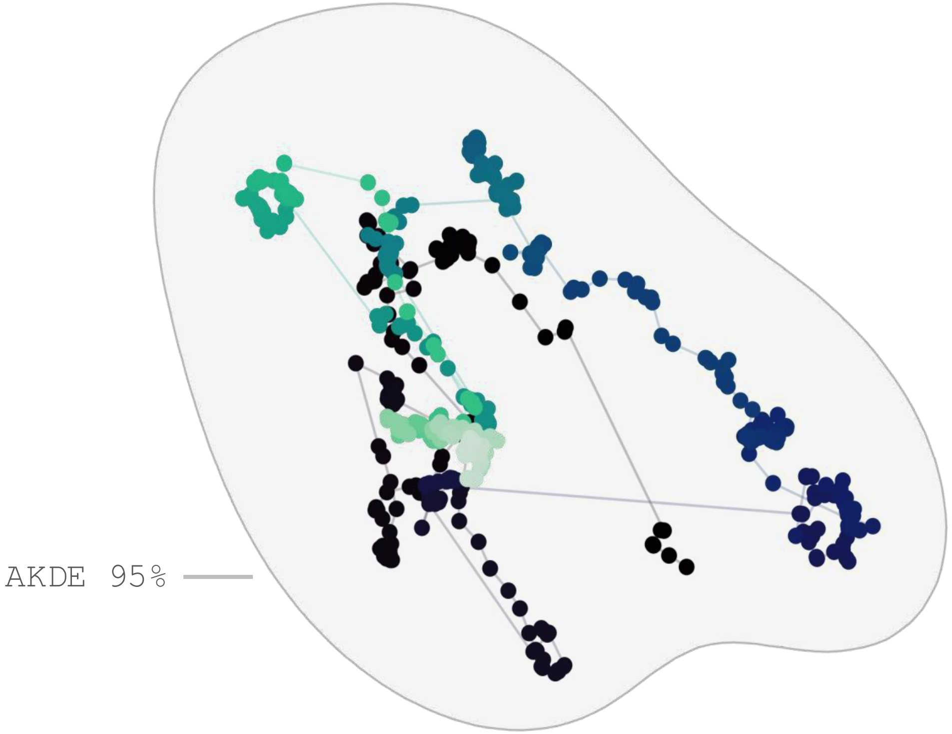
Silva *et al.* (2021)

HR estimation methods

Fleming *et al.* (2015)

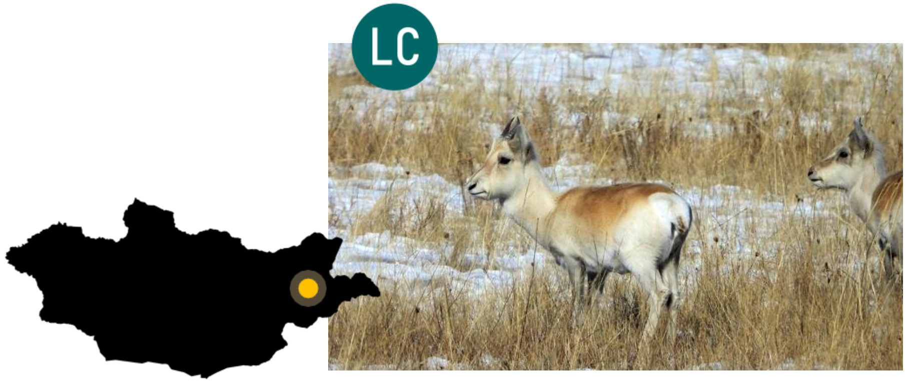
Autocorrelated kernel density estimator (AKDE):

- Re-derived KDE that explicitly assumes the data represents a sample from a *nonstationary, autocorrelated, continuous movement process*



Duration ≈ 1 year

τ_p ≈ 2.4 months



Mongolian Gazelle

(Procapra gutturosa)

HR estimation methods

Fleming *et al.* (2015)

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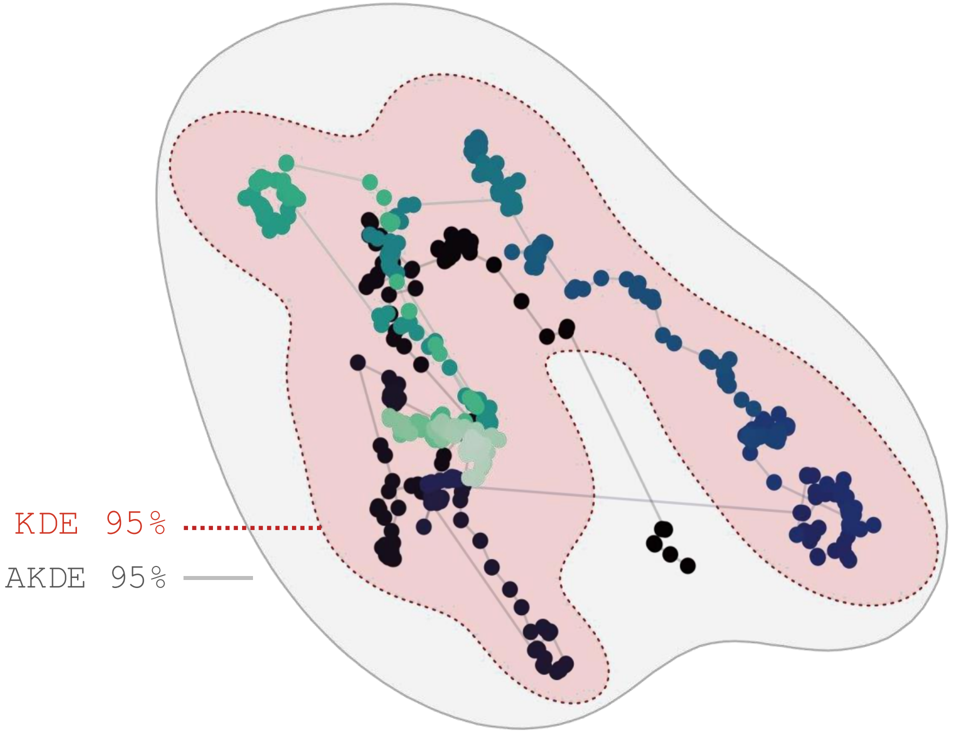

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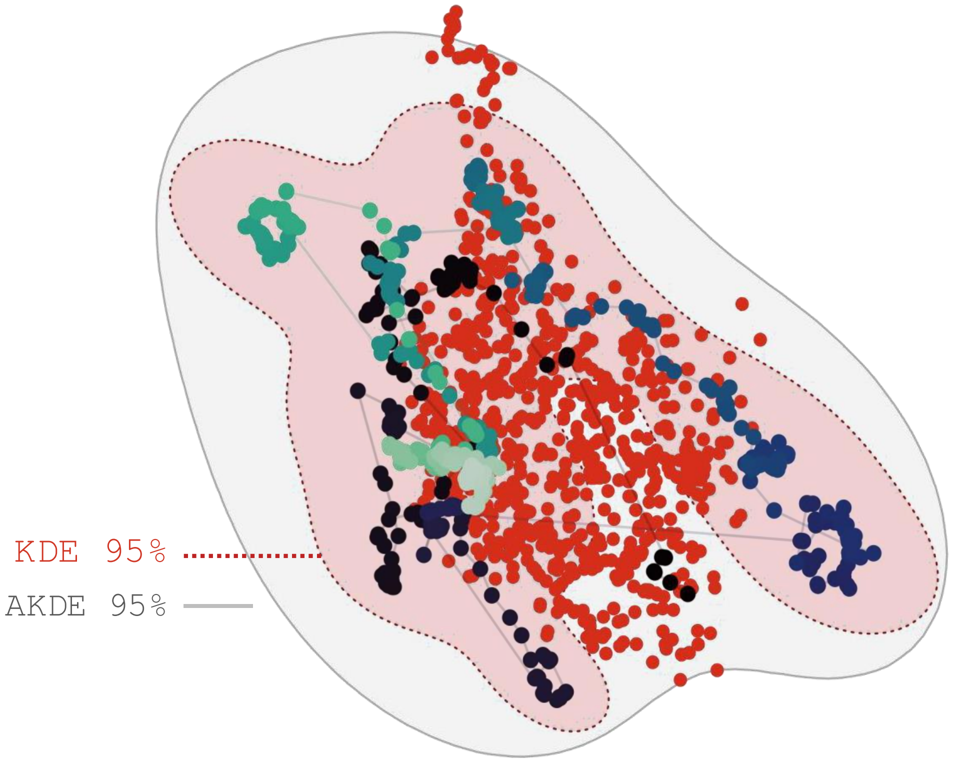


Mongolian Gazelle
(Procapra gutturosa)

Duration ≈ 2 years - - - - - - -> showing an *extra year of data*!

$\tau_p \approx$ 2.4 months

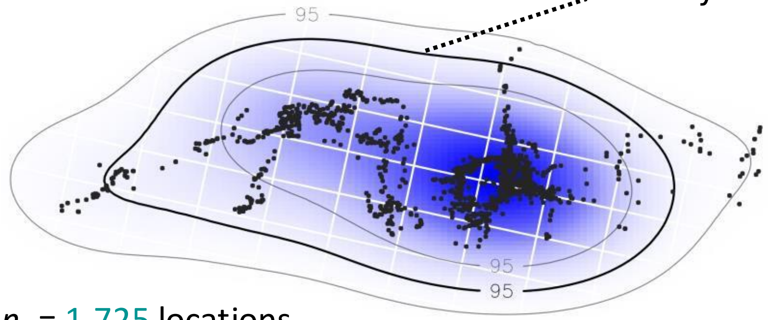
HR estimation methods

Fleming *et al.* (2015)

Autocorrelated kernel density estimator (AKDE):

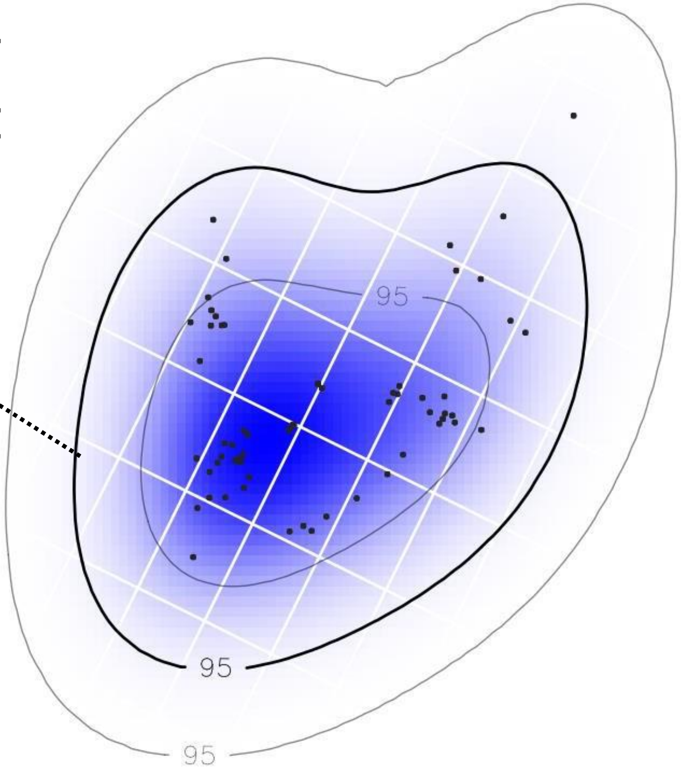
- AKDEs also provide accurate confidence intervals diagnose situations where the data are insufficient provide a reasonable home-range estimate

The 95% contour represents the most likely area where the animal spends 95% of its time.



n = 1,725 locations
N_{ESS} ≈ 15.7 locations

Home range area estimate
770 km² (440 – 1,200)



n = 66 locations
N_{ESS} ≈ 7.1 locations

Home range area estimate
4,500 km² (1,800 – 8,300)

ctmm_range_occurrence.R

Overview

1) What is a home range?
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6) HR estimation workflow
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Sample sizes

It is important to distinguish 2 sample size concepts:

Absolute sample size $\neq$ **Effective** sample size

- *Total number of locations*

n

$T \div \Delta t$

Sampling ***duration***

How long is an animal tracked for?

Sampling ***frequency***

How frequently are locations collected?

Sample sizes

It is important to distinguish 2 sample size concepts:

Absolute sample size ≠ **Effective** sample size

- *Total number of locations*

n

$T \div \Delta t$

Sampling ***duration***

How long is an animal tracked for?

Sampling ***interval***

Time between fixes. Inversely, *how frequently* are locations collected?

N_{area} N_{speed}

Roughly estimated as T/τ_p

T is the sampling duration

τ_p is the average home range crossing time

Sample sizes

- Many biases, including most that affect home range estimation, are exacerbated by *small sample sizes*

For independent data

$$\boldsymbol{n = N_{ESS}}$$

For autocorrelated data

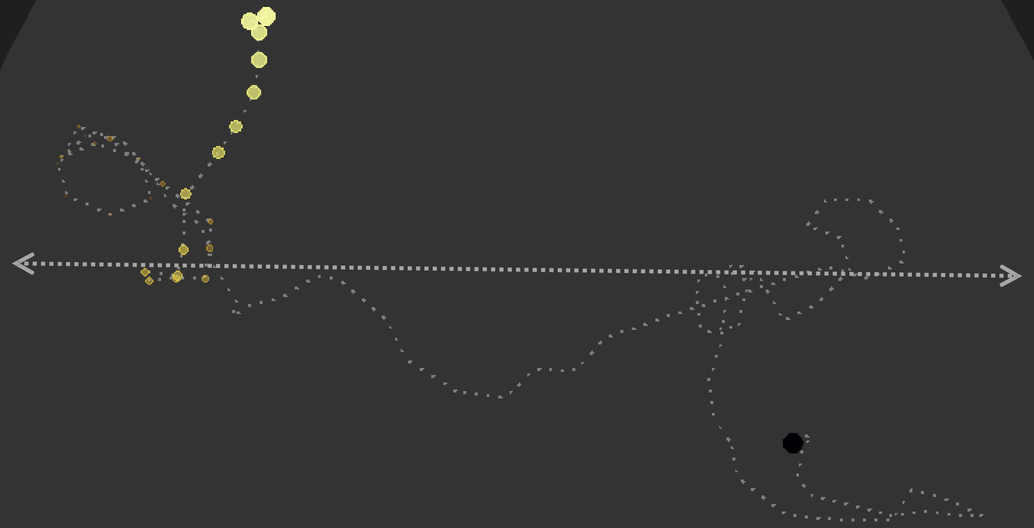
$$\boldsymbol{n \gg N_{ESS}}$$

n = absolute sample size

N_{ESS} = effective sample size (ESS), degrees of freedom (DOF)

τ_p

Position autocorrelation timescale



How long does it take for an animal to traverse the linear extent of its home range?

Duration = 1 day
τ_p = 1 day

τ_p

Position autocorrelation
timescale

τ_p = 1 hour

τ_p = 10 days

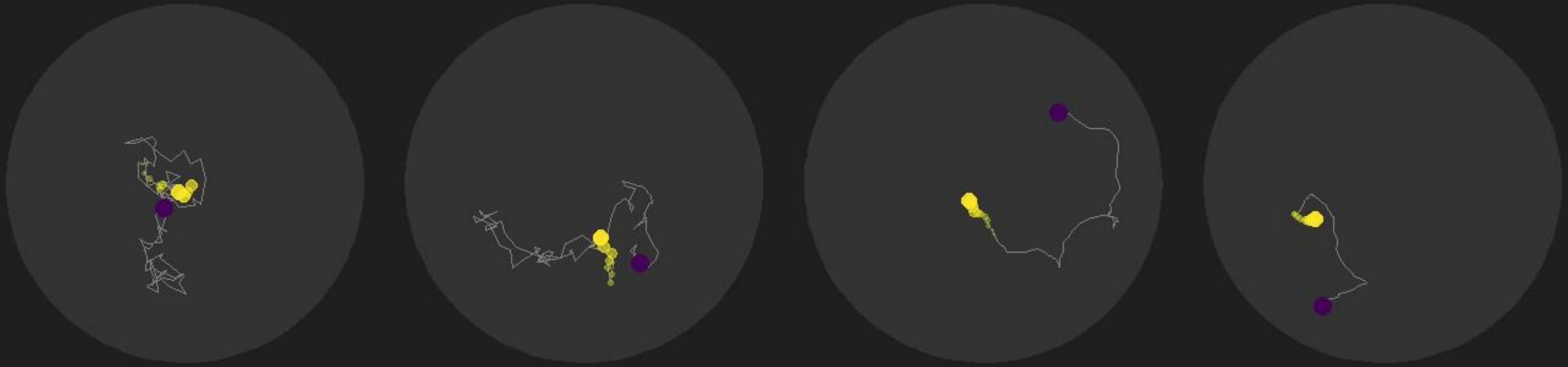
Effective sample size (N_{ESS}) decreases as the
home range crossing time parameter (τ_p) increases.

τ_v

Velocity autocorrelation
timescale

τ_v = 1 minute

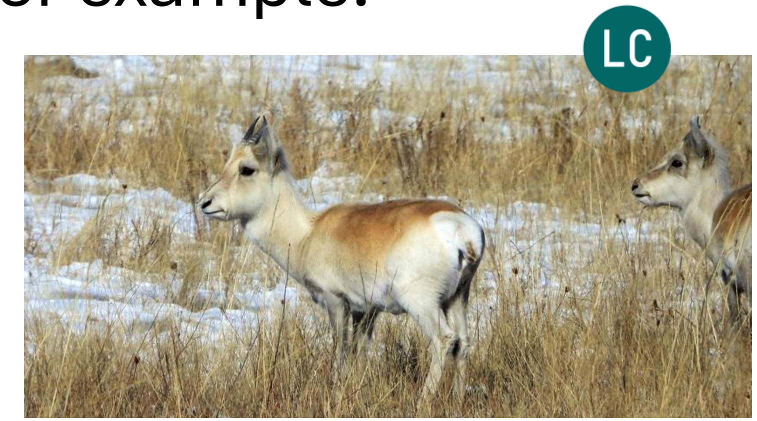
τ_v = 1 day



Tortuosity (winding and twisting) decreases as the *straight-line movement parameter* (τ_v) increases.

Sample sizes

For example:



Mongolian Gazelle
(Procapra gutturosa)

Sampling *duration* ≈ tracked for 1 year (360 days)

Sampling *frequency* ≈ 1 fix every hour

Sample sizes

For example:

Mongolian Gazelle
(Procapra gutturosa)

Absolute sample size, n = 866 locations

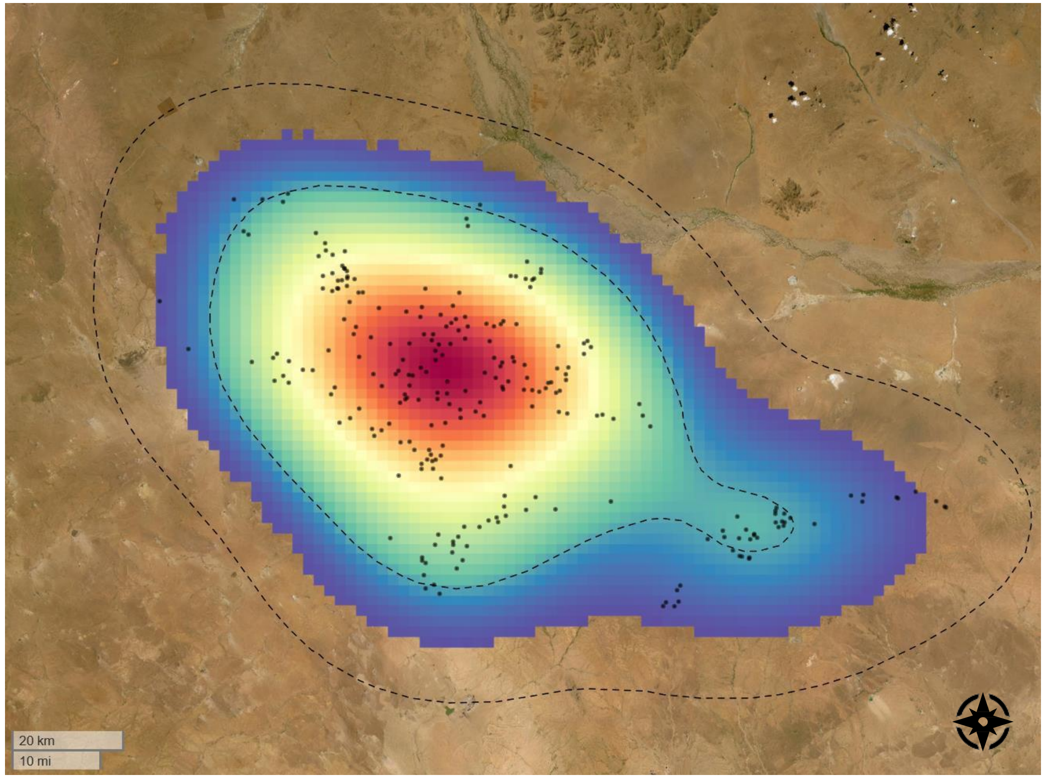
Effective sample size, N_{ESS} ≈ 2.4 locations

$\boxed{\tau_p \approx 170 \text{ days}}$

Home range area estimate
≈ 61,000 km²

Sampling *duration* ≈ tracked for 1 year (360 days)

Sampling *frequency* ≈ 1 fix every hour



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Workflow

1) **Check range residency assumption**
   - Verify if the data is from a range-resident animal
2) **Select movement model**
   - Selecting the best-fit model through model selection
3) **Estimate home range area**
   - Reconstructing range distribution from sampled locations

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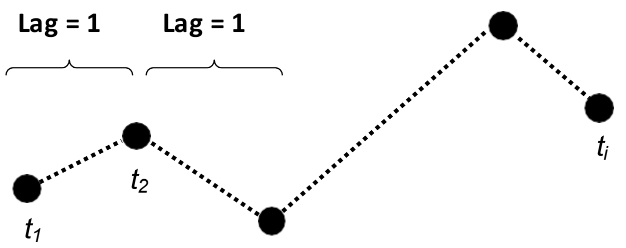
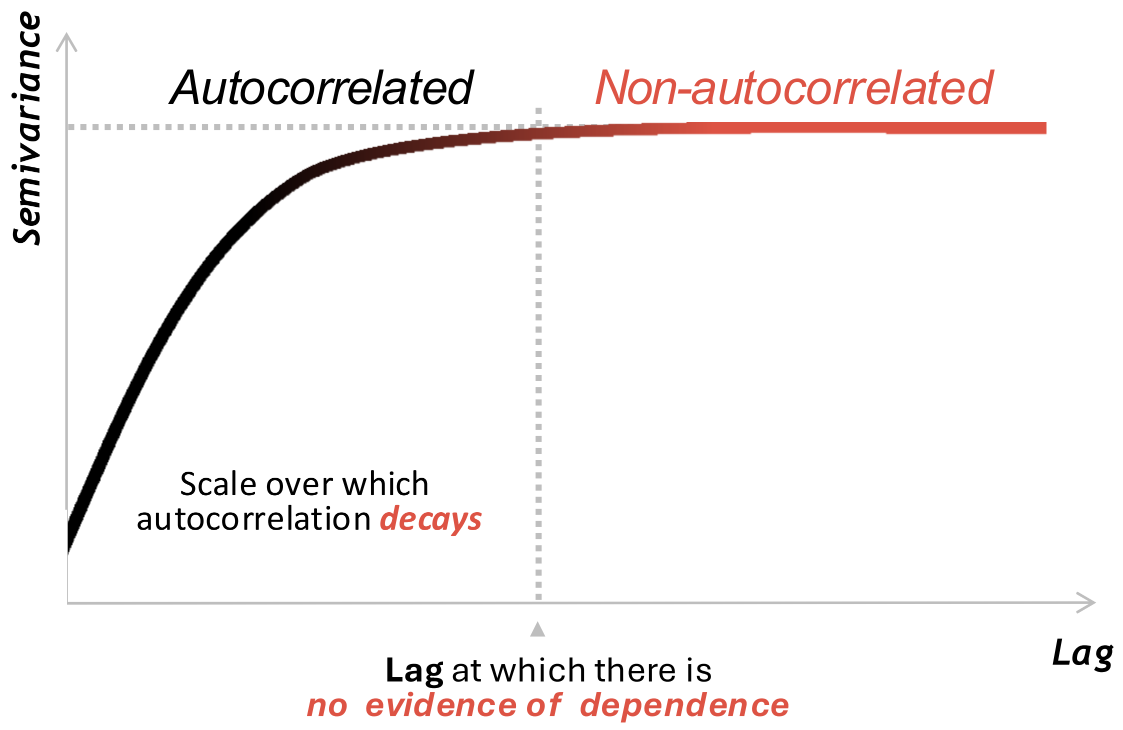
Range residency assumption

How can we measure and visualize autocorrelation?

Semivariograms (or variograms) are used to visualize autocorrelation. We plot lags on the x-axis for all pairs of observation against their semivariance (mean square distance between any 2 locations with a given lag) on the y-axis.





The more similar the pairs of locations per lag, the lower the semivariance for that lag.

Range residency assumption

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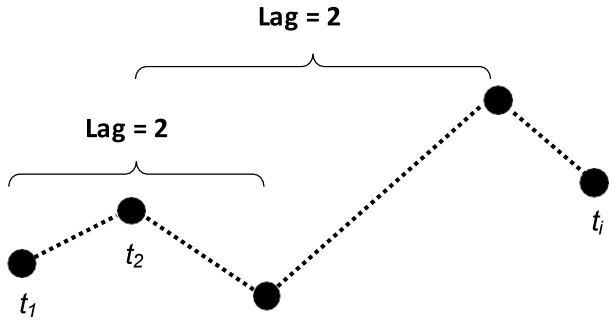

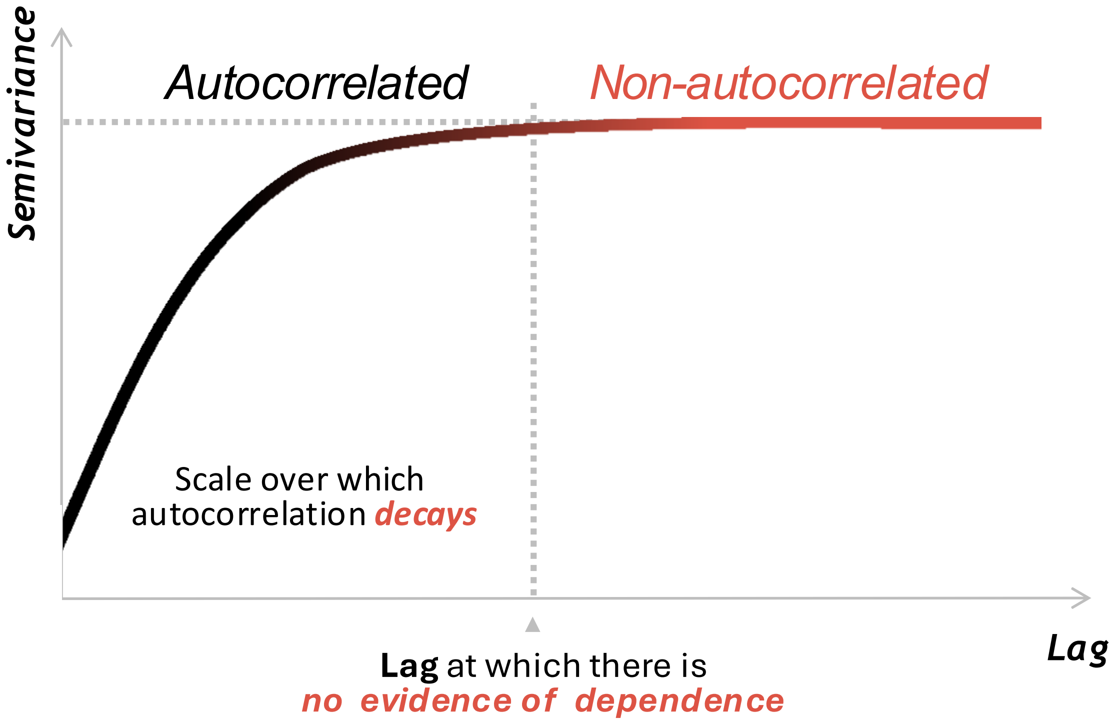

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Selecting movement models

What process best explains the given animal movement dataset?

- **Independent and identically distributed** (IID)
- **Brownian motion** (BM)
- **Ornstein-Uhlenbeck** (OU)
- **Integrated Ornstein-Uhlenbeck** (IOU)
- **Ornstein-Uhlenbeck with Foraging** (OUF)

Conventional methods:

Tracking data → Home range estimate

Assuming *independent process*

Continuous-time methods:

Tracking data → *Movement model* → Home range estimate

Selecting movement models

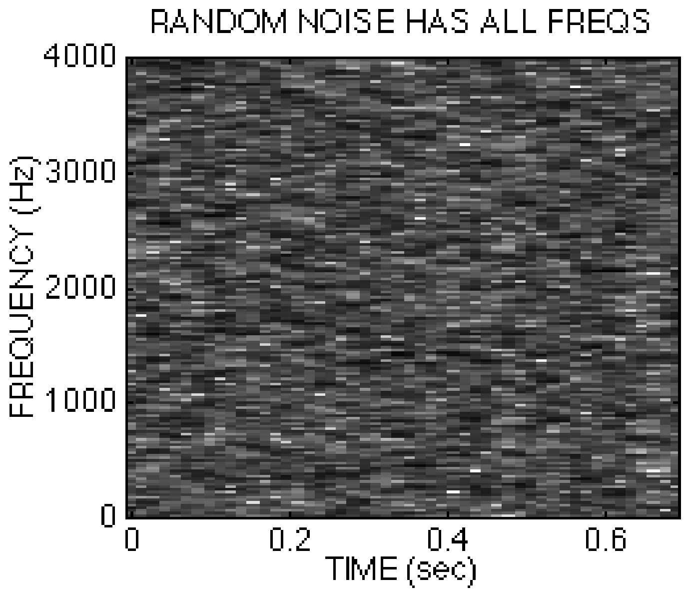
SPATIAL DEPENDENCY

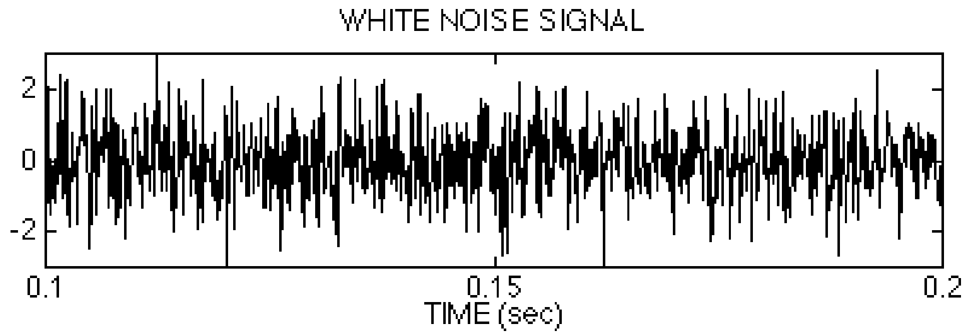
VELOCITY DEPENDENCY

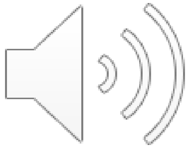
RESTRICTED

Independent and identically distributed (IID):

- Stochastic process where each location has the same probability distribution as the others, and are all mutually independent
  - For example, white noise:







McClellan, Shafer, Yoder

Selecting movement models

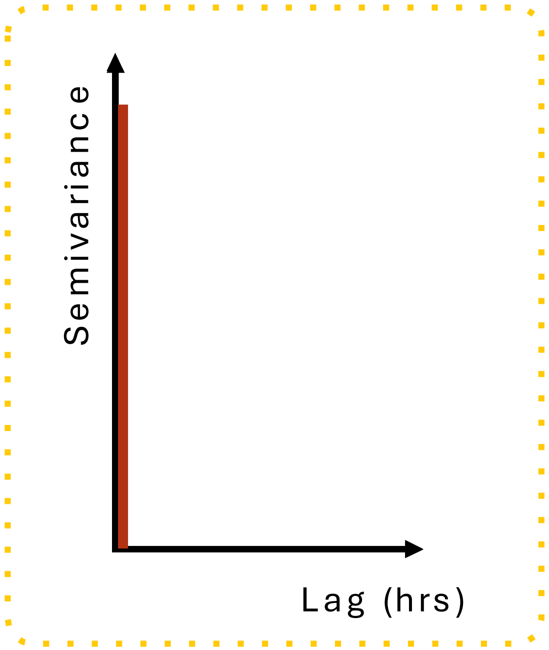
SPATIAL DEPENDENCY

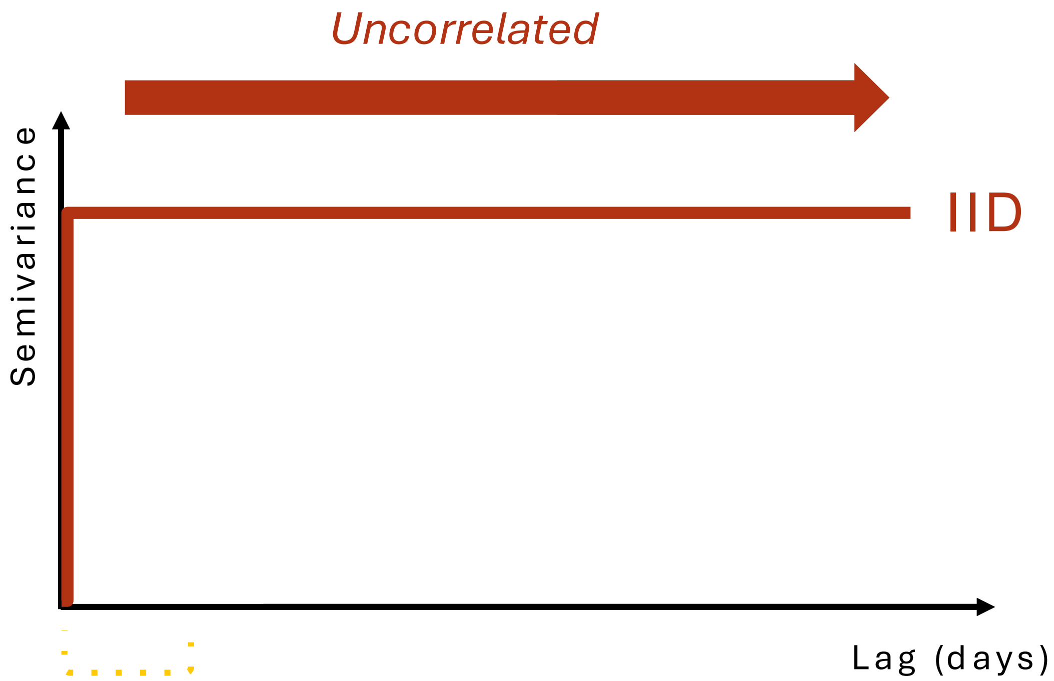
VELOCITY DEPENDENCY

RESTRICTED

Independent and identically distributed (IID):

- How would the variogram of a IID process look?





Selecting movement models

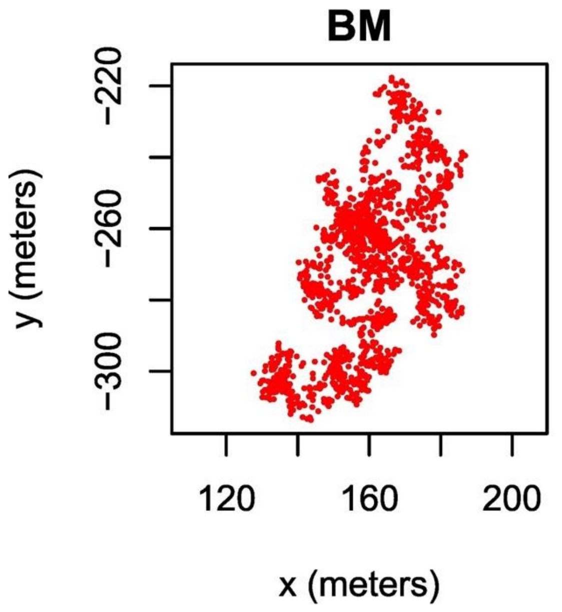
SPATIAL DEPENDENCY

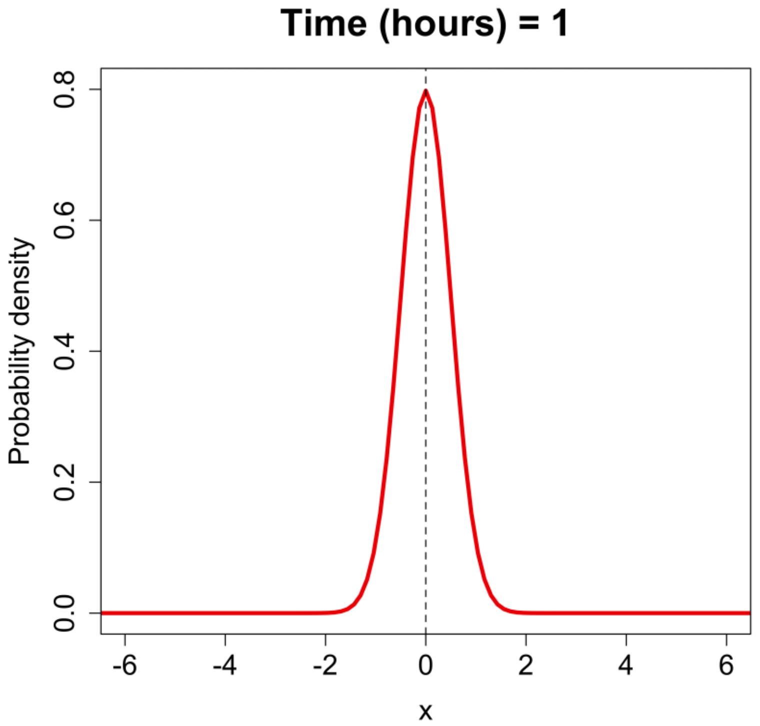
VELOCITY DEPENDENCY

RESTRICTED

Brownian motions (BM):

- Stochastic process with stationary and independent increments (i.e., no "memory"). The future behavior of BM does not depend on its past. Diffusion is constant.





As time goes on, the animal is more likely to be further away from its starting location.

Selecting movement models

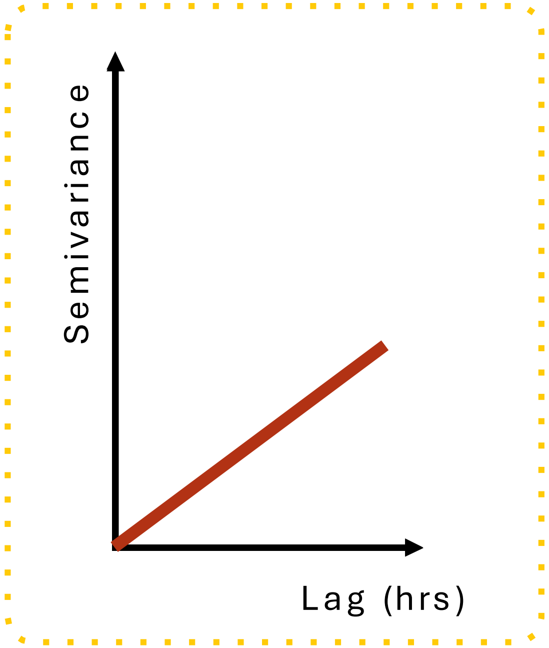
SPATIAL DEPENDENCY

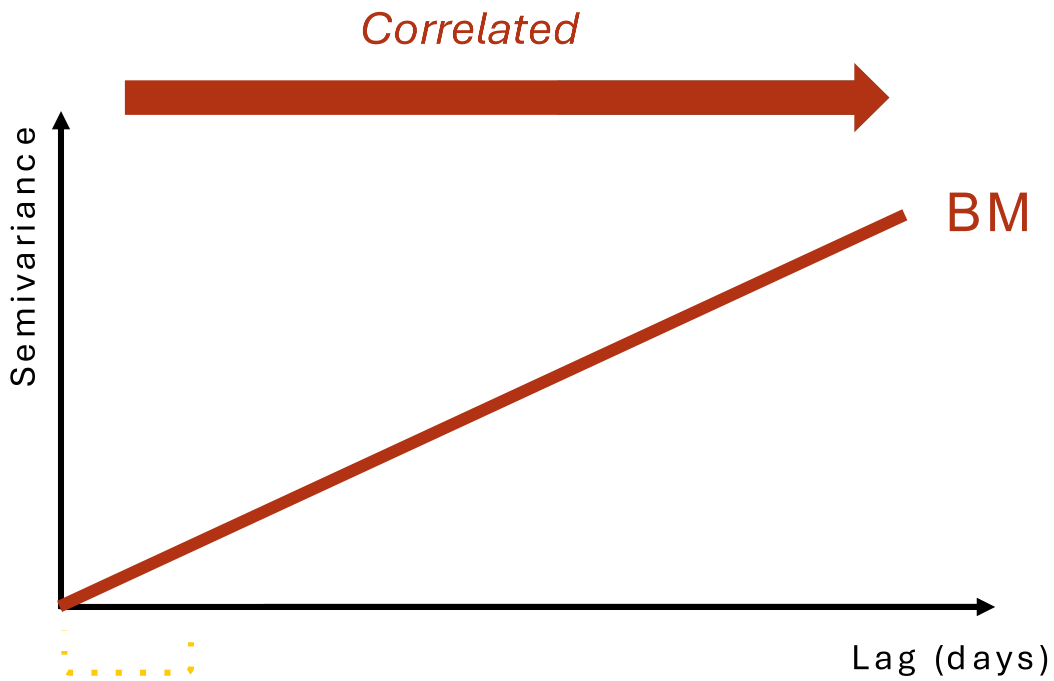
VELOCITY DEPENDENCY

RESTRICTED

Brownian motions (BM):

- How would the variogram of a BM process look?

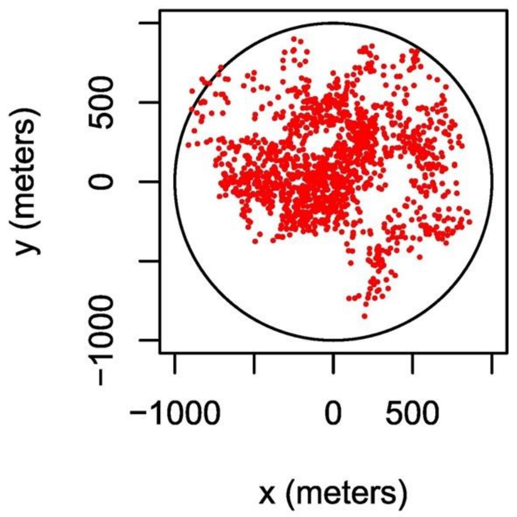




Selecting movement models

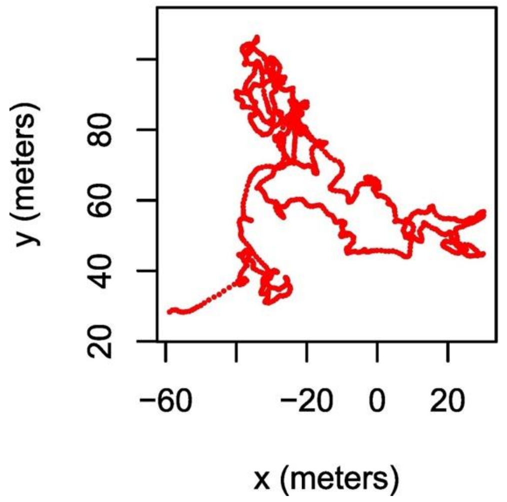
These processes are all modifications of the BM process:

Ornstein-Uhlenbeck **(OU)**



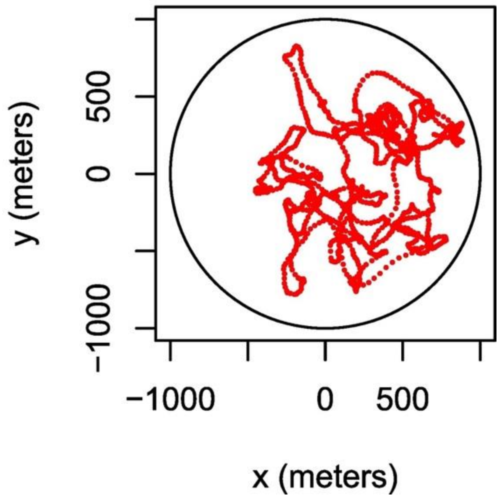
Unlike **BM**, the **OU** process tends towards a central location (*bounded diffusion*).

Integrated OU **(IOU)**



Like **BM**, the **IOU** process exhibits *unbounded diffusion* — but unlike **BM**, it has *persistence of motion*.

OU with Foraging **(OUF)**



Unlike **BM**, the **OUF** process exhibits *bounded diffusion* and *persistence of motion*.

Selecting movement models

SPATIAL DEPENDENCY

VELOCITY DEPENDENCY

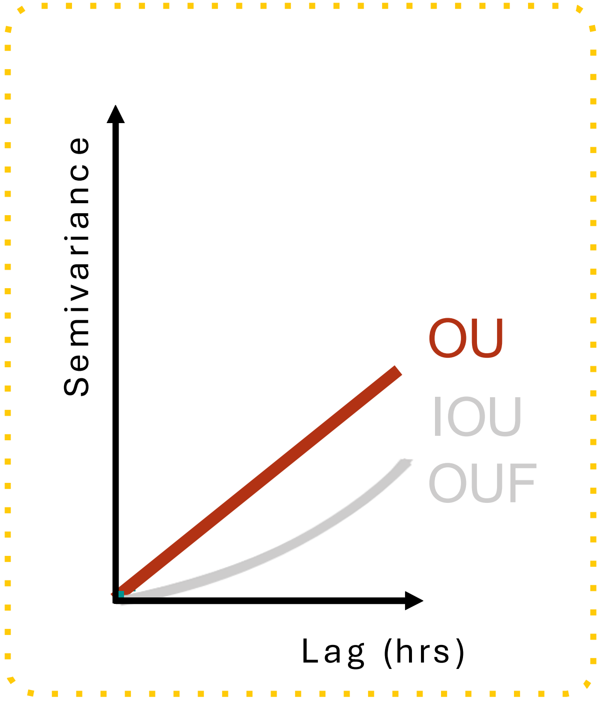
RESTRICTED

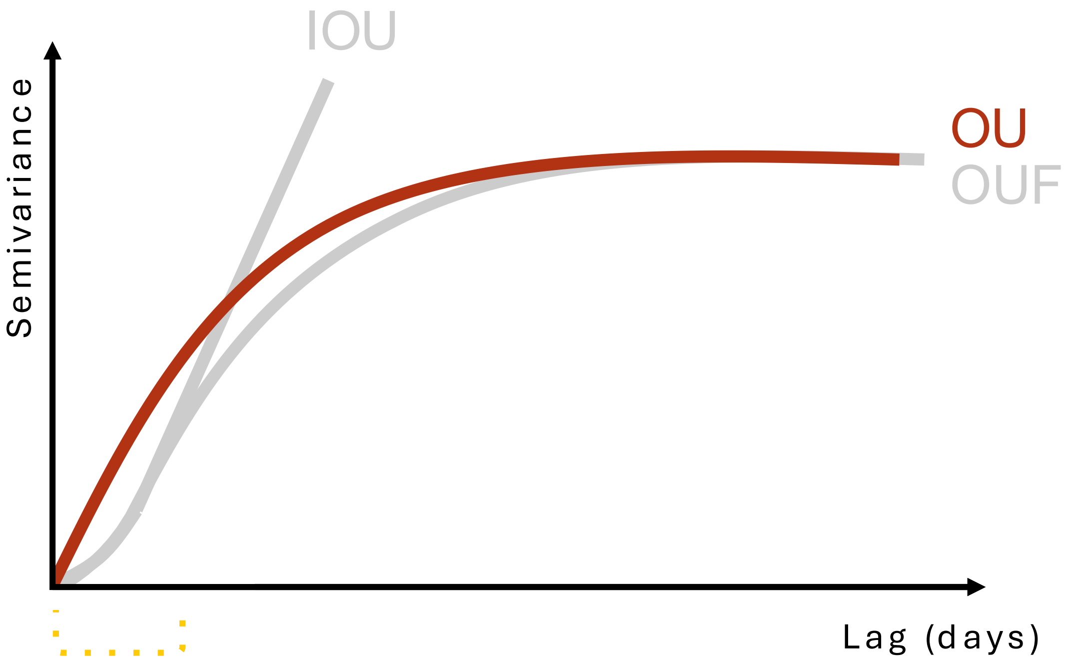
These processes are all related:

Ornstein-Uhlenbeck **(OU)**

Integrated OU **(IOU)**

OU with Foraging **(OUF)**





Selecting movement models

SPATIAL DEPENDENCY

VELOCITY DEPENDENCY

RESTRICTED

These processes are all related:

Ornstein-Uhlenbeck **(OU)**

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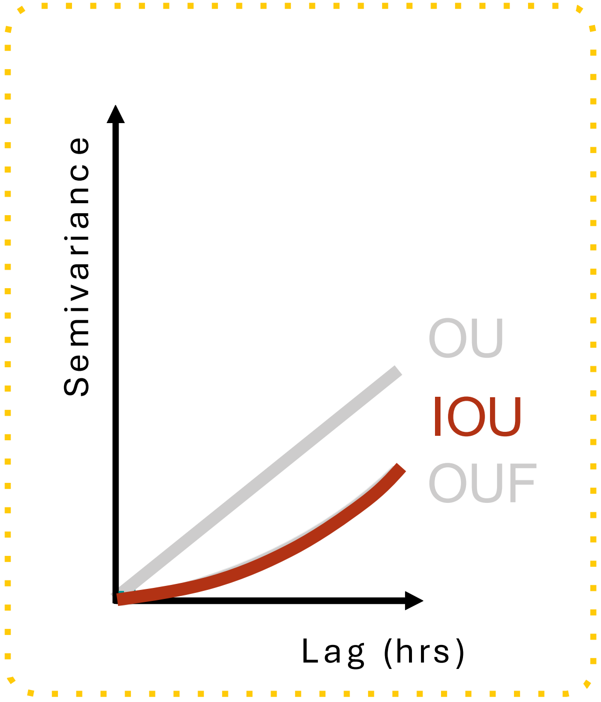
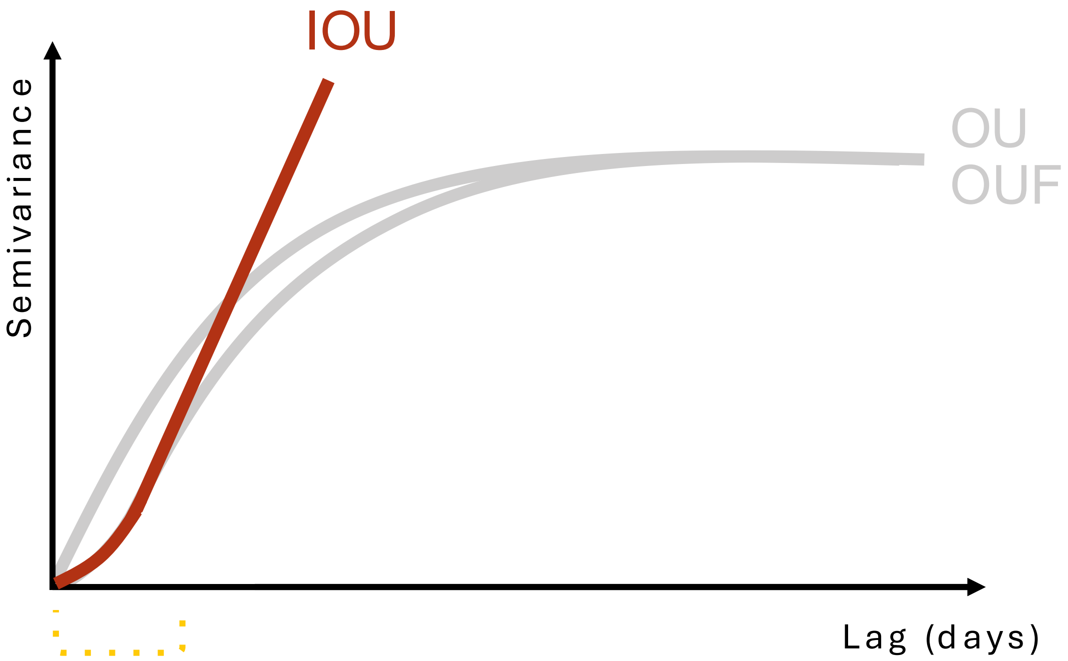




Selecting movement models

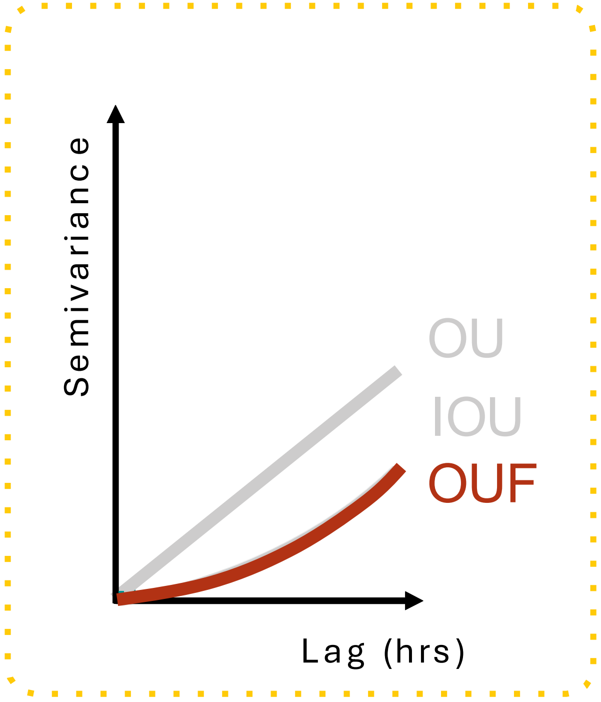
SPATIAL DEPENDENCY

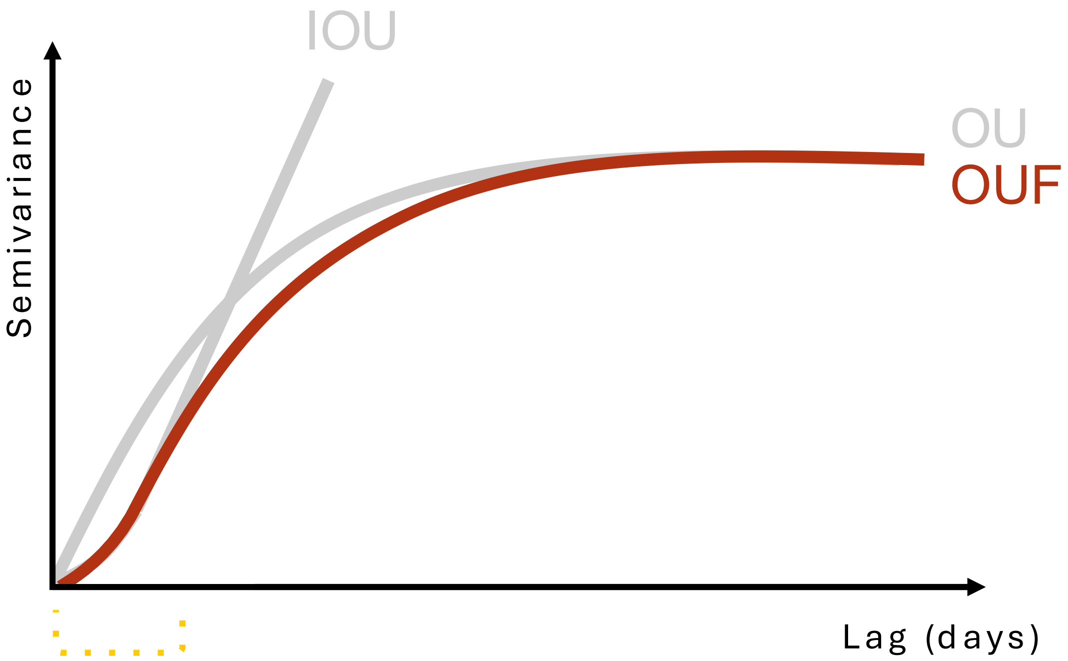
VELOCITY DEPENDENCY

RESTRICTED

These processes are all related:

Ornstein-Uhlenbeck **(OU)** Integrated OU **(IOU)** OU with Foraging **(OUF)**

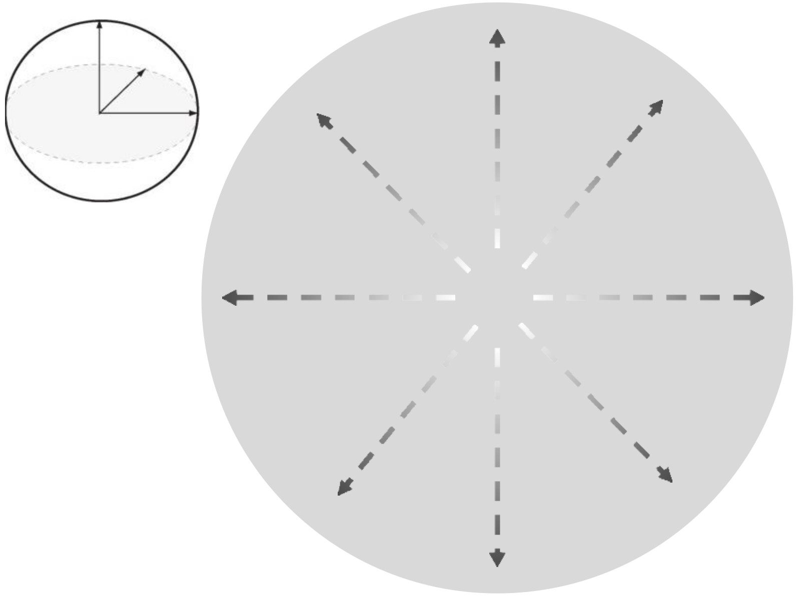




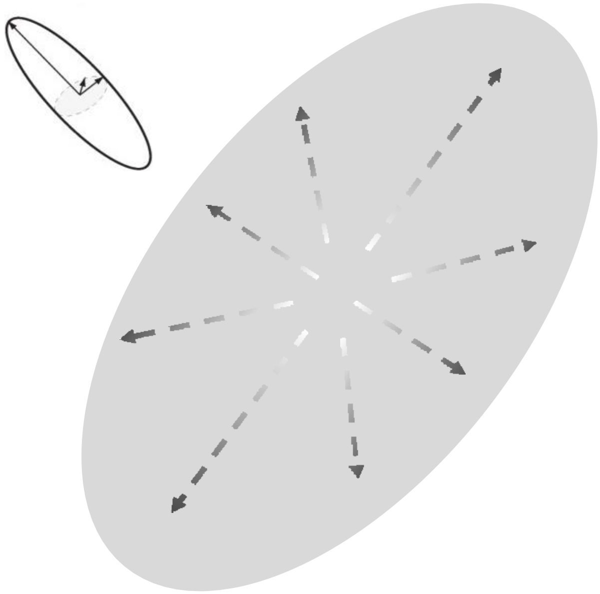
Selecting movement models

Isotropic refers to the properties of a material which are independent of the direction.

Anisotropic is direction-dependent.



Isotropic diffusion (circle)



Anisotropic diffusion (ellipse)

Workflow

1) **Check range residency assumption**
    - Verify if the data is from a range-resident animal
2) **Select movement model**
    - Selecting the best-fit model through model selection
3) **Estimate home range area**
    - Reconstructing range distribution from sampled locations

Estimating HR area

> “ All models are wrong, but some are useful.
>
> Box *et al.* (1987)

Autocorrelation

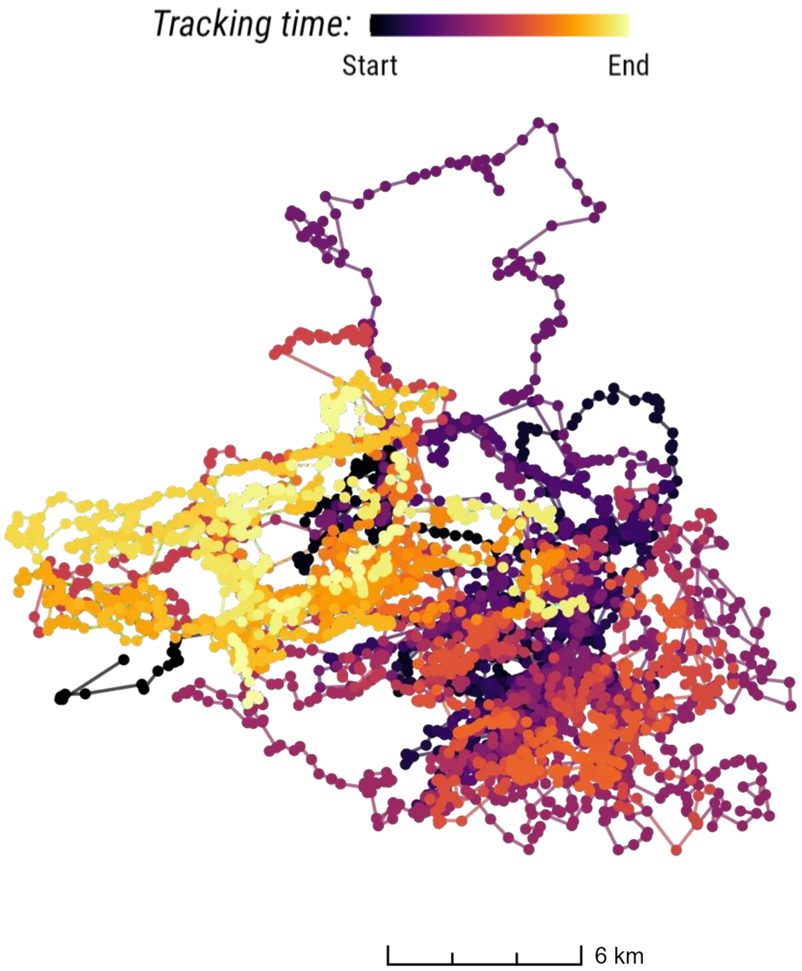
Model	Position	Velocity	Restricted	Parameters
IID	No	No	Yes	τ = NULL
BM	Yes	No	No	$\tau = \infty$
OU	Yes	No	Yes	$\tau = \tau_p$
IOU	Yes	Yes	No	$\tau = \{\infty, \tau_v\}$
OUF	Yes	Yes	Yes	$\tau = \{\tau_p, \tau_v\}$

AKDE explicitly requires a movement model that accounts for ***autocorrelation***.

AKDE reduces to the (**conventional) KDE** in the limit where autocorrelation vanishes, and locations are truly ***independent***.

Estimating HR area

African Buffalo
(*Syncerus caffer*)



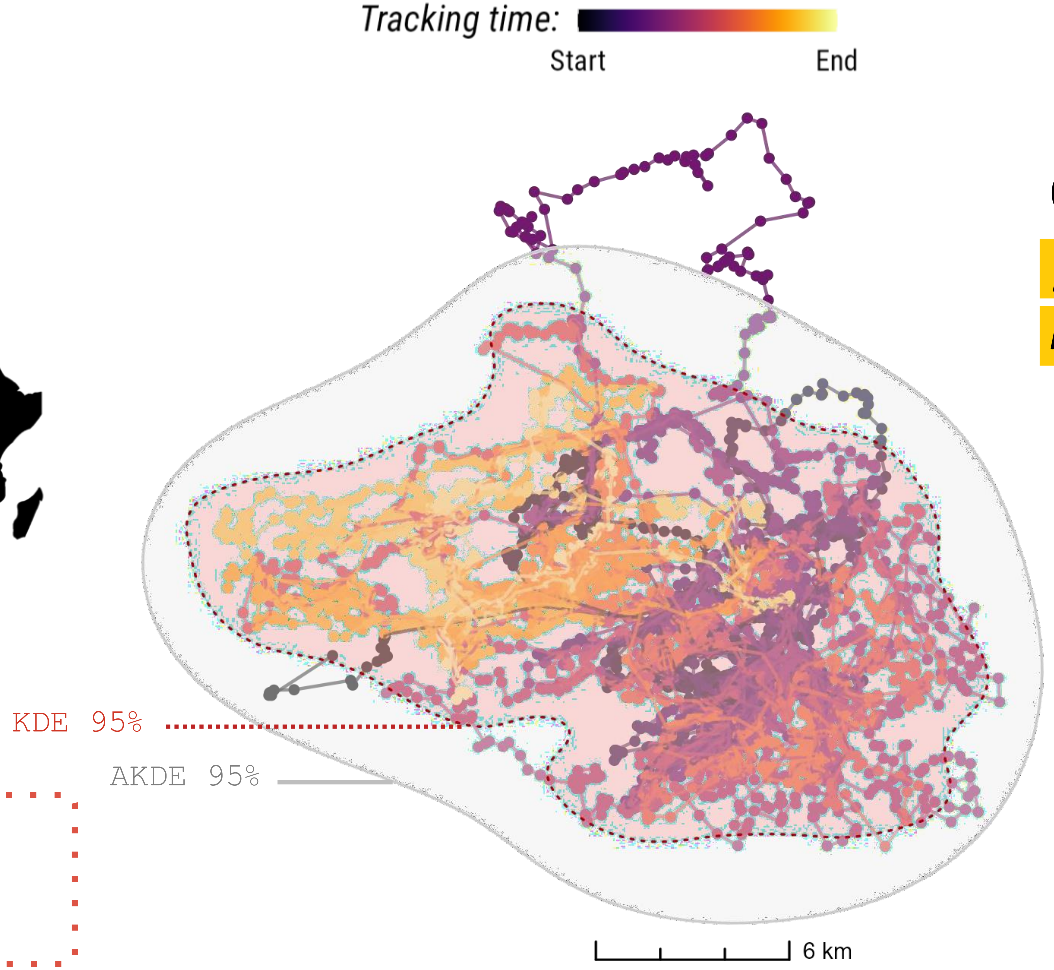
OUF process:

n = 5,677 locations

N ≈ 22.3 locations

Estimating HR area

African Buffalo
(Syncerus caffer)



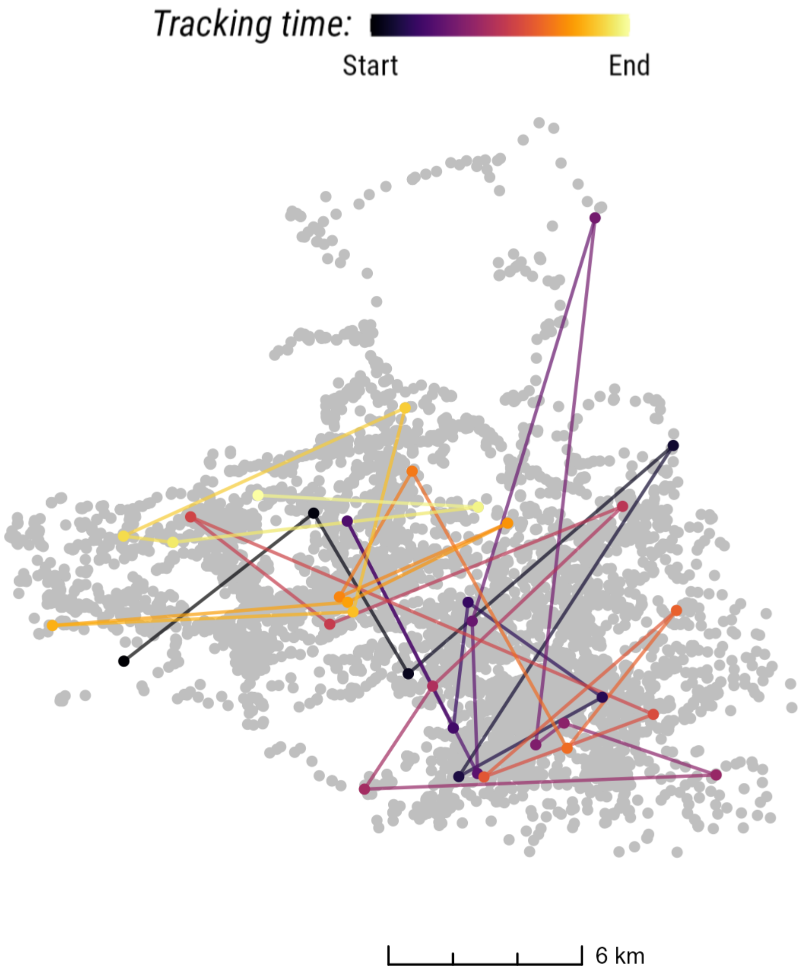
OUF process:

n = 5,677 locations
N ≈ 22.3 locations

Expected home range area underestimation of ~38.4%

Estimating HR area

African Buffalo
(*Syncerus caffer*)



OUF process:

n = 5,677 locations

N ≈ 22.3 locations

IID process:

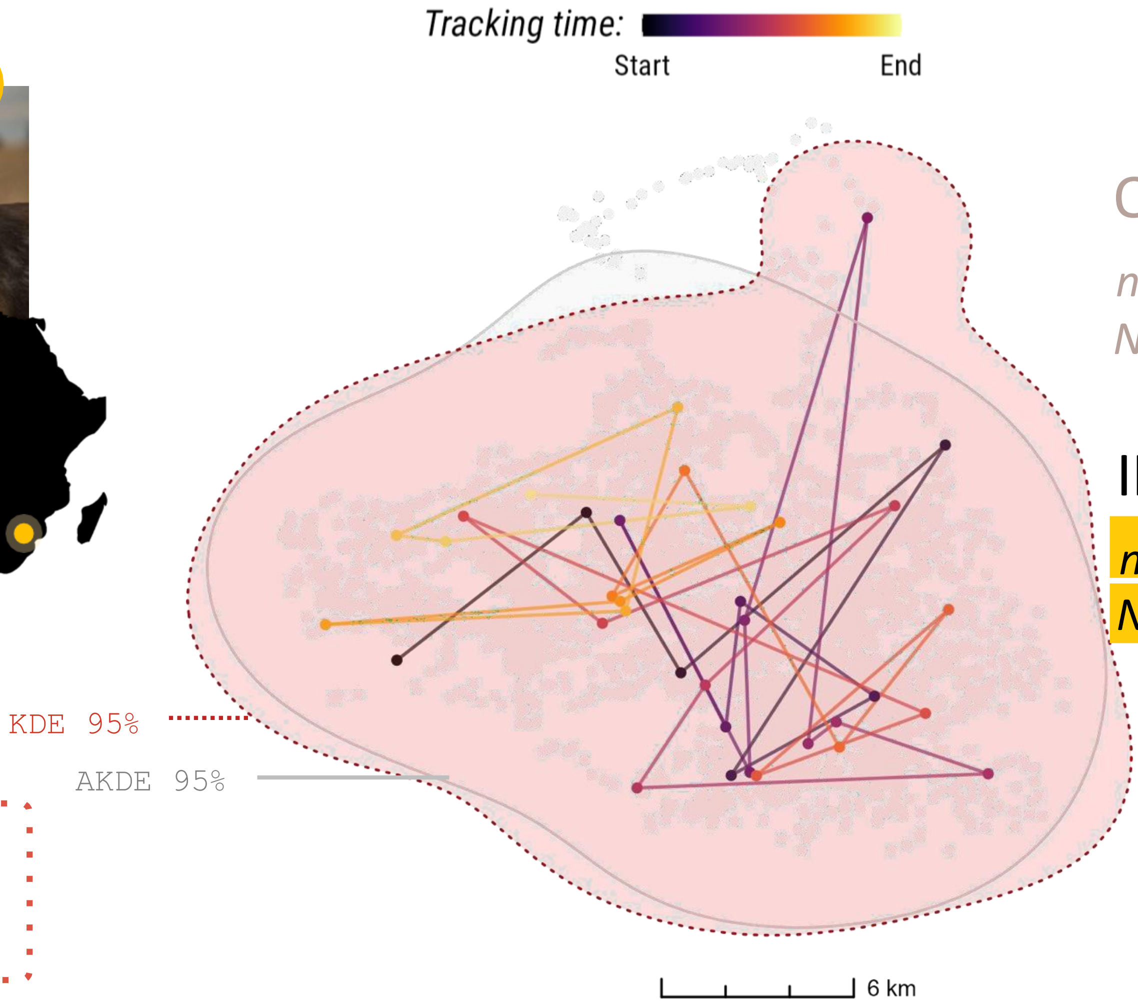
n = 35 locations

N = 35 locations

Data loss of ≈ **99.4%**

Estimating HR area

African Buffalo
(Syncerus caffer)



OUF process:

n = 5,677 locations
N ≈ 22.3 locations

IID process:

n = 35 locations
N = 35 locations

Data loss of ≈ **99.4%**

Expected home range area overestimation of ~14.8%

Overview

1) What is a home range?
2) Why do these concepts matter?
3) Methods for home range estimation
4) Effective sample size
5) HR estimation workflow
6) **Mitigating bias**

Workflow

1) **Check range residency assumption**
   - Verify if the data is from a range-resident animal
2) **Select movement model**
   - Selecting the best-fit model through model selection
3) **Estimate home range area**
   - Reconstructing range distribution from sampled locations
4) **Apply mitigation measures**
   - Accounting for common biases in movement data

Mitigating bias

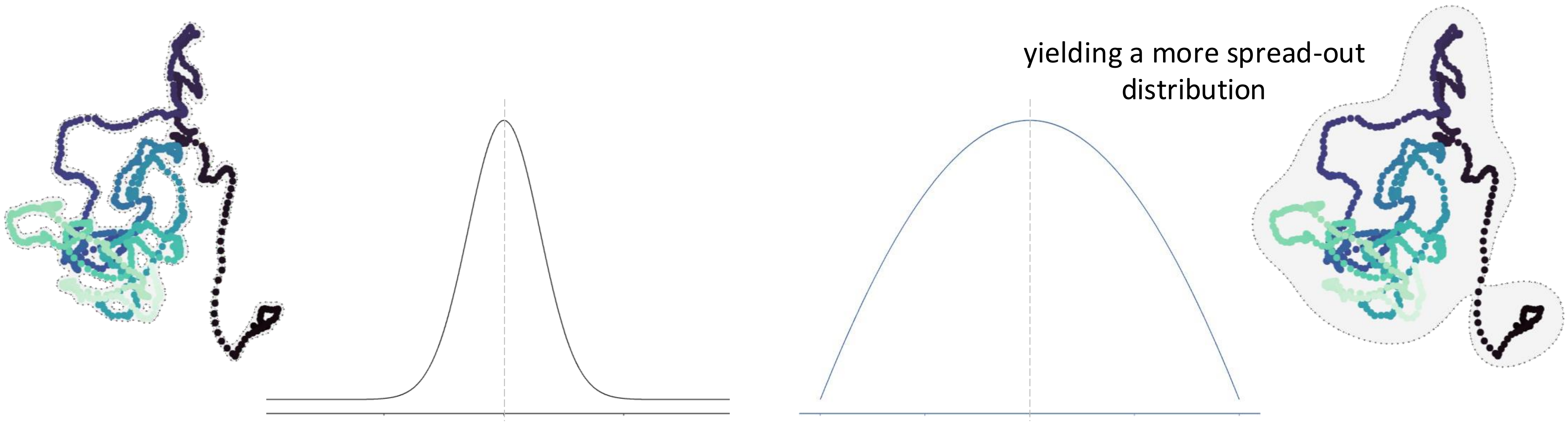
Many biases, including most that affect home range estimates, are exacerbated by *small sample sizes*. Conversely, *large sample sizes* in modern tracking datasets exacerbate autocorrelation.

Bias sources (in order of general importance):		**Mitigation measures**
Unmodelled autocorrelation	→	AKDE
Oversmoothing	→	AKDE_c (default)
Autocorrelation estimation bias	→	pHREML (default) Parametric bootstrapping
Unrepresentative sampling in time	→	Weighted AKDE (wAKDE)

Mitigating bias

Area-corrected AKDE ($\mathbf{AKDE_c}$):

- Deals with *oversmoothing*
- Even when we account for autocorrelation, *GRF-KDEs* remain biased due to the natural tendency of the GRF approximation to oversmooth.



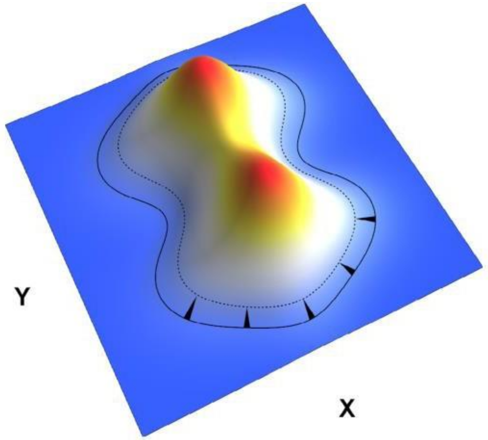
Mitigating bias

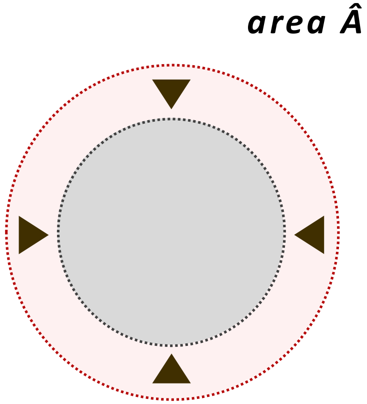
Area-corrected AKDE (**AKDE$_c$**):

- Deals with *oversmoothing*
- Even when we account for autocorrelation, *GRF-KDEs* remain biased due to the natural tendency of the GRF approximation to oversmooth.

Derived an *improved (A)KDE* that pulls the contours of the location distribution estimate inward towards the data without distorting its shape.

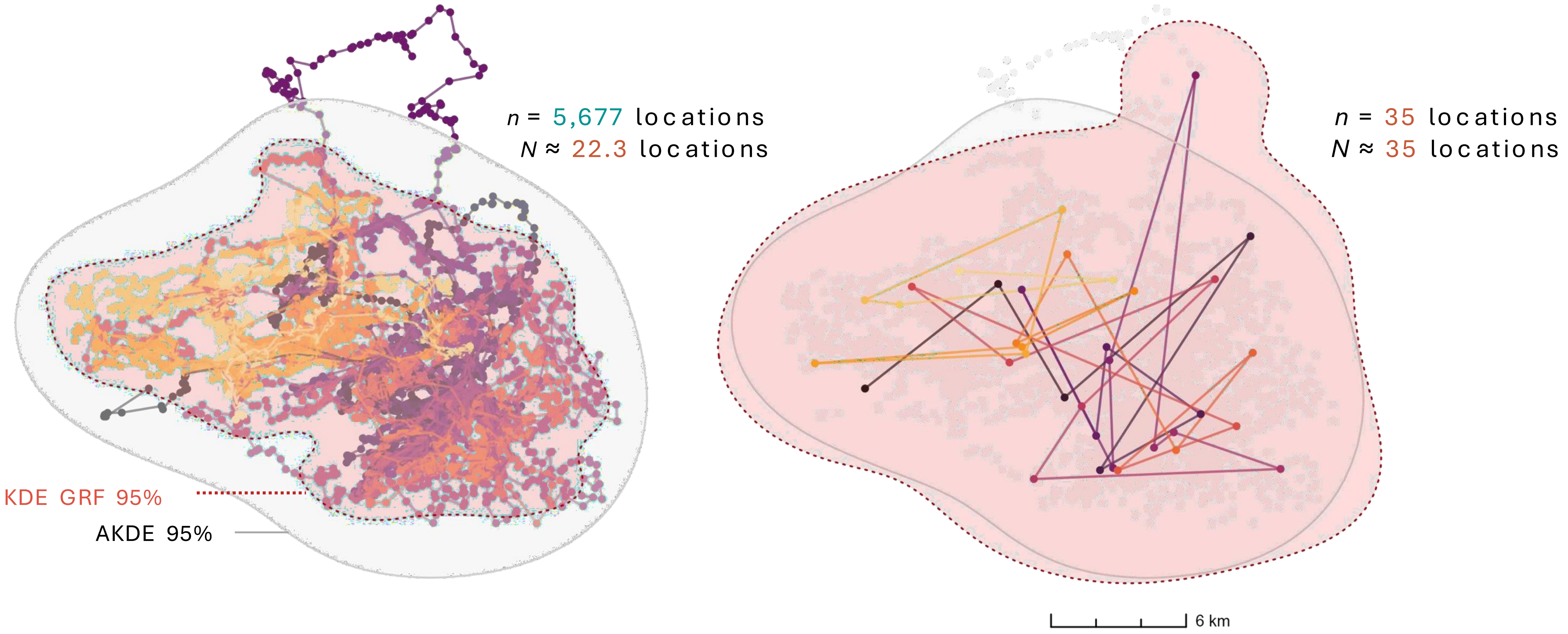
Fleming & Calabrese (2017)





Mitigating bias

- The oversmoothing (positive) bias can be *masked* by the often-stronger negative bias caused by the unmodeled autocorrelation



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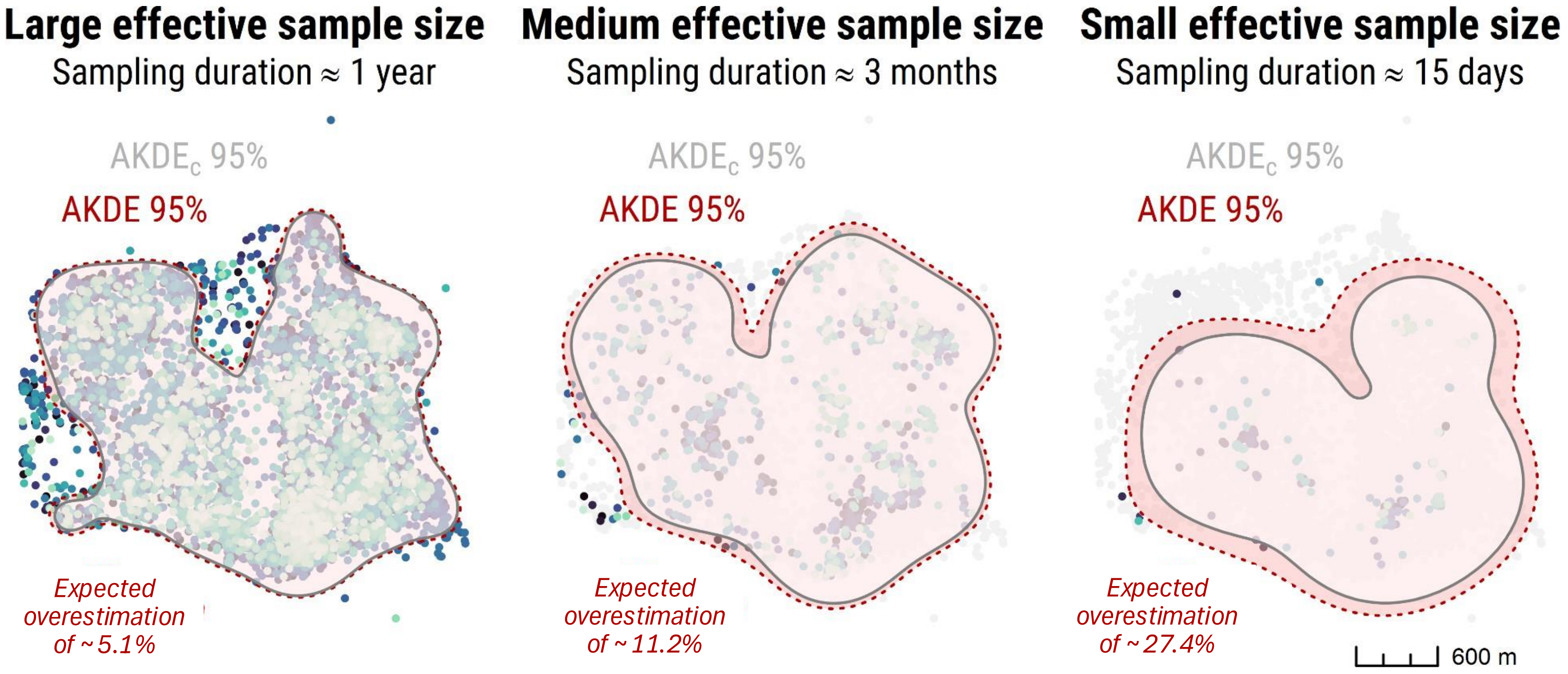
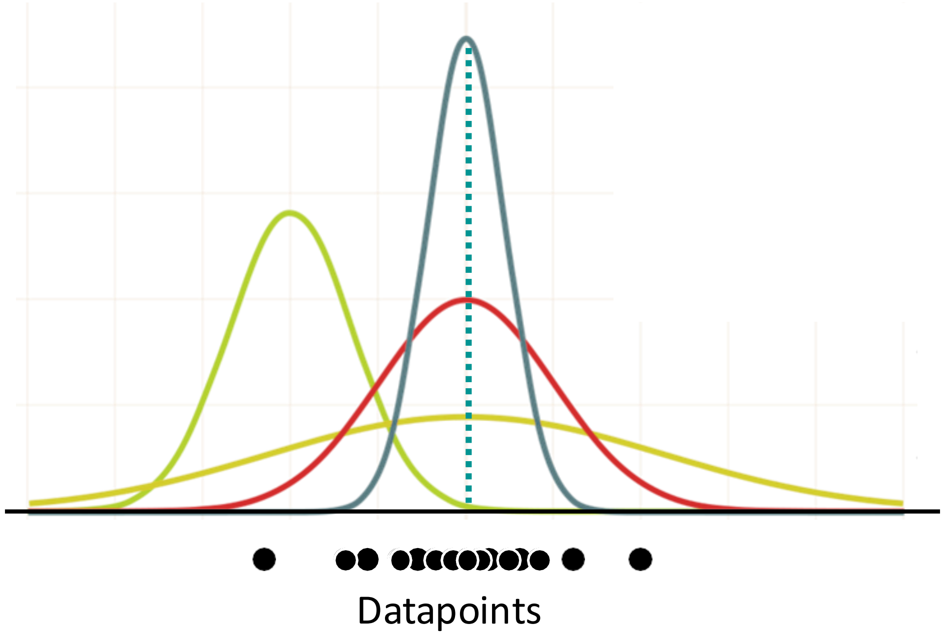


Mitigating bias

Maximum likelihood (ML):

- Standard approach to fitting movement models
- Goal of ML is to select the parameter values that make the observed data most probable

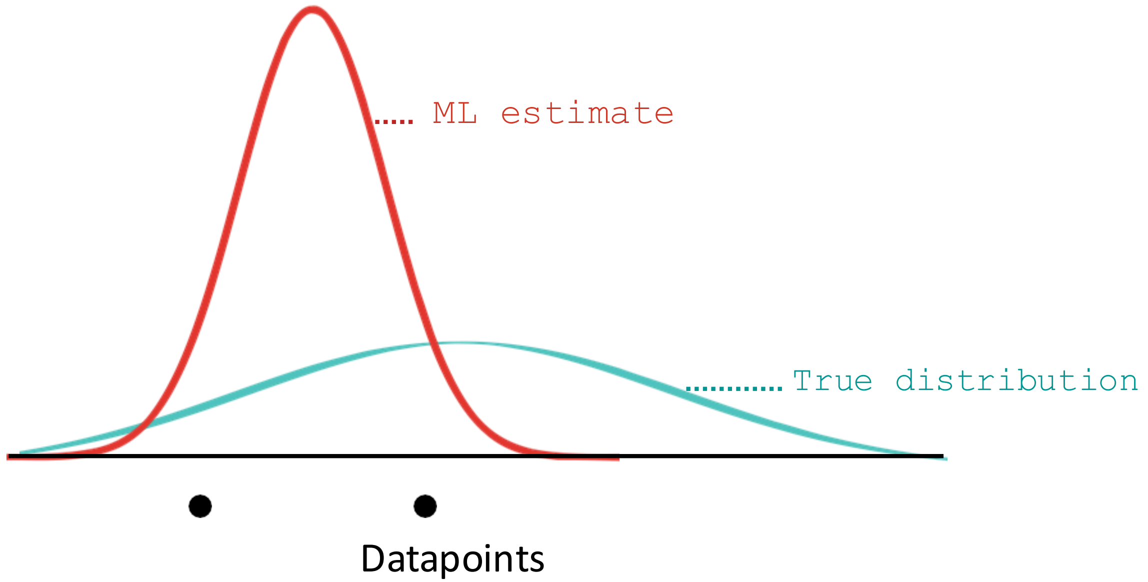


Mitigating bias

Cressie (1993)

Maximum likelihood (ML):

- Unfortunately, ML performs poorly at *small sample sizes*



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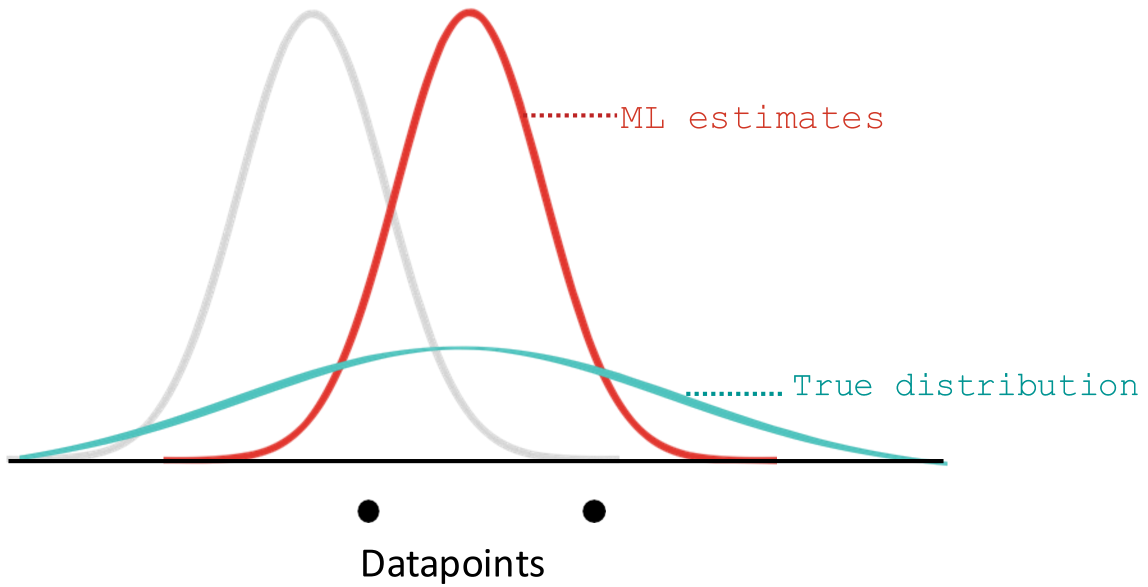

Mitigating bias

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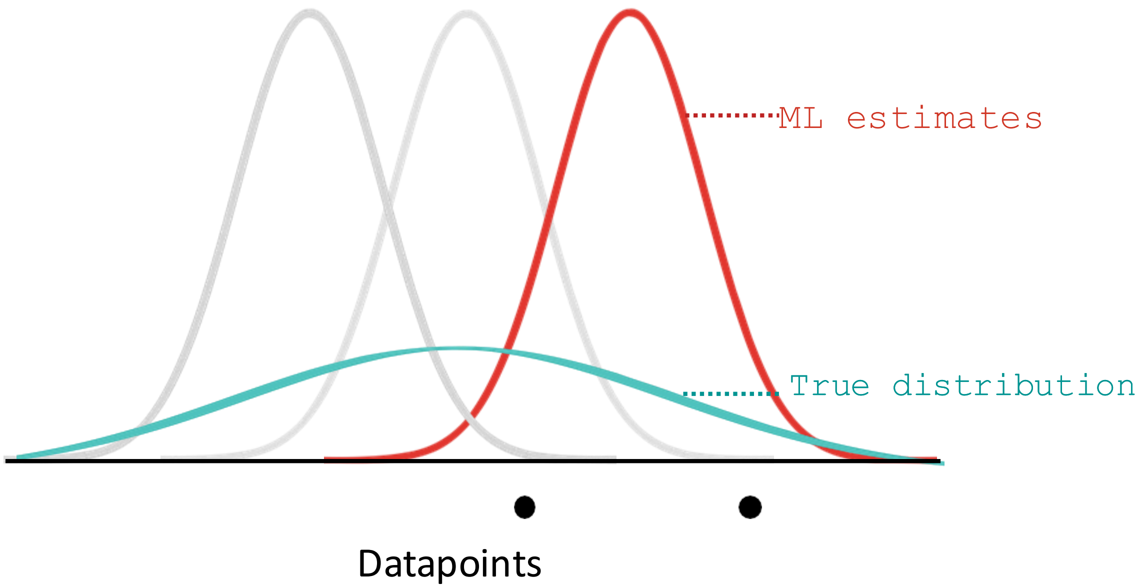
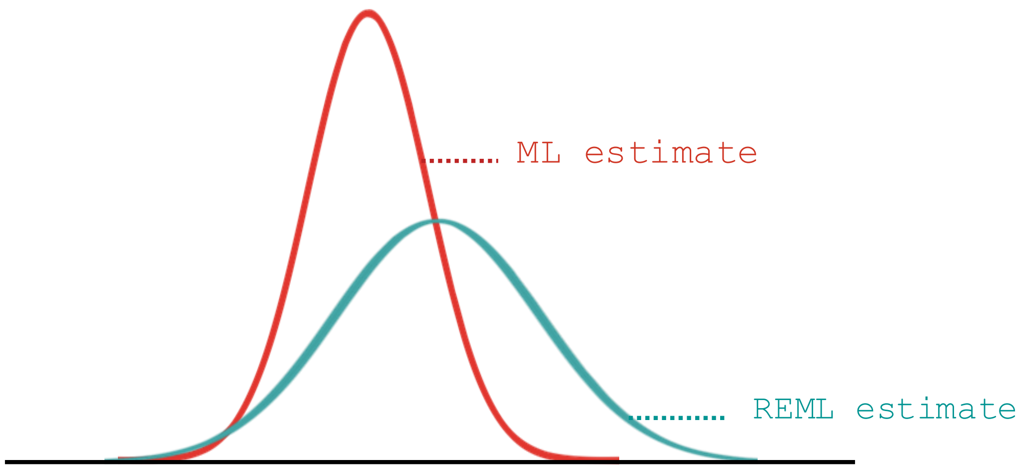


Mitigating bias

Bartlett (1937)

Residual ML (REML):

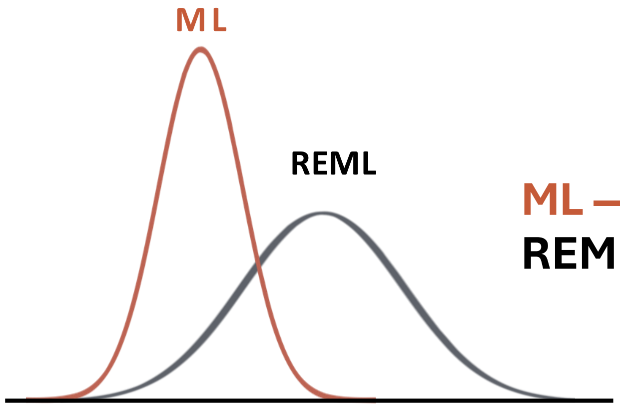
- Widely used method for reducing bias in *ML variance estimation*, by maximizing the likelihood or residuals rather than the data
- In simple cases, REML is unbiased, while more generally, REML reduces bias



Mitigating bias

pHREML AKDE:

- Deals with *autocorrelation estimation bias*
- For optimal performance, we need to estimate autocorrelation *correctly*



ML — underestimates at ***small*** **effective sizes**.
REML — is unreliable ***small*** **effective sample sizes**.

Mitigating bias

pHREML AKDE:

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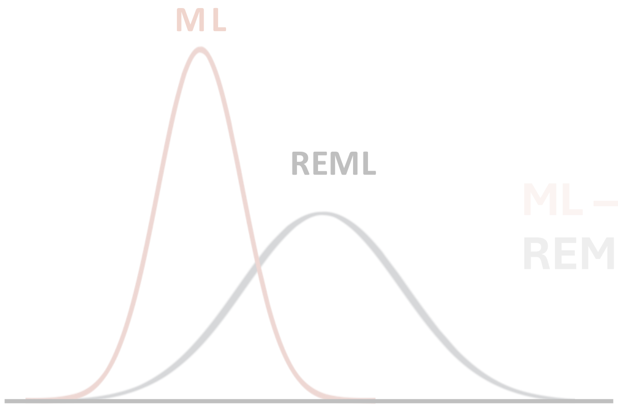

ML — underestimates at ***small*** **effective sizes**.
REML — is unreliable ***small*** **effective sample sizes**.

As such, we consider other parameter estimation methods:

Focus on:

small *effective* sample sizes — perturbative REML (**pREML**)
small *absolute* sample sizes — Hybrid REML (**HREML**)
small *absolute* and *effective* sample sizes — perturbative Hybrid REML (**pHREML**)

Fleming et al. (2019)

Mitigating bias

pHREML AKDE:

- Deals with *autocorrelation estimation bias*
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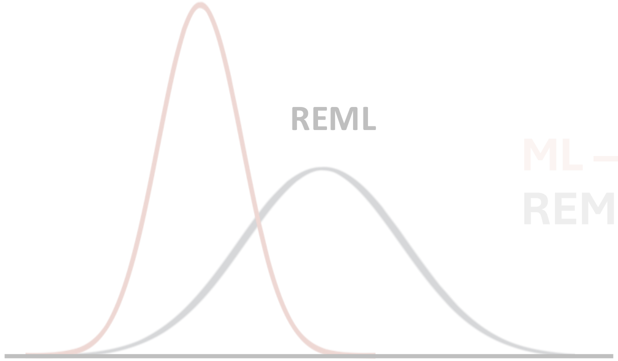

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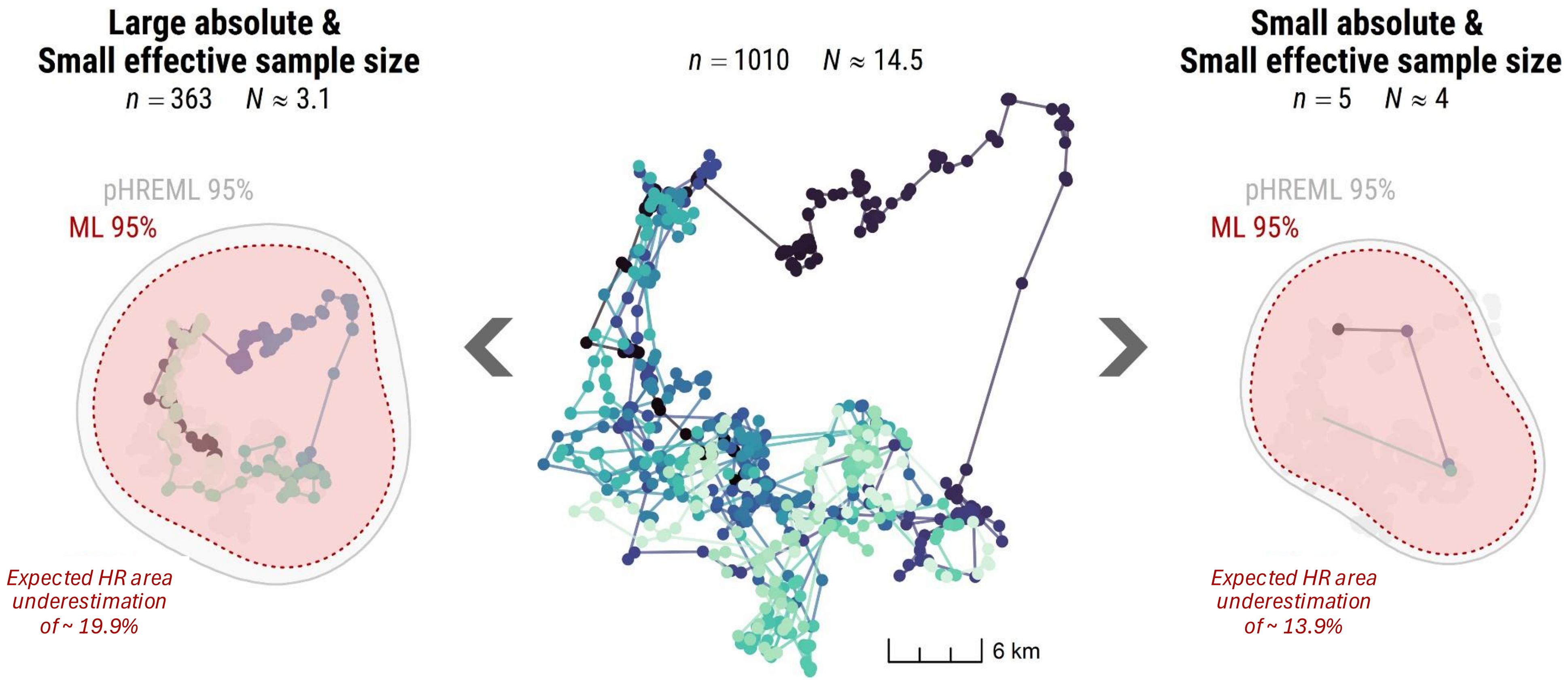
small *effective* sample sizes — perturbative REML (**pREML**)
small *absolute* sample sizes — Hybrid REML (**HREML**)
small *absolute* and *effective* sample sizes — perturbative Hybrid REML (**pHREML**)

Fleming et al. (2019)

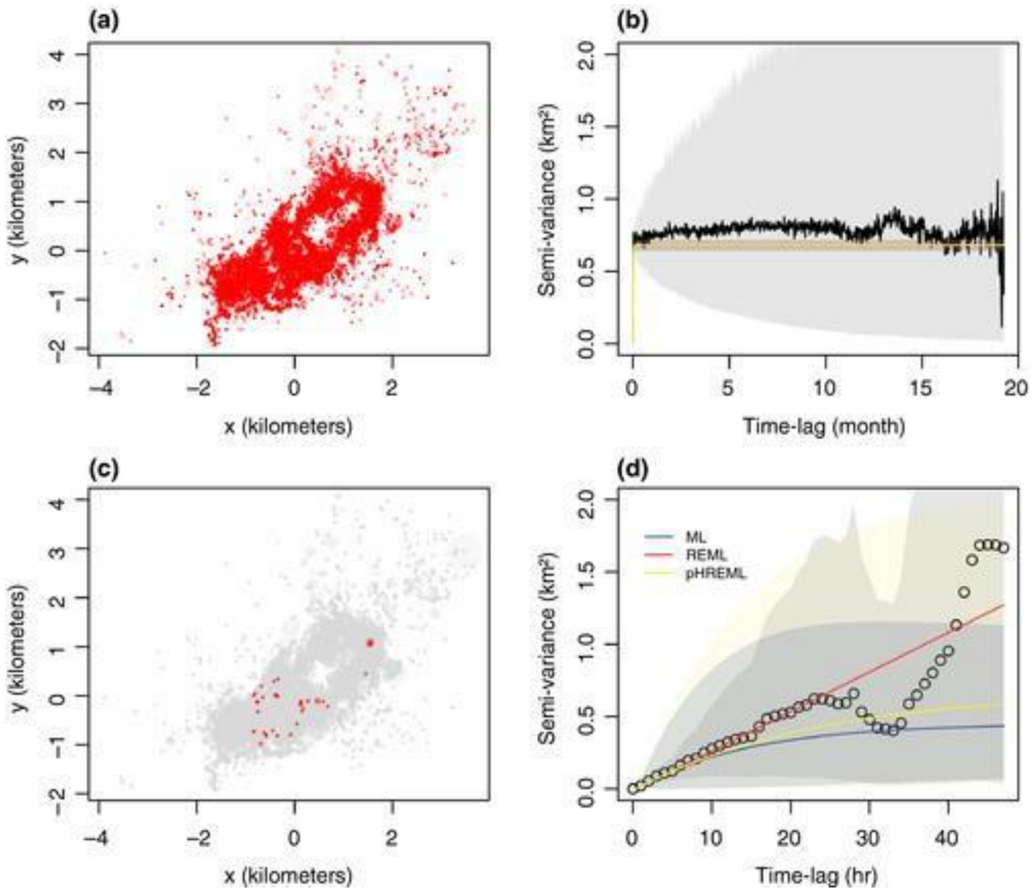
Mitigating bias

pHREML AKDE:



Mitigating bias

Tracking data: 1-hr intervals for *19 months*,
reduced to *2 days*.



Lowland tapir
(Tapirus terrestris)

Mitigating bias

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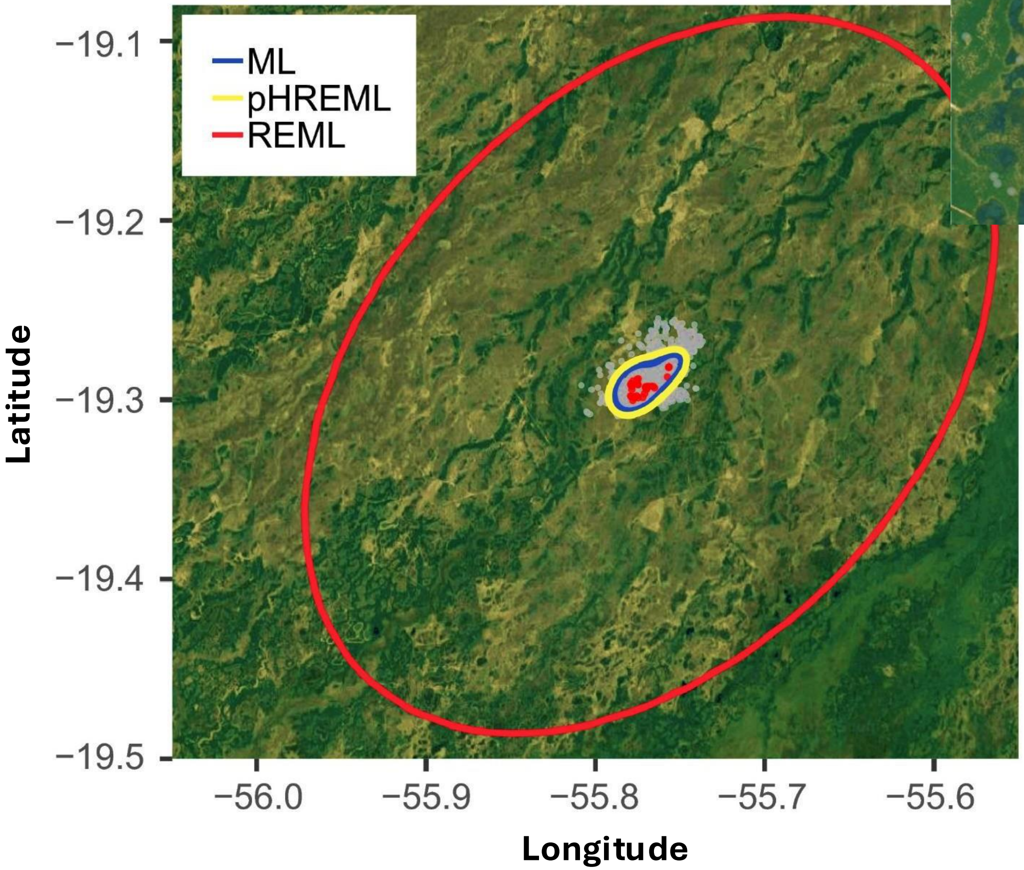

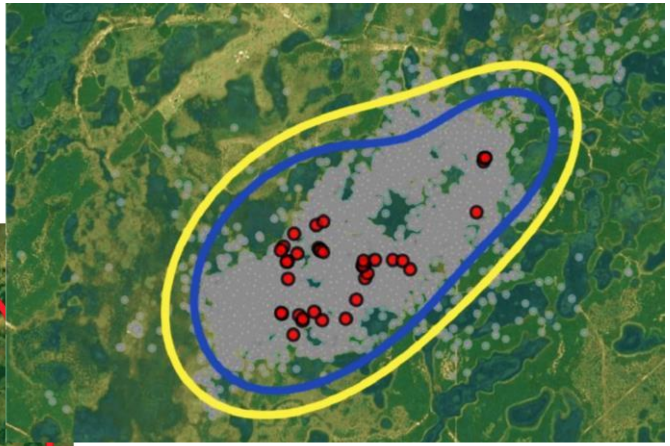

ML
pHREML
REML

Lowland tapir
(Tapirus terrestris)

Mitigating bias

Weighted AKDE (wAKDE):

- Deals with *unrepresentative sampling in time*
- Many real-world issues can lead to irregular sampling:
  - Duty-cycling tags to avoid wasting battery
  - Acceleration-informed sampling
  - Device malfunction
  - Habitat-related signal loss
  - … among other causes
- Shifting sampling schedules (based on behavioral or seasonal patterns) is also a common strategy

Mitigating bias

Weighted AKDE (wAKDE):

- wAKDE optimally upweights observations that occur during *under-sampled times*, while optimally downweighing observations occurring during *over-sampled times*

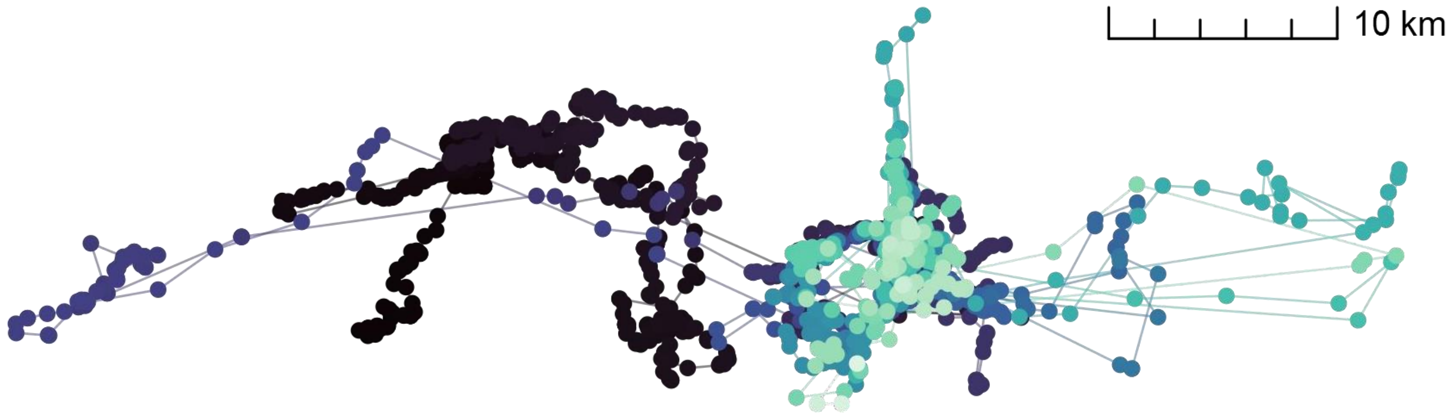

Fig. African buffalo dataset (nicknamed "Pepper") with an ***irregular*** sampling schedule.

Mitigating bias

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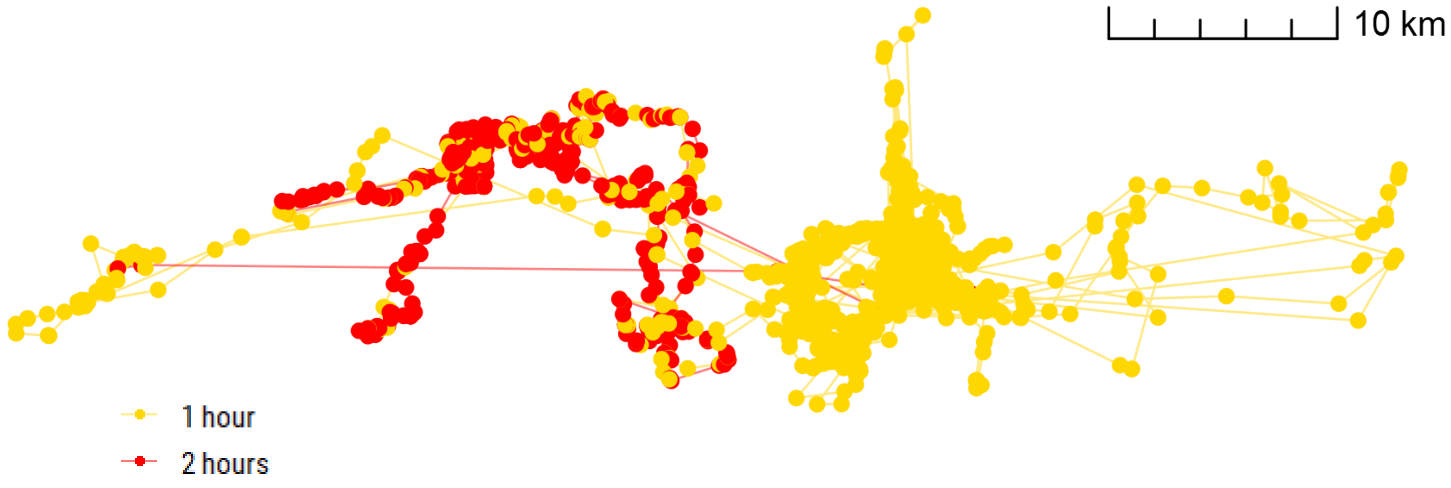

Fig. African buffalo dataset (nicknamed "Pepper") with an ***irregular*** sampling schedule. Sampling rate shifted from 1 fix every hour to 1 fix every 2 hours.

Mitigating bias

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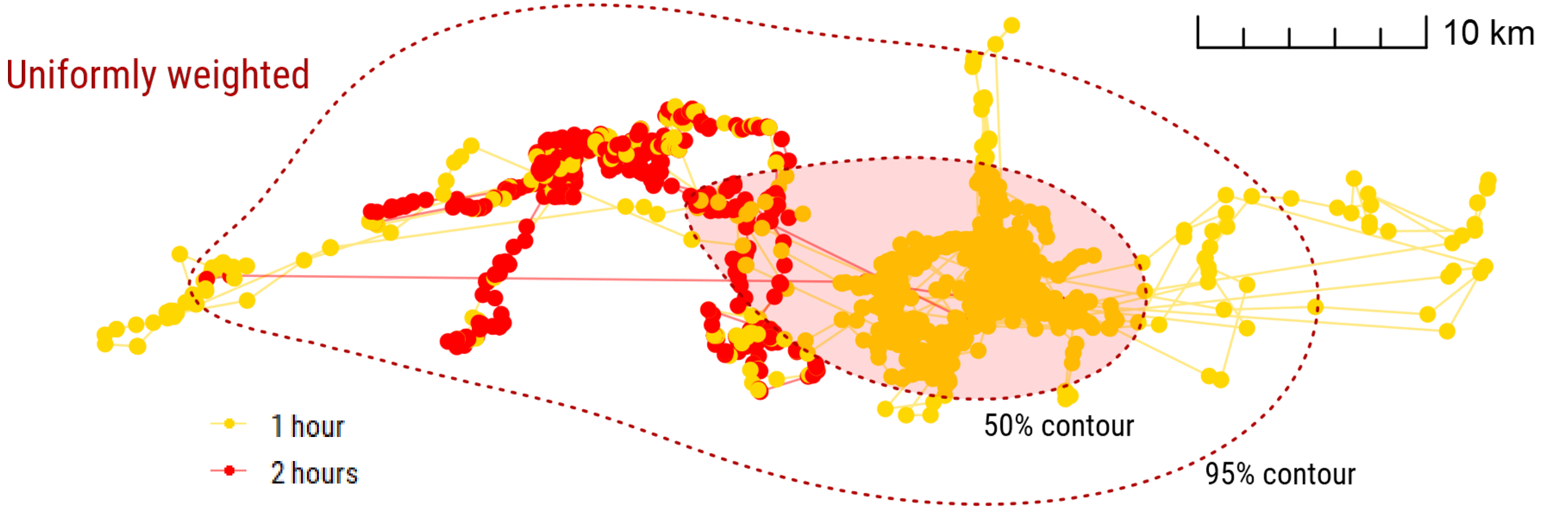

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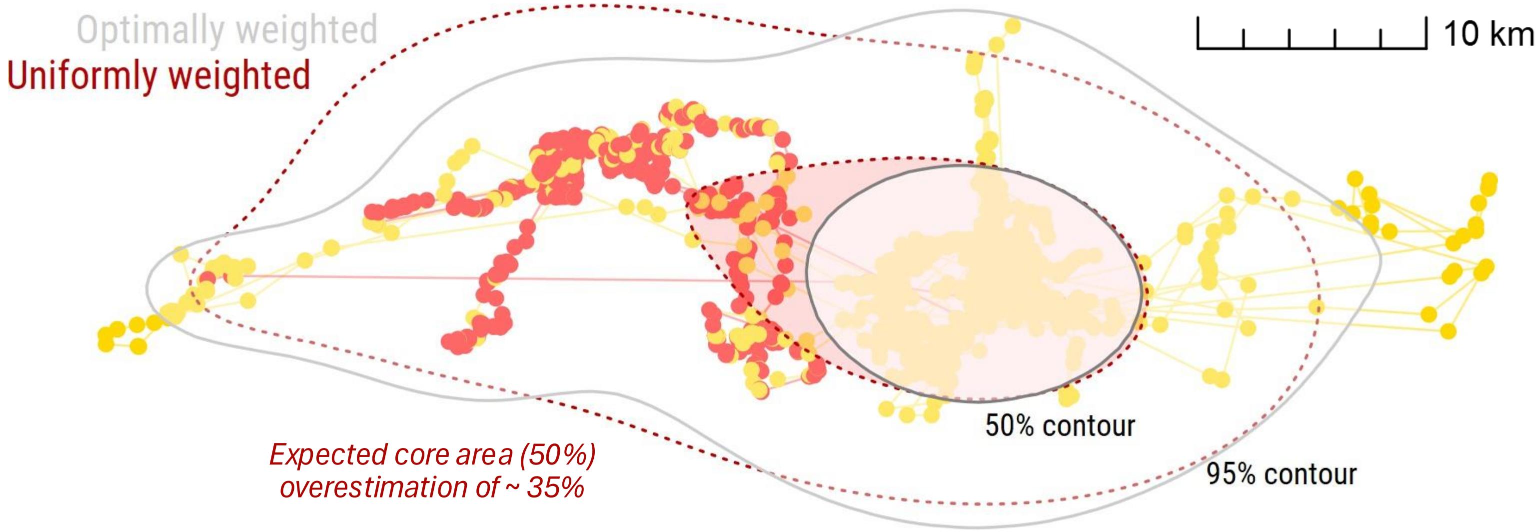


Fig. African buffalo dataset (nicknamed "Pepper") with an ***irregular*** sampling schedule.
Sampling rate shifted from 1 fix every hour to 1 fix every 2 hours.

Mitigating bias

Parametric bootstrapping AKDE:

- Deals with *very low effective sample size*

Bartlett (1937)

Residual ML (or REML)

Fleming *et al.* (2019)

perturbative Hybrid REML (pHREML)

+

Efron & Efron (1982)

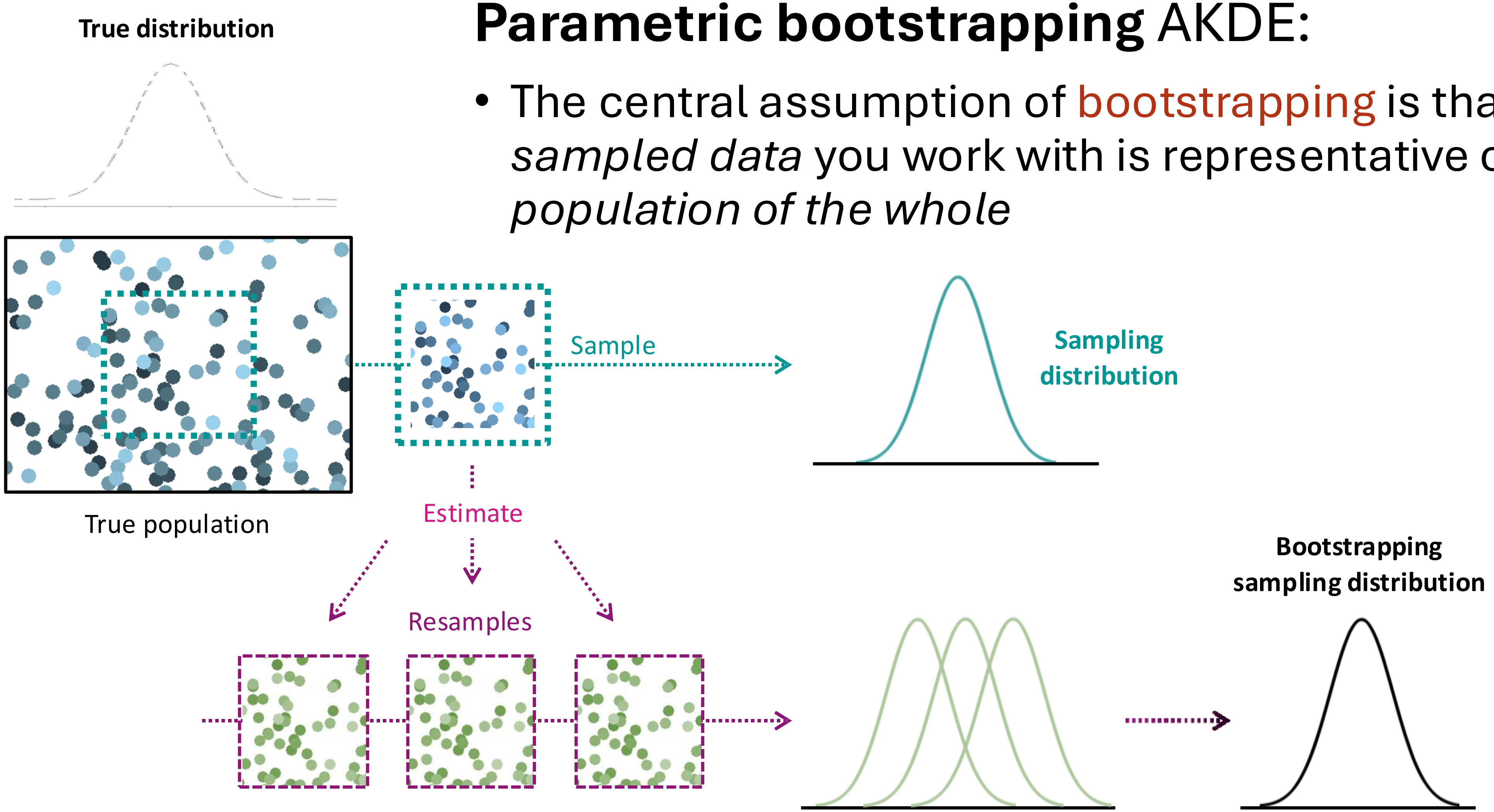
Parametric bootstrapping

The parametric bootstrap estimates the bias and variance of an estimator by *approximating* the sampling distribution of the true movement model with that of the best-fit model.

Mitigating bias

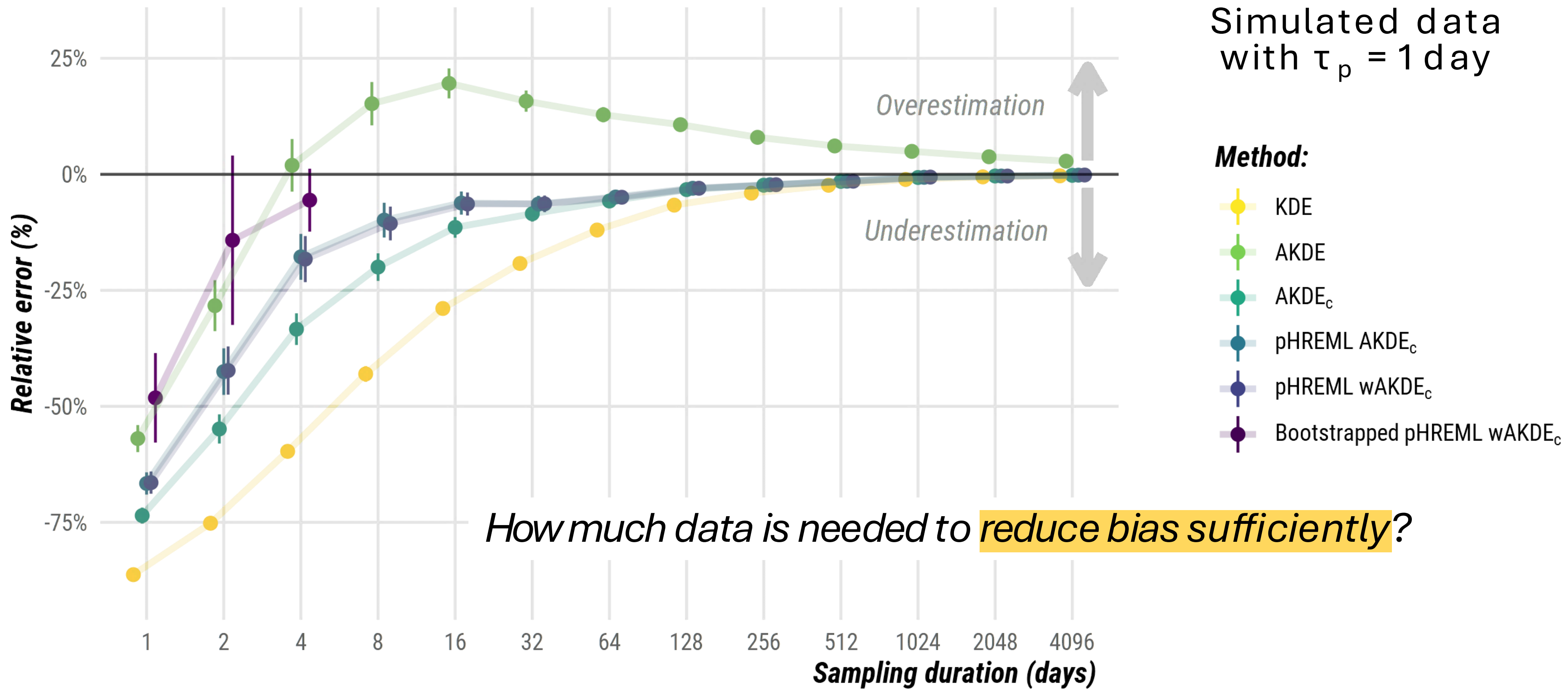
Parametric bootstrapping AKDE:

- The central assumption of bootstrapping is that the *sampled data* you work with is representative of the *population of the whole*



Mitigating bias

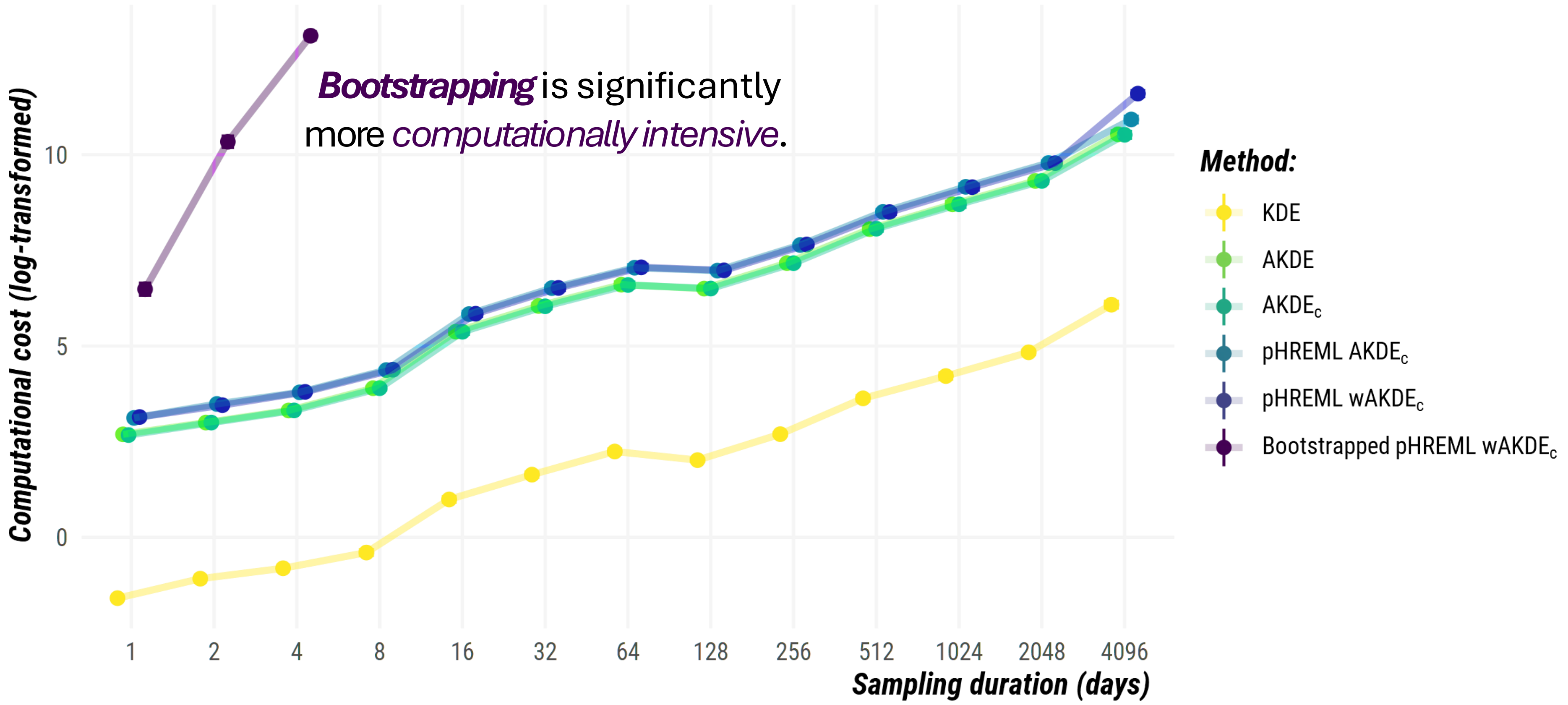
AKDE family relative error vs. sampling duration:



How much data is needed to reduce bias sufficiently?

Mitigating bias

AKDE family computational cost vs. sampling duration:



ctmm_akde.R

ctmm_effective_sample_sizes.R

Thank you

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- Funded by and supported by:

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CENTER FOR ADVANCED SYSTEMS UNDERSTANDING
UNIVERSITY OF MARYLAND 18 56